

Ordinary Differential Equations

► 1. INTRODUCTION

A great many applied problems involve rates, that is, derivatives. An equation containing derivatives is called a *differential equation*. If it contains partial derivatives, it is called a *partial differential equation*; otherwise it is called an *ordinary differential equation*. In this chapter we shall consider methods of solving ordinary differential equations which occur frequently in applications. Let us look at a few examples.

Newton's second law in vector form is $\mathbf{F} = m\mathbf{a}$. If we write the acceleration as $d\mathbf{v}/dt$, where $\mathbf{v}$ is the velocity, or as $d^2\mathbf{r}/dt^2$, where $\mathbf{r}$ is the displacement, we have a differential equation (or a set of differential equations, one for each component). Thus any mechanics problem in which we want to describe the motion of a body (automobile, electron, or satellite) under the action of a given force, involves the solution of a differential equation or a set of differential equations.

The rate at which heat Q escapes through a window or from a hot water pipe is proportional to the area A and to the rate of change of temperature with distance in the direction of flow of heat. Thus we have

$$(1.1) \quad \frac{dQ}{dt} = kA \frac{dT}{dx}$$

(k is called the thermal conductivity and depends on the material through which the heat is flowing). Here we have two different derivatives in the differential equation. In such a problem we might know either dT/dx or dQ/dt and solve the differential equation to find either T as a function of x , or Q as a function of t . (See Problems 2.23 to 2.25.)

Consider a simple series circuit (Figure 1.1) containing a resistance R , a capacitance C , an inductance L , and a source of emf V . If the current flowing around the circuit at time t is $I(t)$ and the charge on the capacitor is $q(t)$, then $I = dq/dt$. The voltage across R is RI , the voltage across C is q/C , and the voltage across L is

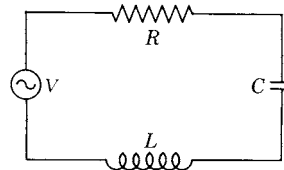


Figure 1.1

$L(dI/dt)$. Then at any time we must have

$$(1.2) \quad L \frac{dI}{dt} + RI + \frac{q}{C} = V.$$

If we differentiate this equation with respect to t and substitute $dq/dt = I$, we have

$$(1.3) \quad L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{I}{C} = \frac{dV}{dt}$$

as the differential equation satisfied by the current I in a simple series circuit with given L , R , and C , and a given $V(t)$.

There are many more examples of physical problems leading to differential equations; we shall consider some of them later in the text and problems. You might find it interesting at this point to browse through the problems to see the wide range of topics giving rise to differential equations.

The *order* of a differential equation is the order of the highest derivative in the equation. Thus the equations

$$(1.4) \quad \begin{aligned} y' + xy^2 &= 1, \\ xy' + y &= e^x, \\ \frac{dv}{dt} &= -g, \\ L \frac{dI}{dt} + RI &= V, \end{aligned}$$

are first-order equations, while (1.3) and

$$m \frac{d^2 r}{dt^2} = -kr$$

are second-order equations. A *linear* differential equation (with x as independent and y as dependent variable) is one of the form

$$a_0 y + a_1 y' + a_2 y'' + a_3 y''' + \cdots = b,$$

where the a 's and b are either constants or functions of x . The first equation in (1.4) is not linear because of the y^2 term; all the other equations we have mentioned so far are linear. Some other examples of nonlinear equations are:

$$\begin{aligned} y' &= \cot y && \text{(not linear because of the term } \cot y); \\ yy' &= 1 && \text{(not linear because of the product } yy'); \\ y'^2 &= xy && \text{(not linear because of the term } y'^2). \end{aligned}$$

Many of the differential equations which occur in applied problems are linear and of the first or second order; we shall be particularly interested in these.

A *solution* of a differential equation (in the variables x and y) is a relation between x and y which, if substituted into the differential equation, gives an identity.

► **Example 1.** The relation

$$(1.5) \quad y = \sin x + C$$

is a solution of the differential equation

$$(1.6) \quad y' = \cos x$$

because if we substitute (1.5) into (1.6) we get the identity $\cos x = \cos x$.

► **Example 2.** The equation $y'' = y$ has solutions $y = e^x$ or $y = e^{-x}$ or $y = Ae^x + Be^{-x}$ as you can verify by substitution.

If we integrate $y' = f(x)$, the expression for y , namely $y = \int f(x) dx + C$, contains one arbitrary constant of integration. If we integrate $y'' = g(x)$ twice to get $y(x)$, then y contains two independent integration constants. We might expect that in general the solution of a differential equation of the n th order would contain n independent arbitrary constants. Note that in Example 1 above, the solution of the first-order equation $y' = \cos x$ contained one arbitrary constant C , and in Example 2 the solution $y = Ae^x + Be^{-x}$ of the second-order equation $y'' = y$ contained two arbitrary constants A and B .

Any *linear* differential equation of order n has a solution containing n independent arbitrary constants, from which *all* solutions of the differential equation can be obtained by letting the constants have particular values. This solution is called the *general* solution of the linear differential equation.

(This may not be true for nonlinear equations; see Section 2.)

In applications, we usually want a *particular* solution, that is, one which satisfies the differential equation and some other requirements as well. Here are some examples of this.

► **Example 3.** Find the distance which an object falls under gravity in t seconds if it starts from rest.

Let x be the distance the object has fallen in time t . The acceleration of the object is g , the acceleration of gravity. Then we have

$$(1.7) \quad \frac{d^2x}{dt^2} = g.$$

Integrating, we get

$$(1.8) \quad \frac{dx}{dt} = gt + \text{const.} = gt + v_0,$$

$$(1.9) \quad x = \frac{1}{2}gt^2 + v_0t + x_0,$$

where v_0 and x_0 are the values of v and x at $t = 0$. Now (1.9) is the *general* solution of (1.7) (because it is a solution of a second-order linear differential equation and contains two independent arbitrary constants). We want the *particular* solution for which $v_0 = 0$ (since the object starts from rest), and $x_0 = 0$ (since the distance the object has fallen is zero at $t = 0$). Then the desired particular solution is

$$(1.10) \quad x = \frac{1}{2}gt^2.$$

- **Example 4.** Find the solution of $y'' = y$ which passes through the origin and through the point $(\ln 2, \frac{3}{4})$.

The general solution of the differential equation is

$$y = Ae^x + Be^{-x}$$

(see Example 2). If the given points satisfy the equation of the curve, we must have

$$\begin{aligned} 0 &= A + B \quad \text{or} \quad A = -B, \\ \frac{3}{4} &= Ae^{\ln 2} + Be^{-\ln 2} = A \cdot 2 + B \cdot \frac{1}{2} = 2A - \frac{1}{2}A = \frac{3}{2}A. \end{aligned}$$

Thus we get

$$A = -B = \frac{1}{2},$$

and the desired particular solution is

$$y = \frac{1}{2}(e^x - e^{-x}) = \sinh x.$$

The given conditions which are to be satisfied by the particular solution are called *boundary conditions*, or when they are conditions at $t = 0$ they may be called *initial conditions*. For linear equations, the desired particular solution can be found from the general solution by determining the values of the constants as we did in Example 4. (For nonlinear equations, see Section 2.)

As you study methods of solving various types of differential equations in the following sections, you may wonder whether you can use computer solutions and not bother to learn these techniques. Just as for indefinite integrals (see Chapter 5, Section 1), there may be various forms for the solution of a differential equation, and your computer may not give the one you need. In order to make intelligent use of computer solutions, you need to know something about what to expect, and an effective way of gaining this knowledge is to solve some equations by hand. (See Example 1, Section 3.) By comparing your solutions with computer solutions, you will learn what you can (and cannot) expect from your computer.

The graphing capabilities of your computer are very useful in differential equations. Consider a first-order equation, say $y' = f(x, y)$. If the solution of this differential equation is $y = y(x)$, the differential equation gives the slope y' of the solution curve at each point (x, y) . Suppose, for a large number of points, we draw a short line (or vector) centered on each point and with the slope y' at that point. (This would be a big job by hand, but your computer does it easily.) This plot is called a *slope field*, or a *direction field*, or a *vector field*. From such a diagram we can see the general trend of the solution curves even without solving the equation.

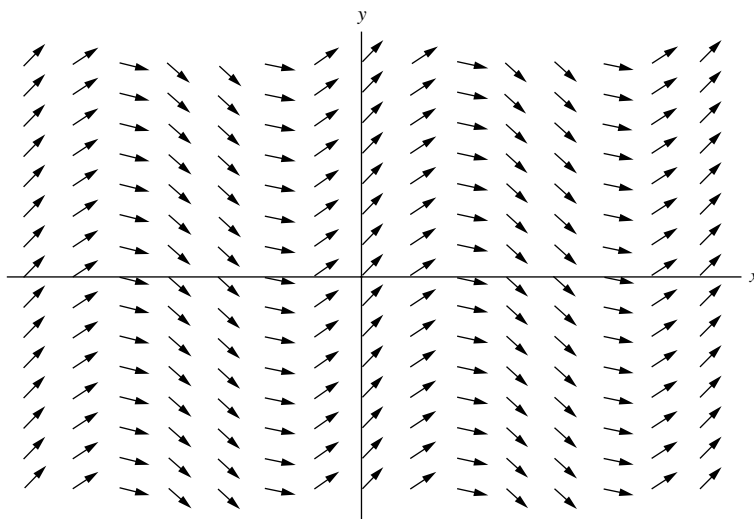


Figure 1.2

► **Example 5.** In Figure 1.2, we have plotted a “slope field” for the differential equation $y' = \cos x$. Note how you can trace the general shape of the solution curves, even without knowing from Example 1 that their equations are $y = \sin x + C$.

► PROBLEMS, SECTION 1

1. Verify the statement of Example 2. Also verify that $y = \cosh x$ and $y = \sinh x$ are solutions of $y'' = y$.
2. Solve Example 4 using the general solution $y = a \sinh x + b \cosh x$.
3. Verify that $y = \sin x$, $y = \cos x$, $y = e^{ix}$, and $y = e^{-ix}$ are all solutions of $y'' = -y$.
4. Find the distance which an object moves in time t if it starts from rest and has an acceleration $d^2x/dt^2 = ge^{-kt}$. Show that for small t the result is approximately (1.10), and for very large t , the speed dx/dt is approximately constant. The constant is called the terminal speed. (This problem corresponds roughly to the motion of a parachutist.)
5. Find the position x of a particle at time t if its acceleration is $d^2x/dt^2 = A \sin \omega t$.
6. A substance evaporates at a rate proportional to the exposed surface. If a spherical mothball of radius $\frac{1}{2}$ cm has radius 0.4 cm after 6 months, how long will it take:
 - (a) For the radius to be $\frac{1}{4}$ cm?
 - (b) For the volume of the mothball to be half of what it was originally?
7. The momentum p of an electron at speed v near the speed c of light increases according to the formula $p = mv/\sqrt{1-v^2/c^2}$, where m is a constant (mass of the electron). If an electron is subject to a constant force F , Newton's second law describing its motion is

$$\frac{dp}{dt} = \frac{d}{dt} \frac{mv}{\sqrt{1-v^2/c^2}} = F.$$

Find $v(t)$ and show that $v \rightarrow c$ as $t \rightarrow \infty$. Find the distance traveled by the electron in time t if it starts from rest.

► 2. SEPARABLE EQUATIONS

Every time you evaluate an integral

$$(2.1) \quad y = \int f(x) dx,$$

you are solving a differential equation, namely

$$(2.2) \quad y' = \frac{dy}{dx} = f(x).$$

This is a simple example of an equation which can be written with only y terms on one side of the equation and only x terms on the other:

$$(2.3) \quad dy = f(x) dx.$$

Whenever we can separate the variables this way, we call the equation *separable*, and we get the solution by just integrating each side of the equation.

- **Example 1.** The rate at which a radioactive substance decays is proportional to the remaining number of atoms. If there are N_0 atoms at $t = 0$, find the number at time t .

The differential equation for this problem is

$$(2.4) \quad \frac{dN}{dt} = -\lambda N.$$

(The proportionality constant λ is called the decay constant.) This is a separable equation; we write it as $dN/N = -\lambda dt$. Then integrating both sides, we get $\ln N = -\lambda t + \text{const.}$ Since we are given $N = N_0$ at $t = 0$, we see that the constant is $\ln N_0$. Solving for N , we have

$$(2.5) \quad N = N_0 e^{-\lambda t}.$$

(For further discussion of radioactive decay problems, see Section 3, Example 2, and Problems 2.19b and 3.19 to 3.21.)

- **Example 2.** Solve the differential equation

$$(2.6) \quad xy' = y + 1.$$

To separate variables, we divide both sides of (2.6) by $x(y + 1)$; this gives

$$(2.7) \quad \frac{y'}{y + 1} = \frac{1}{x} \quad \text{or} \quad \frac{dy}{y + 1} = \frac{dx}{x}.$$

Integrating each side of (2.7), we have

$$(2.8) \quad \ln(y + 1) = \ln x + \text{const.} = \ln x + \ln a = \ln(ax).$$

(We have called the constant of integration $\ln a$ for simplicity.) Then (2.8) gives the solution of (2.6), namely

$$(2.9) \quad y + 1 = ax.$$

This general solution represents a *family* of curves in the (x, y) plane, one curve for each value of the constant a . Or we may call the general solution (2.9) a *family of solutions* of the differential equation (2.6). Finding a particular solution means selecting one particular curve from the family.

Orthogonal Trajectories In Figure 2.1, the straight lines through $(0, -1)$ are the family of curves given by the solutions (2.9) of the differential equation (2.6). They might represent, for example, the lines of electric force due to an electric charge at $(0, -1)$. The circles in Figure 2.1 are then curves of constant electrostatic potential (called equipotentials—see Chapter 6, Sections 5 and 6). Note that the lines of force intersect the equipotential curves at right angles; each family of curves is called a set of *orthogonal trajectories* of the other family. It is often of interest to find the orthogonal trajectories of a given family of curves. Let us do this for the family (2.9). (In this case we know in advance that our answer will be the set of circles in Figure 2.1.)

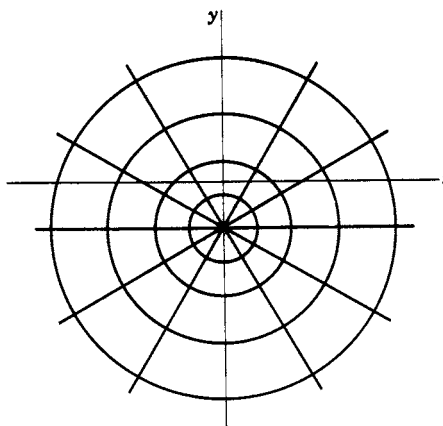


Figure 2.1

First we find the slope of a line of the family (2.9), namely,

$$(2.10) \quad y' = a.$$

For each a this gives the slope of *one* line. We want a formula (as a function of x and y) which gives the slope, at any point of the plane, of the line through that point. To obtain this, we eliminate a between (2.9) and (2.10) to get

$$(2.11) \quad y' = \frac{y+1}{x}.$$

[Or, given (2.6) rather than (2.9), we could simply solve for y' .] Now recall from analytic geometry that the slopes of two perpendicular lines are negative reciprocals. Then at each point we want the slope of the orthogonal trajectory curve to be the negative reciprocal of the slope of the line given by (2.11). Thus,

$$(2.12) \quad y' = -\frac{x}{y+1}$$

gives the slope of the orthogonal trajectories, and we solve (2.12) to obtain the equation of the orthogonal trajectory curves. Now (2.12) is separable; we obtain

$$\begin{aligned} (y+1) dy &= -x dx, \\ \frac{1}{2}y^2 + y &= -\frac{1}{2}x^2 + C, \\ x^2 + y^2 + 2y &= 2C, \\ x^2 + (y+1)^2 &= 2C+1. \end{aligned}$$

This is, as we expected, a family of circles with centers at the point $(0, -1)$.

Nonlinear Differential Equations We have said that for linear differential equations of order n there is always a general solution containing n independent constants, and *all* solutions can be obtained by specializing the constants. You should be aware that this may not be true for some nonlinear equations, and routine

methods of solution (including computer) may sometimes give partially incorrect or incomplete solutions. It is beyond our scope to discuss this in detail (see differential equations books), but here are some examples.

- **Example 3.** Solve the differential equation $y' = \sqrt{1 - y^2}$ and computer plot the slope field and a set of solution curves. Find particular solutions satisfying (a) $y = 0$ when $x = 0$, and (b) $y = 1$ when $x = 0$.

We separate variables and integrate to get

$$\frac{dy}{\sqrt{1 - y^2}} = dx, \quad \arcsin y = x + \alpha, \quad y = \sin(x + \alpha).$$

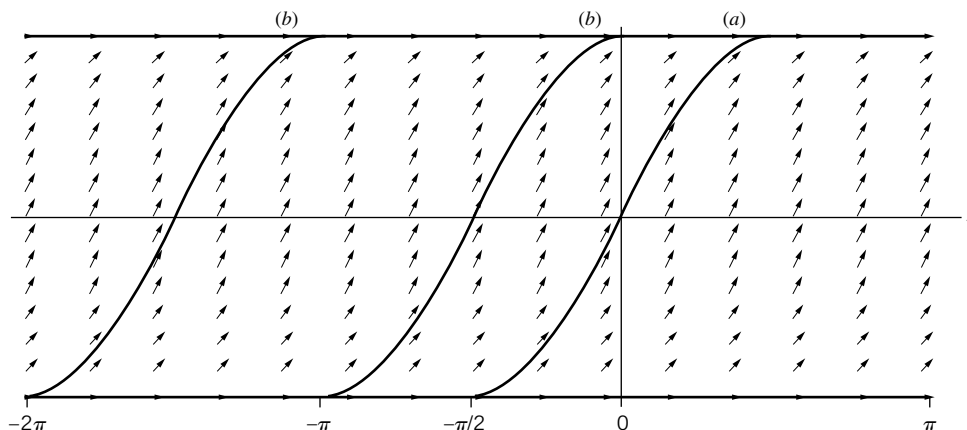


Figure 2.2

A computer gives the same answer. However if we look at either a computer plot of the slope field (Figure 2.2), or the differential equation itself, we see that the slope y' is always non-negative. Thus the solution of the given differential equation includes only the parts of the sine curves with non-negative slopes (Figure 2.2). A second difficulty is that part of the solution is missing. From either the slope field (Figure 2.2) or directly from the differential equation we can see that $y \equiv 1$ and $y \equiv -1$ are solutions not obtainable from the sine solution by any choice of α . (These are sometimes called *singular* solutions.) The fact that we did not find these solutions by separation of variables should not surprise us when we note that in separating variables we divided by $\sqrt{1 - y^2}$ and this step is not valid if $y^2 = 1$.

Now for the particular solution (a) passing through $(0, 0)$, the sine solution gives either $y = \sin x$ or $y = \sin(x + \pi) = -\sin x$. But since we know that y' is non-negative, only the $y = \sin x$ solution is correct in the vicinity of $x = 0$. In fact (Figure 2.2), $y = \sin x$ is a correct particular solution from $x = -\pi/2$ to $x = \pi/2$. We could construct a continuous solution from $-\infty$ to ∞ by letting $y = -1$ from $-\infty$ to $-\pi/2$, $y = \sin x$ from $-\pi/2$ to $\pi/2$, and $y = 1$ from $\pi/2$ to ∞ . Thus for case (a) [solution passing through the origin] we find just *one* particular solution.

For particular solution (b) [passing through $(0, 1)$], we find either $y = \sin(x + \frac{\pi}{2}) = \cos x$, or $y \equiv 1$; the $\cos x$ solution is valid from $x = -\pi$ to $x = 0$. As in (a), we can extend it by using parts of $y = -1$ and $y = 1$; this is one particular solution. But there are an infinite number of other particular solutions passing through $(0, 1)$ obtained by moving this one solution any distance to the left (Figure 2.2).

► PROBLEMS, SECTION 2

For each of the following differential equations, separate variables and find a solution containing one arbitrary constant. Then find the value of the constant to give a particular solution satisfying the given boundary condition. Computer plot a slope field and some of the solution curves.

- | | |
|---|---|
| 1. $xy' = y,$ | $y = 3$ when $x = 2.$ |
| 2. $x\sqrt{1-y^2}dx + y\sqrt{1-x^2}dy = 0,$ | $y = \frac{1}{2}$ when $x = \frac{1}{2}.$ |
| 3. $y' \sin x = y \ln y,$ | $y = e$ when $x = \pi/3.$ |
| 4. $(1+y^2)dx + xydy = 0,$ | $y = 0$ when $x = 5.$ |
| 5. $xy' - xy = y,$ | $y = 1$ when $x = 1.$ |
| 6. $y' = \frac{2xy^2 + x}{x^2y - y},$ | $y = 0$ when $x = \sqrt{2}.$ |
| 7. $ydy + (xy^2 - 8x)dx = 0,$ | $y = 3$ when $x = 1.$ |
| 8. $y' + 2xy^2 = 0,$ | $y = 1$ when $x = 2.$ |
| 9. $(1+y)y' = y,$ | $y = 1$ when $x = 1.$ |
| 10. $y' - xy = x,$ | $y = 1$ when $x = 0.$ |
| 11. $2y' = 3(y-2)^{1/3},$ | $y = 3$ when $x = 1.$ |
| 12. $(x+xy)y' + y = 0,$ | $y = 1$ when $x = 1.$ |

In Problems 13 to 15, find a solution (or solutions) of the differential equation not obtainable by specializing the constant in your solution of the original problem. *Hint:* See Example 3.

13. Problem 2. 14. Problem 8. 15. Problem 11.

16. By separation of variables, find a solution of the equation $y' = \sqrt{y}$ containing one arbitrary constant. Find a particular solution satisfying $y = 0$ when $x = 0$. Show that $y \equiv 0$ is a solution of the differential equation which cannot be obtained by specializing the arbitrary constant in your solution above. Computer plot a slope field and some of the solution curves. Show that there are an infinite number of solution curves passing through any point on the x axis, but just one through any point for which $y > 0$. *Hint:* See Example 3. Problems 17 and 18 are physical problems leading to this differential equation.
17. The speed of a particle on the x axis, $x \geq 0$, is always numerically equal to the square root of its displacement x . If $x = 0$ when $t = 0$, find x as a function of t . Show that the given conditions are satisfied if the particle remains at the origin for any arbitrary length of time t_0 and then moves away; find x for $t > t_0$ for this case.
18. Let the rate of growth dN/dt of a colony of bacteria be proportional to the square root of the number present at any time. If there are no bacteria present at $t = 0$, how many are there at a later time? Observe here that the routine separation of variables solution gives an unreasonable answer, and the correct answer, $N \equiv 0$, is not obtainable from the routine solution. (You have to think, not just follow rules!)

19. (a) Consider a light beam traveling downward into the ocean. As the beam progresses, it is partially absorbed and its intensity decreases. The rate at which the intensity is decreasing with depth at any point is proportional to the intensity at that depth. The proportionality constant μ is called the *linear absorption coefficient*. Show that if the intensity at the surface is I_0 , the intensity at a distance s below the surface is $I = I_0 e^{-\mu s}$. The linear absorption coefficient for water is of the order of 10^{-2} ft^{-1} (the exact value depending on the wavelength of the light and the impurities in the water). For this value of μ , find the intensity as a fraction of the surface intensity at a depth of 1 ft, 50 ft, 500 ft, 1 mile. When the intensity of a light beam has been reduced to half its surface intensity ($I = \frac{1}{2} I_0$), the distance the light has penetrated into the absorbing substance is called the *half-value thickness* of the substance. Find the half-value thickness in terms of μ . Find the half-value thickness for water for the value of μ given above.
- (b) Note that the differential equation and its solution in this problem are mathematically the same as those in Example 1, although the physical problem and the terminology are different. In discussing radioactive decay, we call λ the *decay constant*, and we define the *half-life* T of a radioactive substance as the time when $N = \frac{1}{2} N_0$ (compare half-value thickness). Find the relation between λ and T .
20. Consider the following special cases of the simple series circuit [Figure 1.1 and equation (1.2)].
- (a) RC circuit (that is, $L = 0$) with $V = 0$; find q as a function of t if q_0 is the charge on the capacitor at $t = 0$.
- (b) RL circuit (that is, no capacitor; this means $1/C = 0$) with $V = 0$; find $I(t)$ given $I = I_0$ at $t = 0$.
- (c) Again note that these are the same differential equations as in Problem 19 and Example 1. The terminology is again different; we define the time constant τ for a circuit as the time required for the charge (or current) to fall to $1/e$ times its initial value. Find the time constant for the circuits (a) and (b). If the same equation, say $y = y_0 e^{-at}$, represented either radioactive decay or light absorption or an RC or RL circuit, what would be the relations among the half-life, the half-value thickness, and the time constant?
21. Suppose the rate at which bacteria in a culture grow is proportional to the number present at any time. Write and solve the differential equation for the number N of bacteria as a function of time t if there are N_0 bacteria when $t = 0$. Again note that (except for a change of sign) this is the same differential equation and solution as in the preceding problems.
22. Solve the equation for the rate of growth of bacteria if the rate of increase is proportional to the number present but the population is being reduced at a constant rate by the removal of bacteria for experimental purposes.
23. Heat is escaping at a constant rate [dQ/dt in (1.1) is constant] through the walls of a long cylindrical pipe. Find the temperature T at a distance r from the axis of the cylinder if the inside wall has radius $r = 1$ and temperature $T = 100$ and the outside wall has $r = 2$ and $T = 0$.
24. Do Problem 23 for a spherical cavity containing a constant source of heat. Use the same radii and temperatures as in Problem 23.

25. Show that the thickness of the ice on a lake increases with the square root of the time in cold weather, making the following simplifying assumptions. Let the water temperature be a constant 10°C , the air temperature a constant -10° , and assume that at any given time the ice forms a slab of uniform thickness x . The rate of formation of ice is proportional to the rate at which heat is transferred from the water to the air. Let $t = 0$ when $x = 0$.
26. An object of mass m falls from rest under gravity subject to an air resistance proportional to its speed. Taking the y axis as positive down, show that the differential equation of motion is $m(dv/dt) = mg - kv$, where k is a positive constant. Find v as a function of t , and find the limiting value of v as t tends to infinity; this limit is called the *terminal speed*. Can you find the terminal speed directly from the differential equation without solving it? *Hint:* What is dv/dt after v has reached an essentially constant value?
- Consider the following specific examples of this problem.
- (a) A person drops from an airplane with a parachute. Find a reasonable value of k .
- (b) In the Millikan oil drop experiment to measure the charge of an electron, tiny electrically charged drops of oil fall through air under gravity or rise under the combination of gravity and an electric field. Measurements can be made only after they have reached terminal speed. Find a formula for the time required for a drop starting at rest to reach 99% of its terminal speed.
27. According to Newton's law of cooling, the rate at which the temperature of an object changes is proportional to the difference between its temperature and that of its surroundings. A cup of coffee at 200° in a room of temperature 70° is stirred continually and reaches 100° after 10 min. At what time was it at 120° ?
28. A glass of milk at 38° is removed from the refrigerator and left in a room at temperature 70° . If the temperature of the milk is 54° after 10 min, what will its temperature be in half an hour? (See Problem 27.)
29. A solution containing 90% by volume of alcohol (in water) runs at 1 gal/min into a 100-gal tank of pure water where it is continually mixed. The mixture is withdrawn at the rate of 1 gal/min. When will it start coming out 50% alcohol?
30. If P dollars are left in the bank at interest I percent per year compounded continuously, find the amount A at time t . *Hint:* Find dA , the interest on A dollars for time dt .

Find the orthogonal trajectories of each of the following families of curves. In each case, sketch or computer plot several of the given curves and several of their orthogonal trajectories. Be careful to eliminate the constant from y' for the original curves; this constant takes different values for different curves of the original family, and you want an expression for y' which is valid for all curves of the family crossed by the orthogonal trajectory you are trying to find. See equations (2.10) to (2.12).

31. $x^2 + y^2 = \text{const.}$
32. $y = kx^2$.
33. $y = kx^n$. (Assume that n is a given number; the different curves of the family have different values of k .)
34. $xy = k$.
35. $(y - 1)^2 = x^2 + k$.

► 3. LINEAR FIRST-ORDER EQUATIONS

A first-order equation contains y' but no higher derivatives. A *linear* first-order equation means one which can be written in the form

$$(3.1) \quad y' + Py = Q,$$

where P and Q are functions of x . To see how to solve (3.1), let us first consider the simpler equation when $Q = 0$. The equation

$$(3.2) \quad y' + Py = 0 \quad \text{or} \quad \frac{dy}{dx} = -Py$$

is separable. As in Section 2, we obtain the solution as follows:

$$(3.3) \quad \begin{aligned} \frac{dy}{y} &= -P dx, \\ \ln y &= -\int P dx + C, \\ y &= e^{-\int P dx + C} = Ae^{-\int P dx} \end{aligned}$$

where $A = e^C$. Let us simplify the notation for future use; we write

$$(3.4) \quad I = \int P dx.$$

Then

$$(3.5) \quad \frac{dI}{dx} = P$$

and we can write (3.3) as $y = Ae^{-I}$ or

$$(3.6) \quad ye^I = A.$$

We can now see how to solve (3.1). If we differentiate (3.6) with respect to x and use (3.5), we get

$$(3.7) \quad \frac{d}{dx}(ye^I) = y'e^I + ye^I \frac{dI}{dx} = y'e^I + ye^I P = e^I(y' + Py),$$

which is the left-hand side of (3.1) multiplied by e^I . (We call e^I an integrating factor—see Section 4.) Thus, we can write (3.1) (times e^I) as

$$(3.8) \quad \frac{d}{dx}(ye^I) = e^I(y' + Py) = Qe^I.$$

Since Q and e^I are functions of x only, we can now integrate both sides of (3.8) with respect to x to get

$$(3.9) \quad \left. \begin{aligned} ye^I &= \int Qe^I dx + c, & \text{or} \\ y &= e^{-I} \int Qe^I dx + ce^{-I}, \end{aligned} \right\} \quad \text{where} \quad I = \int P dx.$$

This is the general solution of (3.1). Note that it contains one arbitrary constant as expected for a first-order linear equation. The term ce^{-I} is a solution of equation (3.2); the first term in y is one particular solution of (3.1). Borrowing notation which we shall use in Section 6, let's call the term $ce^{-I} = y_c$ and the particular solution $= y_p$. Then $y_p + y_c$ is a solution of (3.1) for any value of c . Also note that $y_p e^I = \int Qe^I dx$ is an indefinite integral which, as we know (see Chapter 5, Section 1), has infinitely many answers differing from each other by constants of integration. Thus the particular solution obtained by you and by your computer may not be the same (see Example 1 and Problems).

► **Example 1.** Solve $(1 + x^2)y' + 6xy = 2x$. In the form of (3.1), this is

$$y' + \frac{6x}{1+x^2}y = \frac{2x}{1+x^2}.$$

From (3.9), we get

$$\begin{aligned} I &= \int \frac{6x}{1+x^2} dx = 3 \ln(1+x^2) \\ e^I &= e^{3 \ln(1+x^2)} = (1+x^2)^3 \\ ye^I &= \int \frac{2x}{1+x^2} (1+x^2)^3 dx = \int 2x(1+x^2)^2 dx = \frac{1}{3}(1+x^2)^3 + c \\ y &= \frac{1}{3} + \frac{c}{(1+x^2)^3}. \end{aligned}$$

A computer gives the answer

$$y = \frac{3x^2 + 3x^4 + x^6}{3(1+x^2)^3} + \frac{A}{(1+x^2)^3}.$$

Let us show that the answers agree (see comments just after (3.9)). If we put $A = c + 1/3$ in the computer solution above and combine terms, we get

$$y = \frac{3x^2 + 3x^4 + x^6 + 1}{3(1+x^2)^3} + \frac{c}{(1+x^2)^3} = \frac{(1+x^2)^3}{3(1+x^2)^3} + \frac{c}{(1+x^2)^3},$$

which, after cancelling, is our solution above. We see that the computer program chose a more complicated particular solution y_p which differed from our y_p by a multiple of $y_c = 1/(1+x^2)^3$. Always be aware of the possibility of simplifying a particular solution by adding a multiple of y_c .

► **Example 2.** Radium decays to radon which decays to polonium. If at $t = 0$, a sample is pure radium, how much radon does it contain at time t ?

$$\begin{aligned} \text{Let } N_0 &= \text{number of radium atoms at } t = 0, \\ N_1 &= \text{number of radium atoms at time } t, \\ N_2 &= \text{number of radon atoms at time } t, \\ \lambda_1 \text{ and } \lambda_2 &= \text{decay constants for Ra and Rn.} \end{aligned}$$

As in Section 2, we have for radium

$$\frac{dN_1}{dt} = -\lambda_1 N_1, \quad N_1 = N_0 e^{-\lambda_1 t}.$$

The rate at which radon is being created is the rate at which radium is decaying, namely $\lambda_1 N_1$ or $\lambda_1 N_0 e^{-\lambda_1 t}$. But the radon is also decaying at the rate $\lambda_2 N_2$. Hence, we have

$$\begin{aligned}\frac{dN_2}{dt} &= \lambda_1 N_1 - \lambda_2 N_2, \quad \text{or} \\ \frac{dN_2}{dt} + \lambda_2 N_2 &= \lambda_1 N_1 = \lambda_1 N_0 e^{-\lambda_1 t}.\end{aligned}$$

This equation is of the form (3.1), and we solve it as follows:

$$\begin{aligned}(3.10) \quad I &= \int \lambda_2 dt = \lambda_2 t, \\ N_2 e^{\lambda_2 t} &= \int \lambda_1 N_0 e^{-\lambda_1 t} e^{\lambda_2 t} dt + c \\ &= \lambda_1 N_0 \int e^{(\lambda_2 - \lambda_1)t} dt + c = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1} e^{(\lambda_2 - \lambda_1)t} + c,\end{aligned}$$

if $\lambda_1 \neq \lambda_2$. (For the case $\lambda_1 = \lambda_2$, see Problem 19.) Since $N_2 = 0$ at $t = 0$ (we assumed pure Ra at $t = 0$), we must have

$$0 = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1} + c \quad \text{or} \quad c = -\frac{\lambda_1 N_0}{\lambda_2 - \lambda_1}.$$

Substituting this value of c into (3.10) and solving for N_2 , we get

$$N_2 = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}).$$

► PROBLEMS, SECTION 3

Using (3.9), find the general solution of each of the following differential equations. Compare a computer solution and, if necessary, reconcile it with yours. *Hint:* See comments just after (3.9), and Example 1.

- | | |
|--|---|
| 1. $y' + y = e^x$ | 2. $x^2 y' + 3xy = 1$ |
| 3. $dy + (2xy - xe^{-x^2}) dx = 0$ | 4. $2xy' + y = 2x^{5/2}$ |
| 5. $y' \cos x + y = \cos^2 x$ | 6. $y' + y/\sqrt{x^2 + 1} = 1/(x + \sqrt{x^2 + 1})$ |
| 7. $(1 + e^x)y' + 2e^x y = (1 + e^x)e^x$ | 8. $(x \ln x)y' + y = \ln x$ |
| 9. $(1 - x^2)y' = xy + 2x\sqrt{1 - x^2}$ | 10. $y' + y \tanh x = 2e^x$ |
| 11. $y' + y \cos x = \sin 2x$ | 12. $\frac{dx}{dy} = \cos y - x \tan y$ |
| 13. $dx + (x - e^y) dy = 0$ | 14. $\frac{dy}{dx} = \frac{3y}{3y^{2/3} - x}$ |

Hint: For Problems 12 to 14, solve for x in terms of y .

15. Water with a small salt content (5 lb in 1000 gal) is flowing into a very salty lake at the rate of $4 \cdot 10^5$ gal per hr. The salty water is flowing out at the rate of 10^5 gal per hr. If at some time (say $t = 0$) the volume of the lake is 10^9 gal, and its salt content is 10^7 lb, find the salt content at time t . Assume that the salt is mixed uniformly with the water in the lake at all times.
16. Find the general solution of (1.2) for an RL circuit ($1/C = 0$) with $V = V_0 \cos \omega t$ ($\omega = \text{const.}$).
17. Find the general solution of (1.3) for an RC circuit ($L = 0$), with $V = V_0 \cos \omega t$.
18. Do Problems 16 and 17 using $V = V_0 e^{i\omega t}$, and find the solutions for 16 and 17 by taking real parts of the complex solutions.
19. If $\lambda_1 = \lambda_2 = \lambda$ in (3.10), then $\int e^{(\lambda_2 - \lambda_1)t} dt = \int dt$. Find N_2 for this case.
20. Extend the radioactive decay problem (Example 2) one more stage, that is, let λ_3 be the decay constant of polonium and find how much polonium there is at time t .
21. Generalize Problem 20 to any number of stages.
22. Find the orthogonal trajectories of the family of curves $x = y + 1 + ce^y$. (See the instructions above Problem 2.31.)
23. Find the orthogonal trajectories of the family of curves $y = -e^{x^2} \operatorname{erf} x + Ce^{x^2}$. *Hint:* See Chapter 11, equation (9.1) for definition of $\operatorname{erf} x$, and Chapter 4, Section 12, for differentiation of an integral. Solve for x in terms of y .

► 4. OTHER METHODS FOR FIRST-ORDER EQUATIONS

Separable equations and linear equations are the two types of first-order equations you are most apt to meet in elementary applications. However, we shall also mention briefly a few other methods of solving special first-order equations. You will find more details in the problems and in most differential equations books.

The Bernoulli Equation The differential equation

$$(4.1) \quad y' + Py = Qy^n,$$

where P and Q are functions of x , is known as the Bernoulli equation. It is not linear but is easily reduced to a linear equation. We make the change of variable

$$(4.2) \quad z = y^{1-n}.$$

Then

$$(4.3) \quad z' = (1-n)y^{-n}y'.$$

Next multiply (4.1) by $(1-n)y^{-n}$ and make the substitutions (4.2) and (4.3) to get

$$\begin{aligned} (1-n)y^{-n}y' + (1-n)Py^{1-n} &= (1-n)Q, \\ z' + (1-n)Pz &= (1-n)Q. \end{aligned}$$

This is now a first-order linear equation which we can solve as we did the linear equations above. (See Section 7 for an example of a physical problem in which we need to solve a Bernoulli equation.)

Exact Equations; Integrating Factors Recall from Chapter 6, Section 8, that the expression $P(x, y) dx + Q(x, y) dy$ is an *exact differential* [that is, the differential of a function $F(x, y)$] if

$$(4.4) \quad \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

If (4.4) holds, then there is a function $F(x, y)$ such that

$$(4.5) \quad P = \frac{\partial F}{\partial x}, \quad Q = \frac{\partial F}{\partial y}, \quad P dx + Q dy = dF.$$

In Chapter 6 we considered ways of finding F when (4.4) holds. The differential equation

$$(4.6) \quad P dx + Q dy = 0 \quad \text{or} \quad y' = -\frac{P}{Q}$$

is called *exact* if (4.4) holds. In this case

$$P dx + Q dy = dF = 0,$$

and the solution of (4.6) is then

$$(4.7) \quad F(x, y) = \text{const.}$$

We find F as in Chapter 6, Section 8.

An equation which is not exact may often be made exact by multiplying it by an appropriate factor.

► **Example 1.** The equation

$$(4.8) \quad x dy - y dx = 0$$

is not exact [by (4.4)]. But the equation

$$(4.9) \quad \frac{x dy - y dx}{x^2} = \frac{1}{x} dy - \frac{y}{x^2} dx = d\left(\frac{y}{x}\right) = 0,$$

obtained by dividing (4.8) by x^2 , is exact [use (4.4)], and its solution is

$$(4.10) \quad \frac{y}{x} = \text{const.}$$

We multiplied (4.8) by $1/x^2$ to make the equation exact; the factor $1/x^2$ is called an *integrating factor*. To see another example of an integrating factor, look back at Section 3. The expression e^f is an integrating factor for equations (3.1) and (3.2); as you can see in (3.8), multiplying (3.1) by e^f makes it an exact equation.

The method of finding an integrating factor and solving the resulting exact equation is useful mainly in simple cases when we can see the result by inspection. It is not usually worth while to spend much time searching for integrating factors.

Homogeneous Equations A *homogeneous function* of x and y of degree n means a function which can be written as $x^n f(y/x)$. For example, $x^3 - xy^2 = x^3[1 - (y/x)^2]$ is a homogeneous function of degree 3. (Also see Problem 21.) An equation of the form

$$(4.11) \quad P(x, y) dx + Q(x, y) dy = 0,$$

where P and Q are homogeneous functions of *the same degree* is called *homogeneous*. (The term homogeneous is also used in another sense; see Section 5.) If we divide two homogeneous functions of the same degree, the x^n factors cancel and we have a function of y/x . Thus, from (4.11) we can write

$$(4.12) \quad y' = \frac{dy}{dx} = -\frac{P(x, y)}{Q(x, y)} = f\left(\frac{y}{x}\right),$$

and we can say that a differential equation is homogeneous if it can be written as $y' =$ a function of y/x . This suggests that we solve homogeneous equations by making the change of variables $v = y/x$, or

$$(4.13) \quad y = xv.$$

This substitution does, in fact, give us a separable equation in x and v (see Problem 22). We solve it to find a relation between v and x and then put back $v = y/x$ to find the solution of (4.11).

Also see Problem 23 for another way to solve homogeneous equations.

Change of Variables We have solved both Bernoulli equations and homogeneous equations by making changes of variables. Other equations may yield to this method also. If a differential equation contains some combination of the variables x, y (especially if this combination appears more than once), we try replacing this combination by a new variable. See Problems 11, 15, and 16 for examples.

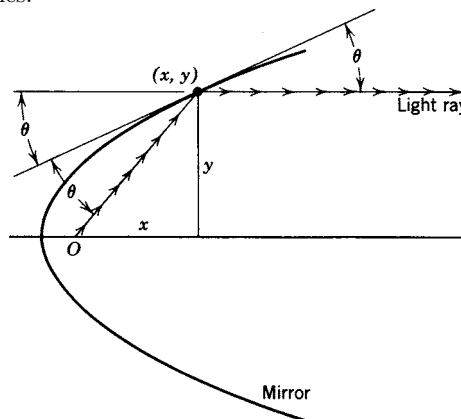
► PROBLEMS, SECTION 4

Use the methods of this section to solve the following differential equations. Compare computer solutions and reconcile differences.

1. $y' + y = xy^{2/3}$
2. $y' + \frac{1}{x}y = 2x^{3/2}y^{1/2}$
3. $3xy^2y' + 3y^3 = 1$
4. $(2xe^{3y} + e^x) dx + (3x^2e^{3y} - y^2) dy = 0$
5. $(x - y) dy + (y + x + 1) dx = 0$
6. $(\cos x \cos y + \sin^2 x) dx - (\sin x \sin y + \cos^2 y) dy = 0$
7. $x^2 dy + (y^2 - xy) dx = 0$
8. $y dy = (-x + \sqrt{x^2 + y^2}) dx$
9. $xy dx + (y^2 - x^2) dy = 0$
10. $(y^2 - xy) dx + (x^2 + xy) dy = 0$
11. $y' = \cos(x + y)$ *Hint:* Let $u = x + y$; then $u' = 1 + y'$.
12. $y' = \frac{y}{x} - \tan \frac{y}{x}$
13. $yy' - 2y^2 \cot x = \sin x \cos x$

14. $(x-1)y' + y - x^{-2} + 2x^{-3} = 0$ 15. $xy' + y = e^{xy}$ *Hint: Let $u = xy$*
16. Solve the differential equation $yy'^2 + 2xy' - y = 0$ by changing from variables y, x , to r, x , where $y^2 = r^2 - x^2$; then $yy' = rr' - x$.
17. If an incompressible fluid flows in a corner bounded by walls meeting at the origin at an angle of 60° , the streamlines of the flow satisfy the equation $2xy dx + (x^2 - y^2) dy = 0$. Find the streamlines.
18. Find the family of orthogonal trajectories of the circles $(x-h)^2 + y^2 = h^2$. (See the instructions above Problem 2.31.)
19. Find the family of curves satisfying the differential equation $(x+y) dy + (x-y) dx = 0$ and also find their orthogonal trajectories.

20. Find the shape of a mirror which has the property that rays from a point O on the axis are reflected into a parallel beam. *Hint: Take the point O at the origin. Show from the figure that $\tan 2\theta = y/x$. Use the formula for $\tan 2\theta$ to express this in terms of $\tan \theta = dy/dx$ and solve the resulting differential equation. (*Hint: See Problem 16.*)*



21. As in text just before (4.11), show that
- $x^2 - 5xy + y^3/x$ is a homogeneous function of degree 2;
 - $x^{-1}(y^4 - x^3y) - xy^2 \sin(x/y)$ is homogeneous of degree 3;
 - $x^2y^3 + x^5 \ln(y/x) - y^6/\sqrt{x^2 + y^2}$ is homogeneous of degree 5;
 - $x^2 + y$, $x + \cos y$, and $y + 1$ are not homogeneous.

See Chapter 4, Section 13, Problem 1 for a more general definition of a homogeneous function of any number of variables.

22. Show that the change of variables (4.13) in (4.11) or (4.12) gives a separable equation. *Hints: Substitute $y = xv$ and $dy = x dv + v dx$ from (4.13) into (4.12) and rearrange terms to get the equation*

$$(a) \quad [f(v) - v] dx = x dv.$$

Alternatively, suppose P and Q are homogeneous of degree n ; that is $P(x, y) = x^n P(1, y/x) = x^n P(1, v)$ and a similar equation for Q . Substitute these results and $dy = x dv + v dx$ into (4.11), divide by x^n , and rearrange terms to get

$$(b) \quad [P(1, v) + Q(1, v)v] dx + Q(1, v)x dv = 0.$$

Write both (a) and (b) with variables separated.

23. Show that $(xP + yQ)^{-1}$ is an integrating factor for (4.11). *Hint: You want to show that $(P dx + Q dy)/(xP + yQ)$ is an exact differential (see Chapter 6, Section 8). Remember that P and Q are homogeneous of the same degree. Divide numerator and denominator by Q and use $P/Q = -f(y/x)$ from (4.12). Now find the needed partial derivatives. *Comment: If $(xP + yQ)$ turns out to be very simple, this may be an easier way to solve a homogeneous equation than the $v = y/x$ substitution (see Problem 24).**

24. Solve Problems 9 and 10 by using an integrating factor as discussed in Problem 23.
25. An equation of the form $y' = f(x)y^2 + g(x)y + h(x)$ is called a *Riccati* equation. If we know one particular solution y_p , then the substitution $y = y_p + \frac{1}{z}$ gives a linear first-order equation for z . We can solve this for z and substitute back to find a solution of the y equation containing one arbitrary constant (see Problem 26). Following this method, check the given y_p , and then solve
- (a) $y' = xy^2 - \frac{2}{x}y - \frac{1}{x^3}, \quad y_p = \frac{1}{x^2};$
- (b) $y' = \frac{2}{x}y^2 + \frac{1}{x}y - 2x, \quad y_p = x;$
- (c) $y' = e^{-x}y^2 + y - e^x, \quad y_p = e^x.$
26. Show that the substitution given in Problem 25 does in general give a solution of the Riccati equation. *Hints:* First show that the substitution $y = y_p + u$ yields the following equation for u : $u' - (g + 2fy_p)u = fu^2$. Note by text equation (4.1) that this is a Bernoulli equation with $n = 2$, so by equation (4.2) we let $z = u^{-1}$. Show that the z equation is the linear first-order equation $z' + (g + 2fy_p)z = -f$. Note that we could have obtained the z equation in one step by substituting $y = y_p + z^{-1}$ in the original equation as claimed in Problem 25.

► 5. SECOND-ORDER LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS AND ZERO RIGHT-HAND SIDE

Because of their importance in applications, we are going to consider carefully the solution of differential equations of the form

$$(5.1) \quad a_2 \frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = 0,$$

where a_2, a_1, a_0 are constants; also we shall consider (Section 6) the corresponding equation when the right-hand side of (5.1) is a function of x . Equations of the form (5.1) are called *homogeneous* because every term contains y or a derivative of y . Equations of the form (6.1) are called *inhomogeneous* because they contain a term which does not depend on y . (Note, however, that this use of the term homogeneous is completely unrelated to its use in Section 4.) Although we shall concentrate on second-order equations, which are the ones that occur most frequently in applications, most of our discussion can be extended immediately to linear equations of higher order with constant coefficients (see Problems 21 to 30).

These problems are pretty simple by hand; you may be able to write down answers faster than you can type the problem into a computer! Remember that a computer may not give an answer in the form you need. To use computer solutions effectively, you need to know what to expect, and you can learn this by studying the following methods and doing some problems by hand. Let us consider an equation of the form (5.1).

► **Example 1.** Solve the equation

$$(5.2) \quad y'' + 5y' + 4y = 0.$$

It is convenient to let D stand for d/dx ; then

$$(5.3) \quad Dy = \frac{dy}{dx} = y', \quad D^2y = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2} = y''.$$

Expressions involving D , such as $D + 1$ or $D^2 + 5D + 4$, are called *differential operators*. (See Problem 31.) In this notation (5.2) becomes

$$(5.4) \quad D^2y + 5Dy + 4y = 0 \quad \text{or} \quad (D^2 + 5D + 4)y = 0.$$

The *algebraic* expression $D^2 + 5D + 4$ can be factored as $(D + 1)(D + 4)$ or $(D + 4)(D + 1)$. You should satisfy yourself that

$$(5.5) \quad (D + 1)(D + 4)y = (D + 4)(D + 1)y = (D^2 + 5D + 4)y$$

when $D = d/dx$, and, in fact, that a similar statement is true for $(D - a)(D - b)$ where a and b are any *constants*. (This is not necessarily true if a and b are functions of x ; see Problem 31.) Then we can write (5.2) or (5.4) as

$$(5.6) \quad (D + 1)(D + 4)y = 0 \quad \text{or} \quad (D + 4)(D + 1)y = 0.$$

To solve (5.4) [or (5.6) which is the same equation rewritten], we shall first solve the simpler equations

$$(5.7) \quad (D + 4)y = 0 \quad \text{and} \quad (D + 1)y = 0.$$

These are separable equations (Section 2) with solutions

$$(5.8) \quad y = c_1 e^{-4x}, \quad y = c_2 e^{-x}.$$

Now if $(D + 4)y = 0$, then

$$(D + 1)(D + 4)y = (D + 1) \cdot 0 = 0,$$

so any solution of $(D + 4)y = 0$ is a solution of the differential equation (5.6) or (5.4). Similarly, any solution of $(D + 1)y = 0$ is a solution of (5.6) or (5.4). Since the two solutions (5.8) are linearly independent [Problem 13; also see Chapter 3, equation (8.5)], a linear combination of them contains two arbitrary constants and so is the general solution. Thus

$$(5.9) \quad y = c_1 e^{-4x} + c_2 e^{-x}$$

is the general solution of (5.4). Note that we can think of the two solutions e^{-4x} and e^{-x} as basis vectors of a 2-dimensional linear vector space (see Chapter 3, Section 14). Then the general solution (5.9) gives all the vectors of that space. (See Problem 21.)

Now we must investigate whether we can solve all second-order linear equations with constant coefficients (and zero right-hand side) by this method. We first wrote the differential equation using D for d/dx , and then factored the D expression to get (5.5). In this last step, we treated D as if it were an algebraic letter instead of d/dx ; this is justified by checking the result (5.5) when $D = d/dx$. Recall from algebra that saying that the algebraic expression $D^2 + 5D + 4$ has the factors $(D + 4)$ and $(D + 1)$ is equivalent to saying that the quadratic equation

$$(5.10) \quad D^2 + 5D + 4 = 0$$

has roots -4 and -1 . The equation (5.10) is called the *auxiliary* (or characteristic) equation for the given differential equation (5.2). From equations (5.6) to (5.9), we

see that to solve a linear second-order equation with constant coefficients, we should first solve the auxiliary equation; if the roots of the auxiliary equation are a and b ($a \neq b$), the general solution of the differential equation is a linear combination of e^{ax} and e^{bx} .

$$(5.11) \quad y = c_1 e^{ax} + c_2 e^{bx} \text{ is the general solution of } (D-a)(D-b)y = 0, \quad a \neq b.$$

(If $a = b$, we get only one solution this way; we shall consider this case shortly.) Recall from algebra that the roots of a quadratic equation (with real coefficients; see Problem 19) can be real and unequal, real and equal, or a complex conjugate pair. The equation (5.2) which we have solved is an example in which the roots are real and unequal. Let us consider the other two cases.

Equal Roots of the Auxiliary Equation If the two roots of the auxiliary equation are equal, then the differential equation can be written

$$(5.12) \quad (D-a)(D-a)y = 0,$$

where a is the value of the two equal roots. From our previous discussion (5.5) to (5.11), we know that one solution of (5.12) is $y = c_1 e^{ax}$. But our previous second solution $y = c_2 e^{bx}$ in (5.11) is not a second solution here since $b = a$. To find the second solution for this case, we let

$$(5.13) \quad u = (D-a)y.$$

Then (5.12) becomes

$$(D-a)u = 0,$$

from which we get

$$(5.14) \quad u = Ae^{ax}.$$

We substitute (5.14) into (5.13) to get

$$(D-a)y = Ae^{ax} \quad \text{or} \quad y' - ay = Ae^{ax}.$$

This is a first-order linear equation which we solve as in Section 3:

$$ye^{-ax} = \int e^{-ax} Ae^{ax} dx = \int A dx = Ax + B.$$

Thus

$$(5.15) \quad y = (Ax + B)e^{ax} \text{ is the general solution of (5.12).}$$

This is the general solution of (5.1) for the case of equal roots of the auxiliary equation. The solution e^{ax} we already know; what is new here is the fact that xe^{ax} is a second (linearly independent; see Problem 14) solution of the differential equation when a is a double root of the auxiliary equation. Equations (5.11) and (5.15) then give the general solution of (5.1) for both unequal and equal roots of the auxiliary equation.

Complex Conjugate Roots of the Auxiliary Equation Suppose the roots of the auxiliary equation are $\alpha \pm i\beta$. These are unequal roots, so by (5.11) the general solution of the differential equation is

$$(5.16) \quad y = Ae^{(\alpha+i\beta)x} + Be^{(\alpha-i\beta)x} = e^{\alpha x}(Ae^{i\beta x} + Be^{-i\beta x}).$$

There are two other very useful forms of (5.16). If we substitute $e^{\pm i\beta x} = \cos \beta x \pm i \sin \beta x$ [see Chapter 2, equation (9.3)] into (5.16), then the parenthesis becomes a linear combination of $\sin \beta x$ and $\cos \beta x$ and we can write (5.16) as

$$(5.17) \quad y = e^{\alpha x}(c_1 \sin \beta x + c_2 \cos \beta x),$$

where c_1 and c_2 are new arbitrary constants. We can also write (5.17) in the form

$$(5.18) \quad y = ce^{\alpha x} \sin(\beta x + \gamma),$$

where c and γ are now the arbitrary constants. An easy way to see that this is correct is to expand $\sin(\beta x + \gamma)$ by the trigonometric addition formula; this gives a linear combination of $\sin \beta x$ and $\cos \beta x$ as in (5.17). Although it is not hard to express any one of the sets of arbitrary constants [A, B in (5.16); c_1, c_2 in (5.17); and c, γ in (5.18)] in terms of either of the other sets, there is seldom any need to do this. In solving actual problems we simply write whichever one of the three forms seems best for the problem at hand and then determine the arbitrary constants in that form from the given data.

► **Example 2.** Solve the differential equation

$$(5.19) \quad y'' - 6y' + 9y = 0.$$

We can write the equation as

$$(5.20) \quad (D^2 - 6D + 9)y = 0 \quad \text{or} \quad (D - 3)(D - 3)y = 0.$$

Since the roots of the auxiliary equation are equal, we know that the solution is of the form (5.15) and we simply write the result

$$(5.21) \quad y = (Ax + B)e^{3x}.$$

- **Example 3.** In Section 16, Chapter 2, we discussed the differential equation for the motion of a mass m oscillating at the end of a spring, and we solved it by guessing the solution. Now let's solve it by the methods of this chapter. The differential equation is [see Chapter 2, equation (16.21)]

$$(5.22) \quad m \frac{d^2 y}{dt^2} = -ky \quad \text{or} \quad \frac{d^2 y}{dt^2} = -\frac{k}{m}y = -\omega^2 y \quad \text{if} \quad \omega^2 = \frac{k}{m}.$$

We can write this differential equation as

$$(5.23) \quad D^2 y + \omega^2 y = 0 \quad \text{or} \quad (D^2 + \omega^2)y = 0$$

where $D = d/dt$. The roots of the auxiliary equation are $D = \pm i\omega$; the solution may be written in any of the three forms, (5.16), (5.17), or (5.18):

$$(5.24) \quad \begin{aligned} y &= Ae^{i\omega t} + Be^{-i\omega t} \\ &= c_1 \sin \omega t + c_2 \cos \omega t \\ &= c \sin(\omega t + \gamma). \end{aligned}$$

An object whose displacement from equilibrium satisfies (5.22) or (5.24) is said to be executing *simple harmonic motion*. (Recall Chapter 7, Section 2.)

Equations (5.24) are general solutions of (5.22), each containing two arbitrary constants. Let us find a particular solution corresponding to given initial conditions.

- **Example 4.** Suppose the mass is held at rest at a distance 10 cm below equilibrium and then suddenly let go. If we agree to call y positive when m is above the equilibrium position, then at $t = 0$, we have $y = -10$, and $dy/dt = 0$. Using the second solution in (5.24), we get

$$\frac{dy}{dt} = c_1 \omega \cos \omega t - c_2 \omega \sin \omega t,$$

so the initial conditions give

$$\begin{aligned} -10 &= c_1 \cdot 0 + c_2 \cdot 1, \\ 0 &= c_1 \omega \cdot 1 - c_2 \omega \cdot 0. \end{aligned}$$

Thus we find

$$c_1 = 0, \quad c_2 = -10,$$

and the particular solution we wanted is

$$(5.25) \quad y = -10 \cos \omega t.$$

You can verify that either of the other solutions in (5.24) gives the same particular solution (5.25) for the same initial conditions (Problem 32).

This solution is pretty unrealistic from the practical viewpoint. Equations (5.24) and (5.25) imply that the mass m , once started, will simply oscillate up and down forever! This is certainly not true; what *will* happen is that the oscillations will gradually die down. The reason for the discrepancy between the physical facts and our mathematical answer is that we have neglected “friction” forces.

► **Example 5.** A fairly reasonable assumption for this problem and many other similar ones is that there is a retarding force proportional to the velocity; let us call this force $-l(dy/dt)$ ($l > 0$). Then (5.22), revised to include this force, becomes

$$(5.26) \quad m \frac{d^2 y}{dt^2} = -ky - l \frac{dy}{dt} \quad (l > 0)$$

or with the abbreviations

$$\omega^2 = \frac{k}{m}, \quad 2b = \frac{l}{m} \quad (b > 0)$$

it is

$$(5.27) \quad \frac{d^2 y}{dt^2} + 2b \frac{dy}{dt} + \omega^2 y = 0.$$

To solve (5.27), we find the roots of the auxiliary equation

$$(5.28) \quad D^2 + 2bD + \omega^2 = 0,$$

which are

$$(5.29) \quad D = \frac{-2b \pm \sqrt{4b^2 - 4\omega^2}}{2} = -b \pm \sqrt{b^2 - \omega^2}.$$

There are three possible types of answer here depending on the relative size of b^2 and ω^2 , and there are three special names given to the corresponding types of motion. We say that the motion is

overdamped if	$b^2 > \omega^2,$
critically damped if	$b^2 = \omega^2,$
underdamped or oscillatory if	$b^2 < \omega^2.$

Let us discuss the corresponding general solutions of the differential equation for the three cases.

Overdamped Motion Since $\sqrt{b^2 - \omega^2}$ is real and less than b , both roots of the auxiliary equation are negative, and the general solution is a linear combination of two negative exponentials:

$$(5.30) \quad y = Ae^{-\lambda t} + Be^{-\mu t}, \quad \text{where} \quad \begin{cases} \lambda = b + \sqrt{b^2 - \omega^2}, \\ \mu = b - \sqrt{b^2 - \omega^2}. \end{cases}$$

Critically Damped Motion Since $b = \omega$, the auxiliary equation has equal roots and the general solution is

$$(5.31) \quad y = (A + Bt)e^{-bt}.$$

In both overdamped and critically damped motion, the mass m is subject to such a large retarding force that it slows down and returns to equilibrium rather than oscillating repeatedly.

Underdamped or Oscillatory Motion In this case $b^2 < \omega^2$ so $\sqrt{b^2 - \omega^2}$ is imaginary. Let $\beta = \sqrt{\omega^2 - b^2}$; then $\sqrt{b^2 - \omega^2} = i\beta$ and the roots (5.29) of the auxiliary equation are $-b \pm i\beta$. The general solution in the form (5.17) is then

$$(5.32) \quad y = e^{-bt}(A \sin \beta t + B \cos \beta t)$$

This result is more in accord with what we know actually happens to the mass m ; because of the factor e^{-bt} , the oscillations in this case decrease in amplitude as time goes on. Also note that the frequency of the damped vibrations, namely $\beta = \sqrt{\omega^2 - b^2}$, is less than the frequency ω of the undamped vibrations.

Although we have stated a rather special physical problem, the mathematics we have just discussed applies to a great variety of problems. First, there are many kinds of mechanical vibrations besides a mass attached to a spring. Think of a tuning fork, a pendulum, the needle on the scale of a measuring device, and as more involved examples, the vibrations of complicated structures such as bridges or airplanes, and the vibrations of atoms in a crystal lattice. In such problems, we need to solve differential equations similar to the ones we have discussed. Differential equations of the same form arise in electricity. Consider equations (1.2) and (1.3) when $V = 0$. Remembering that $I = dq/dt$, we can write (1.2) as

$$(5.33) \quad L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = 0$$

and (1.3) as

$$(5.34) \quad L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{1}{C} I = 0.$$

Both these equations are of the form (5.27) which we have solved. Thus there is an analogy between a series circuit and the motion of a mass m described by (5.26); L corresponds to m , R to the “friction” constant l , and $1/C$ to the spring constant k .

► PROBLEMS, SECTION 5

Solve the following differential equations by the methods discussed above and compare computer solutions.

- | | |
|---------------------------|---------------------------|
| 1. $y'' + y' - 2y = 0$ | 2. $y'' - 4y' + 4y = 0$ |
| 3. $y'' + 9y = 0$ | 4. $y'' + 2y' + 2y = 0$ |
| 5. $(D^2 - 2D + 1)y = 0$ | 6. $(D^2 + 16)y = 0$ |
| 7. $(D^2 - 5D + 6)y = 0$ | 8. $D(D + 5)y = 0$ |
| 9. $(D^2 - 4D + 13)y = 0$ | 10. $y'' - 2y' = 0$ |
| 11. $4y'' + 12y' + 9 = 0$ | 12. $(2D^2 + D - 1)y = 0$ |

Recall from Chapter 3, equation (8.5), that a set of functions is linearly independent if their Wronskian is not identically zero. Calculate the Wronskian of each of the following sets to show that in each case they are linearly independent. For each set, write the differential equation of which they are solutions. Also note that each set of functions is a set of basis functions for a linear vector space (see Chapter 3, Section 14, Example 2) and that the general solution of the differential equation gives all vectors of the vector space.

13. e^{-x}, e^{-4x} 14. $e^{ax}, e^{bx}, a \neq b$ (a, b , real or complex)
15. e^{ax}, xe^{ax} 16. $\sin \beta x, \cos \beta x$
17. $1, x, x^2$ 18. $e^{ax}, xe^{ax}, x^2 e^{ax}$
19. Solve the algebraic equation

$$D^2 + (1 + 2i)D + i - 1 = 0$$

(note the complex coefficients) and observe that the roots are complex but not complex conjugates. Show that the method of solution of (5.6) (case of unequal roots) is correct here, and so find the general solution of

$$y'' + (1 + 2i)y' + (i - 1)y = 0.$$

20. As in Problem 19, solve $y'' + (1 - i)y' - iy = 0$. *Hint:* See Chapter 2, Section 10, for a method of finding the square root of a complex number.
21. By the method used in solving (5.4) to get (5.9), show that the solution of the third-order equation

$$(D - a)(D - b)(D - c)y = 0$$

is

$$y = c_1 e^{ax} + c_2 e^{bx} + c_3 e^{cx}$$

if a, b, c are all different, and find the solutions if two or three of the roots of the auxiliary equation are equal. Generalize the result to higher-order equations. State your results in vector space language [see comment following equation (5.9)].

Use the results of Problem 21 to find the general solutions of the following equations and compare computer solutions.

22. $(D - 1)(D + 3)(D + 5)y = 0$
23. $(D^2 + 1)(D^2 - 1)y = 0$ *Hint:* $D^2 + 1 = (D + i)(D - i)$.
24. $y''' + y = 0$
25. $(D^3 + D^2 - 6D)y = 0$
26. $y''' - 3y'' - 9y' - 5y = 0$
27. $D^2(D - 1)^2(D + 2)^3y = 0$
28. $(D^4 + 4)y = 0$ *Hint:* Find the four 4th roots of -4 (see Chapter 2, Section 10).
29. $(D + 1)^2(D^4 - 16)y = 0$
30. $(D^4 - 1)^2y = 0$
31. Let D stand for d/dx , that is, $Dy = dy/dx$; then

$$D^2y = D(Dy) = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2}, \quad D^3y = \frac{d^3y}{dx^3}, \text{ etc.}$$

D (or an expression involving D) is called a differential operator. Two operators are equal if they give the same results when they operate on y . For example,

$$D(D + x)y = \frac{d}{dx} \left(\frac{dy}{dx} + xy \right) = \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = (D^2 + xD + 1)y$$

so we say that

$$D(D + x) = D^2 + xD + 1.$$

In a similar way show that:

- (a) $(D - a)(D - b) = (D - b)(D - a) = D^2 - (b + a)D + ab$ for constant a and b .
- (b) $D^3 + 1 = (D + 1)(D^2 - D + 1)$.
- (c) $Dx = xD + 1$. (Note that D and x do not commute, that is, $Dx \neq xD$.)
- (d) $(D - x)(D + x) = D^2 - x^2 + 1$, but $(D + x)(D - x) = D^2 - x^2 - 1$.

Comment: The operator equations in (c) and (d) are useful in quantum mechanics; see Chapter 12, Section 22.

- 32. In Example 3, we used the second solution in (5.24), and obtained (5.25) as the particular solution satisfying the given initial conditions. Show that the first and third solutions in (5.24) also give the particular solution (5.25) satisfying the given initial conditions.
- 33. A particle moves along the x axis subject to a force toward the origin proportional to x (say $-kx$). Show that the particle executes simple harmonic motion (Example 3). Find the kinetic energy $\frac{1}{2}mv^2$ and the potential energy $\frac{1}{2}kx^2$ as functions of t and show that the total energy is constant. Find the time averages of the potential energy and the kinetic energy and show that these averages are each equal to one-half the total energy (see average values, Chapter 7, Section 4).
- 34. Find the equation of motion of a simple pendulum (see Chapter 7, Problem 2.13), that is, the differential equation for θ as a function of t . Show that, for small θ , this is approximately a simple harmonic motion equation, and find θ if $\theta = \theta_0$, $d\theta/dt = 0$ when $t = 0$.
- 35. The gravitational force on a particle of mass m inside the earth at a distance r from the center ($r < \text{the radius of the earth } R$) is $F = -mgr/R$ (Chapter 6, Section 8, Problem 21). Show that a particle placed in an evacuated tube through the center of the earth would execute simple harmonic motion. Find the period of this motion.
- 36. Find (in terms of L and C) the frequency of electrical oscillations in a series circuit (Figure 1.1) if $R = 0$ and $V = 0$, but $I \neq 0$. (When you tune a radio, you are adjusting C and/or L to make this frequency equal to that of the radio station.)
- 37. A block of wood is floating in water; it is depressed slightly and then released to oscillate up and down. Assume that the top and bottom of the block are parallel planes which remain horizontal during the oscillations and that the sides of the block are vertical. Show that the period of the motion (neglecting friction) is $2\pi\sqrt{h/g}$, where h is the vertical height of the part of the block under water when it is floating at rest. *Hint:* Recall that the buoyant force is equal to the weight of displaced water.
- 38. Solve the RLC circuit equation [(5.33) or (5.34)] with $V = 0$ as we did (5.27), and write the conditions and solutions for overdamped, critically damped, and underdamped electrical oscillations in terms of the quantities R , L , and C .
- 39. (a) Find numerical values of the constants and computer plot together on the same axes graphs of (5.30), (5.31) and (5.32) in order to compare overdamped, critically damped, and oscillatory motion. *Suggested numbers:* Let $\omega = 1$, and $b = 13/5$, 1 , $5/13$ for the three kinds of motion. Let $y(0) = 1$ and $y'(0) = 0$.
 (b) Repeat the problem with the same set of ω and b values and with $y(0) = 1$, but with $y'(0) = 1$.
 (c) Again repeat, with $y'(0) = -1$.
- 40. The natural period of an undamped system is 3 sec, but with a damping force proportional to the velocity, the period becomes 5 sec. Find the differential equation of motion of the system and its solution.

► 6. SECOND-ORDER LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS AND RIGHT-HAND SIDE NOT ZERO

So far we have considered second-order linear equations with constant coefficients and zero right-hand side (5.1). Such equations describe *free vibrations* or oscillations of mechanical or electrical systems. But often such systems are not free but are subject to an applied force or emf. The vibrations are then called *forced vibrations* and the differential equation describing the system is of the form

$$(6.1) \quad \begin{aligned} a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y &= f(x), \quad \text{or} \\ \frac{d^2 y}{dx^2} + \frac{a_1}{a_2} \frac{dy}{dx} + \frac{a_0}{a_2} y &= F(x). \end{aligned}$$

The function $f(x)$ is often called the forcing function; it represents the applied force or emf. We want to find the general solution of equations of the form (6.1).

► **Example 1.** Consider the equation

$$(6.2) \quad (D^2 + 5D + 4)y = \cos 2x.$$

We already know (from Section 5, Example 1) the general solution of the corresponding equation (5.2) with the right-hand side equal to zero. This solution (5.9) is called the *complementary function*; it is not a solution of (6.2) but is related to it as we shall see. We shall denote the complementary function by y_c . Thus for equation (6.2) the complementary function is

$$(6.3) \quad y_c = Ae^{-x} + Be^{-4x}.$$

Now suppose we know just any solution of (6.2); we call this solution a *particular solution* and denote it by y_p . You can easily verify that

$$(6.4) \quad y_p = \frac{1}{10} \sin 2x$$

is a particular solution of (6.2), and we shall soon consider ways of finding such solutions. Then we have

$$(6.5) \quad (D^2 + 5D + 4)y_p = \cos 2x$$

and from Section 5, Example 1,

$$(6.6) \quad (D^2 + 5D + 4)y_c = 0.$$

Adding (6.5) and (6.6), we find

$$(D^2 + 5D + 4)(y_p + y_c) = \cos 2x + 0 = \cos 2x.$$

Thus

$$(6.7) \quad y = y_c + y_p = Ae^{-x} + Be^{-4x} + \frac{1}{10} \sin 2x$$

is a solution of (6.2). In fact, it is the general solution of (6.2) since it contains two independent arbitrary constants (Problem 27).

Thus we see how to solve (6.1):

The general solution of an equation of the form (6.1) is

$$(6.8) \quad y = y_c + y_p$$

where the complementary function y_c is the general solution of the homogeneous equation (as in Section 5) and y_p is a particular solution of (6.1).

We shall now discuss some ways of finding particular solutions. It is worthwhile to know about this even if you are using a computer to find the solution. When you know what to expect, you are better able to judge whether a computer solution is in the best form for your purposes, and if not, to find a better form. (See problems.)

Inspection If there *is* a very simple particular solution, we may be able to guess and verify it.

► **Example 2.** Consider the equation $y'' - 2y' + 3y = 5$.

It is easy to see that $y_p = \frac{5}{3}$ is a particular solution of this equation since if y is constant, y'' and y' are zero.

► **Example 3.** As a less trivial problem, consider

$$(6.9) \quad y'' - 6y' + 9y = 8e^x.$$

We might suspect that a multiple of e^x is a solution of this equation, and it is easy to verify that $y = 2e^x$ is a solution. But trying the same method for the equation

$$(6.10) \quad y'' + y' - 2y = e^x,$$

we fail to find a particular solution since e^x satisfies

$$y'' + y' - 2y = 0.$$

The method of inspection is very good in simple cases where it gives us an answer quickly, but usually we need other methods.

Successive Integration of Two First-Order Equations This is a straightforward method which can always be used to solve equations of the form (6.1). In practice, however, it often involves more work than various special methods; we shall find it particularly useful in deriving the special methods.

► **Example 4.** Let's solve (6.10) again. We can write this differential equation as

$$(6.11) \quad (D - 1)(D + 2)y = e^x.$$

Let

$$(6.12) \quad u = (D + 2)y.$$

Then the differential equation (6.11) becomes

$$(6.13) \quad (D - 1)u = e^x \quad \text{or} \quad u' - u = e^x.$$

This is a first-order linear differential equation which we solve as in Section 3.

$$(6.14) \quad \begin{aligned} I &= \int -dx = -x, \\ ue^{-x} &= \int e^{-x} e^x dx = x + c_1, \\ u &= xe^x + c_1 e^x. \end{aligned}$$

Then the differential equation for y becomes

$$(D + 2)y = xe^x + c_1 e^x \quad \text{or} \quad y' + 2y = xe^x + c_1 e^x.$$

This is again a linear first-order equation which we solve as follows:

$$(6.15) \quad \begin{aligned} I &= \int 2 dx = 2x, \\ ye^{2x} &= \int e^{2x}(xe^x + c_1 e^x) dx = \frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x} + \frac{1}{3}c_1 e^{3x} + c_2 \\ &= \frac{1}{3}xe^{3x} + c_1' e^{3x} + c_2, \\ y &= \frac{1}{3}xe^x + c_1' e^x + c_2 e^{-2x}. \end{aligned}$$

Notice that here we have obtained the general solution all in one process rather than finding the complementary function plus a particular solution in two separate processes. However, we could have obtained just the particular solution $xe^x/3$ by omitting the arbitrary constant at each integration (these led to the complementary function) and also dropping terms which are already in the complementary function ($-e^x/9$ in this example). Since it is easy to write the complementary function (by Section 5), it saves time to omit those terms when we are finding a particular solution. You may find that your computer gives a more complicated particular solution by including terms of the complementary function in the particular solution. Now that you know to watch for this, you can simplify a computer solution by removing those terms.

Exponential Right-Hand Side Let us consider how to find a particular solution when the right-hand side of (6.1) is $F(x) = ke^{cx}$ where k and c are given constants. Observe that c may be complex; we shall be especially interested in this case later. Let a and b be the roots of the auxiliary equation of (6.1); then (6.1) becomes

$$(6.16) \quad (D - a)(D - b)y = F(x) = ke^{cx}.$$

Let us first suppose that c is not equal to either a or b . Solving (6.16) by successive integration of two first-order equations as in the last paragraph is straightforward (Problem 28) and gives the result that the particular solution in this case is simply a multiple of e^{cx} . It is not necessary to remember the formula for the constant factor or to go through this process each time. Now that we know the form of the particular solution, we simply assume a solution of this form and solve for the constant.

► **Example 5.** Solve the equation

$$(6.17) \quad (D - 1)(D + 5)y = 7e^{2x}.$$

We observe that $c = 2$ is not equal to either of the roots of the auxiliary equation. To find a particular solution we substitute $y_p = Ce^{2x}$ into (6.17) and get

$$y_p'' + 4y_p' - 5y_p = C(4e^{2x} + 8e^{2x} - 5e^{2x}) = 7e^{2x}.$$

Thus we must have $C = 1$, and the general solution of (6.17) is

$$y = Ae^x + Be^{-5x} + e^{2x}.$$

We have already seen in solving (6.11) that if c is equal to either a or b ($a \neq b$), the particular solution is of the form Cxe^{cx} . By the same method used for (6.11), you can easily discover that if $a = b = c$, the particular solution is of the form Cx^2e^{cx} (Problem 28c). In practice, then, we find a particular solution of (6.16) by assuming a solution of the form:

$$(6.18) \quad \begin{cases} Ce^{cx} & \text{if } c \text{ is not equal to either } a \text{ or } b; \\ Cxe^{cx} & \text{if } c \text{ equals } a \text{ or } b, a \neq b; \\ Cx^2e^{cx} & \text{if } c = a = b. \end{cases}$$

Now that we know this, we would solve (6.10) as follows. Substitute

$$y_p = Cxe^x, \quad y_p' = C(xe^x + e^x), \quad y_p'' = C(xe^x + 2e^x)$$

into (6.10) and get

$$y_p'' + y_p' - 2y_p = C(xe^x + 2e^x + xe^x + e^x - 2xe^x) = e^x.$$

Thus we find $C = \frac{1}{3}$ as in (6.15) (but with much less work).

Use of Complex Exponentials In applied problems, the function $F(x)$ on the right-hand side of (6.1) is very often a sine or a cosine representing alternating emf or a periodic force. We *could* find y_p for such a problem either by the method of integrating two successive first-order equations or by replacing the sine or cosine by its complex exponential form and using the method of the last paragraph. There is a still more efficient variation of the latter method which we shall now show.

► **Example 6.** Solve

$$(6.19) \quad y'' + y' - 2y = 4 \sin 2x.$$

Instead of tackling this problem directly, we are first going to solve the equation

$$(6.20) \quad Y'' + Y' - 2Y = 4e^{2ix}.$$

Since $e^{2ix} = \cos 2x + i \sin 2x$ is complex, the solution Y may be complex also. Then if $Y = Y_R + iY_I$, (6.20) is equivalent to two equations

$$(6.21) \quad \begin{aligned} Y_R'' + Y_R' - 2Y_R &= \operatorname{Re} 4e^{2ix} = 4 \cos 2x, \\ Y_I'' + Y_I' - 2Y_I &= \operatorname{Im} 4e^{2ix} = 4 \sin 2x. \end{aligned}$$

Since the second equation in (6.21) is the same as (6.19), we see that the solution of (6.19) is the imaginary part of Y . Thus to find y_p for (6.19), we find Y_p for (6.20) and take its imaginary part. We observe that $2i$ is not equal to either of the roots of the auxiliary equation in (6.20). Following the method of the last paragraph, we assume a solution of the form

$$Y_p = Ce^{2ix}$$

and substitute it into (6.20) to get

$$\begin{aligned} (-4 + 2i - 2)Ce^{2ix} &= 4e^{2ix}, \\ C &= \frac{4}{2i - 6} = \frac{4(-2i - 6)}{40} = -\frac{1}{5}(i + 3), \\ Y_p &= -\frac{1}{5}(i + 3)e^{2ix}. \end{aligned}$$

Taking the imaginary part of Y_p , we find y_p for (6.19):

$$(6.22) \quad y_p = -\frac{1}{5} \cos 2x - \frac{3}{5} \sin 2x.$$

We summarize the method of complex exponentials:

	To find a particular solution of	
(6.23)	first solve	$(D - a)(D - b)y = \begin{cases} k \sin \alpha x, \\ k \cos \alpha x, \end{cases}$
		$(D - a)(D - b)y = ke^{i\alpha x}$
		and then take the real or imaginary part.

Method of Undetermined Coefficients The method we have just discussed of assuming an exponential solution and determining the constant factor C is an example (and in practice the most important case) of the *method of undetermined coefficients*. In (6.18) we outlined the form of y_p to assume for equation (6.16), that is, when the right-hand side of (6.1) is an exponential. It is straightforward but tedious (Problems 29 and 32) to find the corresponding result (6.24) when the right-hand side is an exponential times a polynomial.

A particular solution y_p of $(D - a)(D - b)y = e^{cx}P_n(x)$ where $P_n(x)$ is a polynomial of degree n is

$$(6.24) \quad y_p = \begin{cases} e^{cx}Q_n(x) & \text{if } c \text{ is not equal to either } a \text{ or } b, \\ xe^{cx}Q_n(x) & \text{if } c \text{ equals } a \text{ or } b, a \neq b, \\ x^2e^{cx}Q_n(x) & \text{if } c = a = b, \end{cases}$$

where $Q_n(x)$ is a polynomial of the same degree as $P_n(x)$ with *undetermined coefficients* to be found to satisfy the given differential equation. Note that sines and cosines are included in e^{cx} by use of complex exponentials as in (6.19) to (6.23). (Also see Problem 29.)

► **Example 7.** To illustrate using (6.24), let's find a particular solution of

$$(6.25) \quad (D - 1)(D + 2)y = y'' + y' - 2y = 18xe^x.$$

In the notation of (6.24) we have $a = 1$, $b = -2$, $c = 1$; also $P_n(x) = 18x = P_1(x)$ is a polynomial of degree 1. Then Q_1 is a polynomial of degree 1, namely $Ax + B$. Since $c = a \neq b$, we see by (6.24) that the form to assume for a particular solution of (6.25) is

$$y_p = xe^x(Ax + B) = e^x(Ax^2 + Bx).$$

We substitute this into (6.25) and find A and B so that we have an identity.

$$\begin{aligned} y_p' &= e^x(Ax^2 + Bx + 2Ax + B), \\ y_p'' &= e^x(Ax^2 + Bx + 4Ax + 2B + 2A) \\ y_p'' + y_p' - 2y_p &= e^x(6Ax + 3B + 2A) \equiv 18xe^x \end{aligned}$$

To make this an identity, we must have

$$6A = 18, \quad 3B + 2A = 0, \quad \text{or} \quad A = 3, \quad B = -2, \quad \text{so}$$

$$(6.26) \quad y_p = (3x^2 - 2x)e^x.$$

A computer solution may add to this a constant times e^x , but this is an unnecessary complication since e^x is a term in the complementary function.

If the right-hand side of a differential equation is a polynomial, then $c = 0$ in (6.24), and we assume for y_p a polynomial as indicated in (6.24).

► **Example 8.** To solve

$$(6.27) \quad (D - 1)(D + 2)y = y'' + y' - 2y = x^2 - x$$

we assume $y_p = Ax^2 + Bx + C$, and find the particular solution

$$(6.28) \quad y_p = -\frac{1}{2}(x^2 + 1).$$

A computer solution gives the same result.

► PROBLEMS, SECTION 6

Find the general solution of the following differential equations (complementary function + particular solution). Find the particular solution by inspection or by (6.18), (6.23), or (6.24). Also find a computer solution and reconcile differences if necessary, noticing especially whether the particular solution is in simplest form [see (6.26) and the discussion after (6.15)].

1. $y'' - 4y = 10$

2. $(D - 2)^2 y = 16$

3. $y'' + y' - 2y = e^{2x}$

4. $(D + 1)(D - 3)y = 24e^{-3x}$

5. $(D^2 + 1)y = 2e^x$

6. $y'' + 6y' + 9y = 12e^{-x}$

7. $y'' - y' - 2y = 3e^{2x}$

8. $y'' - 16y = 40e^{4x}$

9. $(D^2 + 2D + 1)y = 2e^{-x}$ 10. $(D - 3)^2 y = 6e^{3x}$
11. $y'' + 2y' + 10y = 100 \cos 4x$ *Hint:* First solve $y'' + 2y' + 10y = 100e^{4ix}$.
12. $(D^2 + 4D + 12)y = 80 \sin 2x$ 13. $(D^2 - 2D + 1)y = 2 \cos x$
14. $y'' + 8y' + 25y = 120 \sin 5x$ 15. $5y'' + 12y' + 20y = 120 \sin 2x$
16. $(D^2 + 9)y = 30 \sin 3x$ 17. $y'' + 16y = 16 \cos 4x$
18. $(D^2 + 2D + 17)y = 60e^{-4x} \sin 5x$ *Hint:* First solve $(D^2 + 2D + 17)y = 60e^{(-4+5i)x}$.
19. $(4D^2 + 4D + 5)y = 40e^{-3x/2} \sin 2x$ 20. $y'' + 4y' + 8y = 30e^{-x/2} \cos 5x/2$
21. $5y'' + 6y' + 2y = x^2 + 6x$ 22. $2y'' + y' = 2x$
23. $y'' + y = 2xe^x$ 24. $y'' - 6y' + 9y = 12xe^{3x}$
25. $(D - 3)(D + 1)y = 16x^2 e^{-x}$ 26. $(D^2 + 1)y = 8x \sin x$
27. Verify that (6.4) is a particular solution of (6.2). Verify that another particular solution of (6.2) is

$$y_p = \frac{1}{10} \sin 2x - e^{-x}.$$

Observe that we obtain the same general solution (6.7) whichever particular solution we use [since $(A - 1)$ is just as good an arbitrary constant as A]. Show in general that the difference between two particular solutions of $(a_2 D^2 + a_1 D + a_0)y = f(x)$ is always a solution of the homogeneous equation $(a_2 D^2 + a_1 D + a_0)y = 0$, and thus show that the general solution is the same for all choices of a particular solution.

28. Solve (6.16) by the method used in solving (6.11), for the following three cases, to obtain the result (6.18).
- c is not equal to either a or b ;
 - $a \neq b$, $c = a$;
 - $a = b = c$.
29. Consider the differential equation $(D - a)(D - b)y = P_n(x)$, where $P_n(x)$ is a polynomial of degree n . Show that a particular solution of this equation is given by (6.24) with $c = 0$; that is, y_p is

$$\begin{cases} \text{a polynomial } Q_n(x) \text{ of degree } n \text{ if } a \text{ and } b \text{ are both different from zero;} \\ xQ_n(x) & \text{if } a \neq 0, \text{ but } b = 0; \\ x^2Q_n(x) & \text{if } a = b = 0. \end{cases}$$

Hint: To show that $Q_n(x) = \sum a_n x^n$ is a solution of the differential equation for a given $P_n = \sum b_n x^n$, you have only to show that the coefficients a_n can be found so that $(D - a)(D - b)Q_n(x) \equiv P_n(x)$. Equate coefficients of $x^n, x^{n-1}, \dots$, to see that this is always possible if $a \neq b$. For $b = 0$, the differential equation becomes $(D - a)Dy = P_n$; what is Dy if $y = xQ_n$? Similarly, consider D^2y if $y = x^2Q_n$.

30. (a) Show that

$$\begin{aligned} (D - a)e^{cx} &= (c - a)e^{cx}; \\ (D^2 + 5D - 3)e^{cx} &= (c^2 + 5c - 3)e^{cx}; \\ L(D)e^{cx} &= L(c)e^{cx}, \text{ where } L(D) \text{ is any polynomial in } D; \\ (D - c)xe^{cx} &= e^{cx}; \\ (D - c)^2 x^2 e^{cx} &= 2e^{cx}. \end{aligned}$$

- (b) Define the expression $y = [1/L(D)]u(x)$ to mean a solution of the differential equation $L(D)y = u$. Using part (a), show that

$$\begin{aligned}\frac{1}{D-a}e^{cx} &= \frac{e^{cx}}{c-a}, \quad c \neq a; \\ \frac{1}{D^2+5D-3}e^{cx} &= \frac{e^{cx}}{c^2+5c-3}; \\ \frac{1}{L(D)}e^{cx} &= \frac{e^{cx}}{L(c)}, \quad L(c) \neq 0; \\ \frac{1}{D-c}e^{cx} &= xe^{cx}; \\ \frac{1}{(D-c)^2}e^{cx} &= \frac{1}{2}x^2e^{cx}.\end{aligned}$$

- (c) The expressions $1/L(D)$ in (b) are called inverse operators. They can be used to find particular solutions of differential equations. As an example consider Problem 3. We write

$$\begin{aligned}(D^2 + D - 2)y &= e^{2x}, \\ y &= \frac{1}{D^2 + D - 2}e^{2x} = \frac{e^{2x}}{2^2 + 2 - 2} = \frac{e^{2x}}{4}.\end{aligned}$$

Using inverse operators, find particular solutions of Problems 4 to 20. Be careful to use parts 4 or 5 of (b) if c is a root of the auxiliary equation. For example,

$$\frac{1}{(D-a)(D-c)}e^{cx} = \frac{1}{D-c} \frac{1}{D-a}e^{cx} = \frac{1}{D-c} \frac{e^{cx}}{c-a} = \frac{xe^{cx}}{c-a}.$$

31. (a) Show that

$$\begin{aligned}D(e^{ax}y) &= e^{ax}(D+a)y, \\ D^2(e^{ax}y) &= e^{ax}(D+a)^2y,\end{aligned}$$

and so on; that is, for any positive integral n ,

$$D^n(e^{ax}y) = e^{ax}(D+a)^ny.$$

Thus show that if $L(D)$ is any polynomial in the operator D , then

$$L(D)(e^{ax}y) = e^{ax}L(D+a)y.$$

This is called the *exponential shift*.

- (b) Use (a) to show that

$$\begin{aligned}(D-1)^3(e^xy) &= e^xD^3y, \\ (D^2+D-6)(e^{-3x}y) &= e^{-3x}(D^2-5D)y.\end{aligned}$$

- (c) Replace D by $D-a$, to obtain

$$e^{ax}P(D)y = P(D-a)e^{ax}y.$$

This is called the *inverse exponential shift*.

- (d) Using (c), we can change a differential equation whose right-hand side is an exponential times a polynomial, to one whose right-hand side is just a polynomial. For example, consider $(D^2 - D - 6)y = 10xe^{3x}$; multiplying both sides by e^{-3x} and using (c), we get

$$\begin{aligned} e^{-3x}(D^2 - D - 6)y &= [(D + 3)^2 - (D + 3) - 6]ye^{-3x} \\ &= (D^2 + 5D)ye^{-3x} = 10x. \end{aligned}$$

Show that a solution of $(D^2 + 5D)u = 10x$ is $u = x^2 - \frac{2}{5}x$; then $ye^{-3x} = x^2 - \frac{2}{5}x$ or $y = e^{3x}(x^2 - \frac{2}{5}x)$. Use this method to solve Problems 23 to 26.

32. Using Problems 29 and 31b, show that equation (6.24) is correct.

Several Terms on the Right-Hand Side: Principle of Superposition So far we have brushed over a question which may have occurred to you: What do we do if there are several terms on the right-hand side of the equation involving different exponentials?

► **Example 9.** As an artificial problem to illustrate the ideas, consider the equation

$$(6.29) \quad y'' + y' - 2y = (D - 1)(D + 2)y = [e^x] + [4 \sin 2x] + [x^2 - x].$$

We have already solved differential equations with the same left-hand sides as (6.29) and with right-hand sides equal in turn to each of the three expressions in brackets in (6.29) [see (6.11) to (6.15), (6.19) to (6.22), (6.27), and (6.28)]. Thus we know that

$$\begin{aligned} (D - 1)(D + 2)y &= e^x && \text{has the particular solution } y_{p1} = \frac{1}{3}xe^x; \\ (D - 1)(D + 2)y &= 4 \sin 2x && \text{has the particular solution } y_{p2} = -\frac{1}{5} \cos 2x - \frac{3}{5} \sin 2x; \\ (D - 1)(D + 2)y &= x^2 - x && \text{has the particular solution } y_{p3} = -\frac{1}{2}(x^2 + 1). \end{aligned}$$

Adding these three solutions, we see that

$$(6.30) \quad y_p = y_{p1} + y_{p2} + y_{p3} = \frac{1}{3}xe^x - \frac{1}{5} \cos 2x - \frac{3}{5} \sin 2x - \frac{1}{2}(x^2 + 1)$$

is a particular solution of (6.29).

This is the easiest way of handling a complicated right-hand side: Solve a separate equation for *each different exponential* and add the solutions. The fact that this is correct for a linear equation is often called the *principle of superposition*. As we can see from (6.29) and (6.30), this amounts to a fancy name for the fact that the derivative (of any order) of a sum of terms is equal to the sum of the derivatives of the individual terms. Notice that the principle holds only for *linear* equations; for example, if the equation contained y'^2 , the principle would not hold since $(y'_1 + y'_2)^2$ is not equal to $y'^2_1 + y'^2_2$. In fact, an operator (such as the D operators we have been using) which satisfies the principle of superposition is called a *linear operator*. [See Chapter 3, equation (7.4) and Problem 7.12.] Linear operators are of particular importance because they obey the principle of superposition; for example, $D^2(y_1 + y_2) = y''_1 + y''_2 = D^2y_1 + D^2y_2$, so D^2 is a linear operator. We shall make use of this principle shortly in our discussion of the use of Fourier series in finding particular solutions.

Forced Vibrations Let's return now to the physical problem we considered at the end of Section 5. There we set up and solved the differential equation which describes the free (zero right-hand side, no forcing function) vibrations of a damped oscillator. We commented that the same mathematics applies to a variety of mechanics problems and also to a simple *RLC* series electric circuit. As we know from experiment and as we can see from (5.30), (5.31), and (5.32), the free vibrations we considered in Section 5 die out as time passes. Such oscillations are referred to as *transients*. We next want to consider the vibrations obtained when a periodic force (or emf in the electric case) is applied. This means mathematically that we want to solve (5.27) with a function of t on the right-hand side. The solution will contain the appropriate one of (5.30), (5.31), (5.32); this is the complementary function and it is also the transient since it tends to zero as t tends to infinity. The solution will also contain a particular solution which does not tend to zero as t tends to infinity; this is the *steady-state solution* which we want to find.

► **Example 10.** Let us solve

$$(6.31) \quad \frac{d^2 y}{dt^2} + 2b \frac{dy}{dt} + \omega^2 y = F \sin \omega' t \quad (F = \text{const.}).$$

By the method of complex exponentials, we solve first

$$(6.32) \quad \frac{d^2 Y}{dt^2} + 2b \frac{dY}{dt} + \omega^2 Y = F e^{i\omega' t}.$$

Substitute

$$(6.33) \quad Y_p = C e^{i\omega' t}$$

into (6.32) to get

$$(6.34) \quad \begin{aligned} (-\omega'^2 + 2bi\omega' + \omega^2) C e^{i\omega' t} &= F e^{i\omega' t}, \\ C &= \frac{F}{(\omega^2 - \omega'^2) + 2bi\omega'} = \frac{[(\omega^2 - \omega'^2) - 2bi\omega'] F}{(\omega^2 - \omega'^2)^2 + 4b^2 \omega'^2}. \end{aligned}$$

It is convenient to write the complex number C in the $re^{i\theta}$ form. We have

$$(6.35) \quad |C| = \frac{F}{\sqrt{(\omega^2 - \omega'^2)^2 + 4b^2 \omega'^2}},$$

angle of $C = -\phi$, where ϕ is given by Figure 6.1.

Thus

$$(6.36) \quad C = \frac{F}{\sqrt{(\omega^2 - \omega'^2)^2 + 4b^2 \omega'^2}} e^{-i\phi}$$

and from (6.33)

$$(6.37) \quad Y_p = \frac{F}{\sqrt{(\omega^2 - \omega'^2)^2 + 4b^2 \omega'^2}} e^{i(\omega' t - \phi)}$$

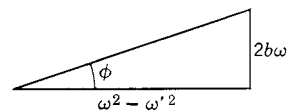


Figure 6.1

To find y_p we take the imaginary part of Y_p :

$$(6.38) \quad y_p = \frac{F}{\sqrt{(\omega^2 - \omega'^2)^2 + 4b^2\omega'^2}} \sin(\omega't - \phi).$$

This is the steady-state solution, so-called because as t increases, the rest of the solution [given by (5.30), (5.31), or (5.32)] becomes negligible. For example, when you turn on an electric light, the current is given by (5.32) plus (6.38). The transient (5.32) tends to zero rapidly and the steady-state solution (6.38) becomes essentially the whole solution.

Resonance We note, by comparing (6.38) and the forcing function in (6.31), that the applied force (or emf) and the solution y (which represents displacement, current, etc.) are *out of phase*; that is, their maximum values do not occur at the same time because of the phase angle ϕ . We also see from (6.38) that for a given forcing frequency ω' , the largest amplitude of y (also of dy/dt , Problem 40) occurs if the natural (undamped) frequency ω is equal to ω' . This situation is often called *resonance*. In the *RLC* series circuit problem, y represents the charge q on the capacitor if the forcing function is the emf, and y represents the current $I = dq/dt$ if the forcing function is the time derivative of the emf. For such a circuit, given the frequency ω' of the applied emf, the current (or charge) will have the largest amplitude when the natural (undamped) frequency ω is equal to ω' . This is almost always called the resonance condition for the electrical case. However, there is another question we could ask here which is of particular interest in mechanics. Given the natural (undamped) frequency ω of the system, what frequency of the forcing function will produce the largest amplitude of y ? In (6.38), we want to maximize the coefficient of the sine; we can instead minimize the square of the denominator of the coefficient; that is, we want to find the value of ω' which minimizes $(\omega^2 - \omega'^2)^2 + 4b^2\omega'^2$ for given ω . Setting the derivative of this function (with respect to ω') equal to zero and solving for ω' , we get

$$(6.39) \quad \begin{aligned} 2(\omega^2 - \omega'^2)(-2\omega') + 8b^2\omega' &= 0, \\ \omega'^2 &= \omega^2 - 2b^2. \end{aligned}$$

Note that this value of ω' is not equal to either the natural undamped frequency ω or the natural damped frequency β where $\beta^2 = \omega^2 - b^2$ [see (5.32)]. However, if we define *resonance* as the situation in which we get the maximum amplitude for y for a given value of ω , then the resonance condition is (6.39). (The maximum amplitude for the velocity—or current in the electrical case—is still obtained for $\omega' = \omega$; Problem 40.) The resonance condition (6.39) is of particular importance in mechanics where we are apt to be interested in the displacement y of a given system under the action of various forces. For example, consider a bridge; we would want to avoid periodic forces with an ω' given by (6.39) since such forces would produce large vibrations. In this case resonance is undesirable. It may in other cases be desirable; for example, when you tune your radio to the frequency of a given station, you are given ω' and you adjust the circuit in your radio to make its natural frequency ω equal to the given ω' .

Use of Fourier Series in Finding Particular Solutions In simple problems, the forcing function in either the electrical or mechanical case is just a sine or cosine and the problem can be solved as we have just done. In more complicated (and realistic) cases, however, the forcing function may very well be some more complicated function; it is often a periodic function, however, and we shall assume this. Suppose, for example, that the periodic emf applied to a circuit is given by one of the graphs in Figure 3.2 of Chapter 7. We learned in Chapter 7 how to expand such a function in a Fourier series. Let us suppose that this has been done, using for definiteness the complex exponential form of the Fourier series. Then we can write (6.1) as

$$(6.40) \quad a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}.$$

We know how to solve the equation

$$(6.41) \quad a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = c_n e^{inx}$$

with the right-hand side equal to any one term of the series. If we now add the solutions of all the equations (6.41) for all n , we have a solution of (6.40) (see *principle of superposition* above).

► **Example 11.** Solve

$$(6.42) \quad \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 10y = f(t),$$

where $f(t)$ is a function of period 2π and

$$f(t) = \begin{cases} 1, & 0 \leq t < \pi, \\ 0, & \pi \leq t < 2\pi. \end{cases}$$

The auxiliary equation is

$$D^2 + 2D + 10 = 0;$$

its roots are

$$D = -1 \pm 3i,$$

so the complementary function is

$$y_c = e^{-t}(A \cos 3t + B \sin 3t).$$

To find a particular solution, we first expand $f(t)$ in a Fourier series; from Chapter 7, equation (7.8), we have

$$(6.43) \quad f(t) = \frac{1}{2} + \frac{1}{i\pi} [e^{it} - e^{-it} + \frac{1}{3}(e^{3it} - e^{-3it}) + \dots].$$

We next write and solve a whole set of differential equations like (6.42) but each having just one term of the series (6.43) on the right-hand side. For the first term (namely $\frac{1}{2}$) we see by inspection that a particular solution of

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 10y = \frac{1}{2}$$

is $y = \frac{1}{20}$. All the other terms of (6.43) are of the form $(1/ik\pi)e^{ikt}$, where k is a positive or negative odd integer. To solve

$$(6.44) \quad \frac{d^2 y}{dt^2} + 2\frac{dy}{dt} + 10y = \frac{1}{ik\pi}e^{ikt},$$

we substitute

$$(6.45) \quad y = Ce^{ikt}$$

into (6.44) and get

$$(-k^2 + 2ik + 10)Ce^{ikt} = \frac{1}{ik\pi}e^{ikt}.$$

Then we have

$$(6.46) \quad C = \frac{1}{ik\pi} \frac{1}{(10 - k^2) + 2ik} = \frac{1}{ik\pi} \frac{(10 - k^2) - 2ik}{(10 - k^2)^2 + 4k^2}.$$

By letting $k = \pm 1, \pm 3, \dots$, and substituting the values of C thus obtained into (6.45), we obtain the solutions of (6.44) for the various k values corresponding to the terms of the series (6.43). The sum of all the solutions corresponding to all the terms is the desired particular solution of (6.42). Thus

$$\begin{aligned} y_p &= \frac{1}{20} + \frac{1}{i\pi} \frac{9 - 2i}{85} e^{it} - \frac{1}{i\pi} \frac{9 + 2i}{85} e^{-it} \\ &\quad + \frac{1}{3i\pi} \frac{1 - 6i}{37} e^{3it} - \frac{1}{3i\pi} \frac{1 + 6i}{37} e^{-3it} + \dots \\ (6.47) \quad &= \frac{1}{20} + \frac{2}{\pi} \frac{9}{85} \left(\frac{e^{it} - e^{-it}}{2i} \right) - \frac{2}{\pi} \frac{2}{85} \left(\frac{e^{it} + e^{-it}}{2} \right) \\ &\quad + \frac{2}{3\pi} \frac{1}{37} \left(\frac{e^{3it} - e^{-3it}}{2i} \right) - \frac{2}{3\pi} \frac{6}{37} \left(\frac{e^{3it} + e^{-3it}}{2} \right) + \dots \\ &= \frac{1}{20} + \frac{2}{85\pi} (9 \sin t - 2 \cos t) + \frac{2}{111\pi} (\sin 3t - 6 \cos 3t) + \dots \end{aligned}$$

is a particular solution of (6.42).

► PROBLEMS, SECTION 6

In Problem 33 to 38, solve the given differential equations by using the principle of superposition [see the solution of equation (6.29)]. For example, in Problem 33, solve three differential equations with right-hand sides equal to the three different brackets. Note that terms with the *same exponential factor* are kept together; thus a polynomial of any degree is kept together in one bracket.

$$33. \quad y'' + y = [x^3 - 1] + [2 \cos x] + [(2 - 4x)e^x]$$

$$34. \quad y'' - 5y' + 6y = 2e^x + 6x - 5$$

$$35. \quad (D^2 - 1)y = \sinh x$$

$$36. \quad (D^2 + 1)y = 2 \sin x + 4x \cos x$$

$$37. \quad (D - 1)^2 y = 4e^x + (1 - x)(e^{2x} - 1)$$

$$38. \quad y'' - 2y' = 9xe^{-x} - 6x^2 + 4e^{2x}$$

39. Find the solutions of (1.2) (put $I = dq/dt$) and (1.3), if $V = V_0 \sin \omega' t$ ($\omega' = \text{const.}$).
40. In (6.38), show that for a given forcing frequency ω' , the displacement y and the velocity dy/dt have their largest amplitude when $\omega = \omega'$.

For a given ω , we have shown in Section 6 that the maximum amplitude of y does not correspond to $\omega' = \omega$. Show, however, that the maximum amplitude of dy/dt for a given ω does correspond to $\omega' = \omega$.

State the corresponding results for an electric circuit in terms of L , R , C .

Solve Problems 41 and 42 by use of Fourier series. Assume in each case that the right-hand side is a periodic function whose values are stated for one period.

41. $y'' + 2y' + 2y = |x|, \quad -\pi < x < \pi.$

42. $y'' + 9y = \begin{cases} x, & 0 < x < 1, \\ 0, & -1 < x < 0. \end{cases}$

43. Consider an equation for damped forced vibrations (mechanical or electrical) in which the right-hand side is a sum of several forces or emfs of different frequencies. For example, in (6.32) let the right-hand side be

$$F_1 e^{i\omega'_1 t} + F_2 e^{i\omega'_2 t} + F_3 e^{i\omega'_3 t}.$$

Write the solution by the principle of superposition. Suppose, for given $\omega'_1, \omega'_2, \omega'_3$, that we adjust the system so that $\omega = \omega'_1$; show that the principal term in the solution is then the first one. Thus the system acts as a “filter” to select vibrations of one frequency from a given set (for example, a radio tuned to one station selects principally the vibrations of the frequency of that station).

► 7. OTHER SECOND-ORDER EQUATIONS

Although second-order linear equations with constant coefficients are the ones used most frequently in applications, there are a few other kinds of second-order equations and methods of solving them which are also important. We shall discuss several of these here, namely (a) equations with y missing; (b) equations with x missing; (c) equations of the form $y'' + f(y) = 0$; (d) Euler-Cauchy equations; (e) reduction of order. For still more methods, see Section 9 (Laplace transforms), Section 12 (Green functions), Problem 12.14b (variation of parameters), and Chapter 12 (special functions, series solutions, ladder operators). You can also find computer solutions but, as we have said, they may not always be in the simplest form or the form you need. Comparing hand solutions can show you what to expect and help you make more efficient use of computer solutions.

To solve either (a) or (b), we make the substitution

$$(7.1) \quad y' = p.$$

Case (a): Dependent variable y missing.

$$(7.2) \quad y' = p, \quad y'' = p'.$$

After these substitutions, an equation of the type (a) is of the first order with p

as the dependent variable and x as the independent variable. First, we solve it for p as a function of x ; then we put back $p = y'$ and solve the resulting first-order equation for y .

Case (b): Independent variable x missing.

$$(7.3) \quad y' = p, \quad y'' = \frac{dp}{dx} = \frac{dp}{dy} \frac{dy}{dx} = p \frac{dp}{dy}.$$

What we are doing here is to change the independent variable from x to y . Observe that there is *one* independent variable in an ordinary differential equation. We were originally thinking of x as the independent variable with y and $p \, dy/dx$ as functions of x . Now we think of y as the independent variable with p a function of y ; (7.3) is just the chain rule (Chapter 4, Section 5) for differentiating a function $p(y)$ with respect to x if y is a function of x . With the substitutions (7.3), a differential equation with x missing becomes a first-order equation with p as the dependent variable and y as the independent variable.

► **Example 1.** In Section 5, we discussed the motion of a mass m subject to a restoring force $-ky$ and a damping force $l(dy/dt)$. Let us now consider a similar problem but with the damping force proportional to the square of the velocity. The differential equation of motion is then [compare (5.26)]

$$(7.4) \quad m \frac{d^2 y}{dt^2} \pm l \left(\frac{dy}{dt} \right)^2 + ky = 0 \quad (l > 0),$$

where the plus or minus sign must be chosen correctly at each stage of the motion so that the retarding force opposes the motion. Let us solve the following special case of this problem. Discuss the motion of a particle which is released from rest at the point $y = 1$ when $t = 0$, and obeys the equation of motion

$$(7.5) \quad 4 \frac{d^2 y}{dt^2} \pm 2 \left(\frac{dy}{dt} \right)^2 + y = 0.$$

This is an example of case (b) (for “ x missing” read “ t missing,” that is, the independent variable missing). Using (7.3) (with x replaced by t), we have

$$(7.6) \quad \begin{aligned} \frac{dy}{dt} &= p, \\ \frac{d^2 y}{dt^2} &= p \frac{dp}{dy}, \end{aligned}$$

so (7.5) becomes

$$(7.7) \quad 4p \frac{dp}{dy} \pm 2p^2 + y = 0 \quad \text{or} \quad \frac{dp}{dy} \pm \frac{1}{2}p = -\frac{1}{4}yp^{-1}.$$

This is a Bernoulli equation [compare (4.1) with y replaced by p , and P and Q functions of y]. We have $n = -1$, and the substitution (4.2) is

$$(7.8) \quad z = p^2.$$

Then

$$\frac{dz}{dy} = 2p \frac{dp}{dy}$$

and (7.7) becomes

$$(7.9) \quad \frac{dz}{dy} \pm z = -\frac{1}{2}y.$$

This is a first-order linear equation; solving it (see Section 3), we get

$$(7.10) \quad \begin{aligned} ze^{\pm y} &= -\frac{1}{2} \int ye^{\pm y} dy = -\frac{1}{2}e^{\pm y}(\pm y - 1) + c, \\ z &= -\frac{1}{2}(\pm y - 1) + ce^{\mp y}. \end{aligned}$$

Since initially $dy/dt = 0$ and $y > 0$, we see from (7.5) that the initial acceleration is in the negative direction; since the particle starts from rest, its velocity for small t is also in the negative direction. Then the damping force must be in the positive direction so we must use the lower sign in (7.10) for the first part of the motion. Thus we have

$$(7.11) \quad z = \frac{1}{2}(y + 1) + ce^y \quad (\text{for small } t).$$

We determine c from the initial conditions $dy/dt = 0$, $y = 1$, at $t = 0$; we have $z = p^2 = (dy/dt)^2 = 0$ when $y = 1$; therefore from (7.11) we get

$$0 = 1 + ce, \quad c = -e^{-1}.$$

Then we have

$$(7.12) \quad z = \left(\frac{dy}{dt}\right)^2 = \frac{1}{2}(y + 1) - e^{y-1} \quad (\text{for small } t).$$

This is a valid solution as long as $dy/dt < 0$ (this is what small t means). Thus the particle initially moves in the negative direction for a while. To continue the problem we would need to find whether it stops and if so where. This means solving a transcendental equation which has to be done by some approximation method. It turns out that when it stops, y is negative; at this point the force $-y$ is in the positive direction and the particle, after stopping, moves in the positive direction. The solution for $(dy/dt)^2$ is then given by (7.10) with the upper sign. After another interval of time, the particle again reverses its motion and we again use the solution (7.11) (with a different c), and so on, the total motion appearing something like a damped vibration. We shall not continue the details further here since we have already accomplished our purpose of illustrating case (b) and the solution of a Bernoulli equation.

Case (c) appears to be very special and is obviously included by (b); however, it is very important to know the easy way to solve it because it so frequently arises in applications. The trick is simply to multiply the equation by y' ; we can then integrate each term.

Case (c): To solve $y'' + f(y) = 0$, multiply by y' .

$$y'y'' + f(y)y' = 0, \quad \text{or} \quad y' dy' + f(y) dy = 0.$$

Then integrate to get

$$(7.13) \quad \frac{1}{2}y'^2 + \int f(y) dy = \text{const.}$$

This equation is separable and so can be solved (except for possible difficulty in evaluating the integrals). We say that the problem is reduced to *quadratures* (indicated integrations); this means that we can write the answer in terms of integrals which may or may not be easy to evaluate!

► **Example 2.** Consider a particle of mass m moving along the x axis under the action of a force $F(x)$. Then the equation of motion is

$$(7.14) \quad m \frac{d^2x}{dt^2} = F(x).$$

If we multiply this equation by $v = dx/dt$ and integrate with respect to t , we get

$$(7.15) \quad \begin{aligned} mv \frac{dv}{dt} &= F(x) \frac{dx}{dt} & \text{or} & \quad mv dv = F(x) dx, \\ \frac{1}{2}mv^2 &= \int F(x) dx + \text{const.} \end{aligned}$$

Recall (Chapter 6, Section 8) that the potential energy of a particle is the negative of the work done by the force. Thus

$$(7.16) \quad \frac{1}{2}mv^2 - \int F(x) dx$$

is the kinetic energy plus the potential energy; equation (7.15) expresses the law of conservation of energy for this problem. This energy equation is often of more interest than the equation of motion (x as a function of t) and so it is useful to be able to find it directly, as we have done, without solving the differential equation for x . Equation (7.15) is known as a *first integral* of the differential equation since we have integrated a second-order equation *once* to get it.

Case (d): An equation of the form

$$(7.17) \quad a_2 x^2 \frac{d^2 y}{dx^2} + a_1 x \frac{dy}{dx} + a_0 y = f(x)$$

(called an Euler or Cauchy equation) can be reduced to a linear equation with constant coefficients by changing the independent variable from x to z where

$$(7.18) \quad x = e^z.$$

For then we have (see Problem 14 and also Chapter 4, Section 11)

$$(7.19) \quad x \frac{dy}{dx} = \frac{dy}{dz} \quad \text{and} \quad x^2 \frac{d^2 y}{dx^2} = \frac{d^2 y}{dz^2} - \frac{dy}{dz}.$$

Substituting (7.18) and (7.19) into (7.17) gives

$$(7.20) \quad a_2 \frac{d^2 y}{dz^2} + (a_1 - a_2) \frac{dy}{dz} + a_0 y = f(e^z).$$

This is a linear equation with constant coefficients which can be solved by the methods of Sections 5 and 6.

It is worth noting that the solutions of (7.17) when $f(x) = 0$ are often powers of x , so a way to solve this case is to assume $y = x^k$ and solve the resulting quadratic equation for k . However, if the values of k turn out to be complex, or equal, or if $f(x) \neq 0$, you may find it easier to use (7.18) which reduces the problem to a familiar one. (See Problems 15 to 23.)

Case (e): Reduction of order. To find a second solution of

$$(7.21) \quad y'' + f(x)y' + g(x)y = 0$$

given one solution $u(x)$, substitute

$$(7.22) \quad y = u(x)v(x)$$

into (7.21) and solve for $v(x)$.

You can verify that when you substitute (7.22) into (7.21), the coefficient of $v(x)$ is $u'' + f(x)u' + g(x)u$. This expression is equal to zero because we assumed that $u(x)$ is a solution of (7.21). Then the equation for $v'(x)$ is a separable first-order equation (Problem 24).

► **Example 3.** Solve $x^3 y'' + xy' - y = 0$, given that $u = x$ is a solution.

We let $y = uv = xv$. Then $y' = xv' + v$, $y'' = xv'' + 2v'$, and the differential equation becomes

$$\begin{aligned} x^3(xv'' + 2v') + x(xv' + v) - xv &= 0 \quad \text{or} \\ x^4 v'' + (2x^3 + x^2)v' &= 0. \end{aligned}$$

Separating variables and integrating, we find

$$\frac{dv'}{v'} = -\left(\frac{2}{x} + \frac{1}{x^2}\right) dx, \quad \ln v' = -2 \ln x + \frac{1}{x} + \ln K.$$

Solving for v' , integrating again, and writing $y = uv$ gives

$$v' = \frac{K}{x^2} e^{1/x}, \quad v = -K e^{1/x}, \quad y = -K x e^{1/x}.$$

Thus the general solution of the given equation is $y = Ax + Bx e^{1/x}$.

► PROBLEMS, SECTION 7

Solve the following differential equations by method (a) or (b) above.

1. $y'' + yy' = 0$. Find a solution satisfying each of the following sets of initial conditions. If your computer says there is no such solution, don't believe it—do it by hand.

(a) $y(0) = 5, \quad y'(0) = 0$	(b) $y(0) = 2, \quad y'(0) = -2$
(c) $y(0) = 1, \quad y'(0) = -1$	(d) $y(0) = 0, \quad y'(0) = 2$
2. $y'' + 2xy' = 0$ *Hint:* The solution is $y = c_1 \operatorname{erf} x + c_2$; see Chapter 11, Section 9 for the definition of $\operatorname{erf} x$.
3. $2yy'' = y'^2$
4. $xy'' = y' + y'^3$
5. The differential equation of a hanging chain supported at its ends is

$$y''^2 = k^2 (1 + y'^2).$$

Solve the equation to find the shape of the chain.

6. The curvature of a curve in the (x, y) plane is

$$K = y'' (1 + y'^2)^{-3/2}.$$

With $K = \text{const.}$, solve this differential equation to show that curves of constant curvature are circles (or straight lines).

7. Solve $y'' + \omega^2 y = 0$ by method (c) above and compare with the solution as a linear equation with constant coefficients.
8. The force of gravitational attraction on a mass m at distance r from the center of the earth ($r > \text{radius } R \text{ of the earth}$) is mgR^2/r^2 . Then the differential equation of motion of a mass m projected radially outward from the surface of the earth, with initial velocity v_0 , is

$$md^2r/dt^2 = -mgR^2/r^2.$$

Use method (c) above to find v as a function of r if $v = v_0$ initially (that is, when $r = R$). Find the maximum value of r for a given v_0 , that is, the value of r when $v = 0$. Find the *escape velocity*, that is, the smallest value of v_0 for which r can tend to infinity.

9. Show that (7.15) is a separable equation. [You may find it helpful to write $\int F(x) dx = f(x)$.] Thus solve (7.14) in terms of quadratures (that is, indicated integrations) as in Problem 2.

In Problems 10 and 11, solve (7.14) to find $v(x)$ and then $x(t)$ for the given $F(x)$ and initial conditions.

10. $F(x) = m/x^3$, $v = 0$, $x = 1$, at $t = 0$.
11. $F(x) = -2m/x^5$, $v = -1$, $x = 1$, at $t = 0$.
12. In Problem 11, find $v(x)$ if $v = 0$, $x = 1$, at $t = 0$. Then write an integral for $t(x)$.
13. The exact equation of motion of a simple pendulum is $d^2\theta/dt^2 = -\omega^2 \sin \theta$ where $\omega^2 = g/l$. By method (c) above, integrate this equation once to find $d\theta/dt$ if $d\theta/dt = 0$ when $\theta = 90^\circ$. Write a formula for $t(\theta)$ as an integral. See Problem 5.34.
14. Verify (7.19) and (7.20). *Hint:* $dy/dz = (dy/dx)(dx/dz)$; write the first equation of (7.19) as $x D_x = D_z$, and find D_z^2 .
15. If you solve (7.17) when $f(x) = 0$ by assuming a solution $y = x^k$, show that the quadratic equation for k is the same as the auxiliary equation for the z equation (7.20). Thus show (see Section 5) that if the two values of k are equal, the second solution is not a power of x but is $x^k \ln x$. Also show that if k is complex, say $k = a \pm bi$, the solutions are $x^a \cos(b \ln x)$ and $x^a \sin(b \ln x)$ or other equivalent forms [see (5.16) to (5.18)].
16. Solve the following equations either by method (d) above or by assuming $y = x^k$ (or try both methods to compare them). See Problem 15.
 - (a) $x^2 y'' + 3xy' - 3y = 0$
 - (b) $x^2 y'' + xy' - 4y = 0$
 - (c) $x^2 y'' + 7xy' + 9y = 0$
 - (d) $x^2 y'' - xy' + 6y = 0$

Solve the following equations using method (d) above.

17. $x^2 y'' + xy' - 16y = 8x^4$
18. $x^2 y'' + xy' - y = x - x^{-1}$
19. $x^2 y'' - 5xy' + 9y = 2x^3$
20. $x^2 y'' - 3xy' + 4y = 6x^2 \ln x$
21. $x^2 y'' + y = 3x^2$
22. $x^2 y'' + xy' + y = 2x$
23. Solve the two differential equations in Problem 5.11 of Chapter 13.
24. Substitute (7.22) into (7.21) to obtain the equation for $v'(x)$. Show that this equation is separable.

For the following problems, verify the given solution and then, by method (e) above, find a second solution of the given equation.

25. $x^2(2-x)y'' + 2xy' - 2y = 0$, $u = x$
26. $(x^2 + 1)y'' - 2xy' + 2y = 0$, $u = x$
27. $xy'' - 2(x+1)y' + (x+2)y = 0$, $u = e^x$
28. $3xy'' - 2(3x-1)y' + (3x-2)y = 0$, $u = e^x$
29. $x^2 y'' + (x+1)y' - y = 0$, $u = x+1$
30. $x(x+1)y'' - (x-1)y' + y = 0$, $u = x-1$

► 8. THE LAPLACE TRANSFORM

As you will see in Section 9, Laplace transforms are useful in solving differential equations (for other uses see end of Section 9, page 442). Here we want to define the Laplace transform and obtain some needed formulas. We define $L(f)$, the Laplace transform of $f(t)$ [also written $F(p)$ since it is a function of p], by the equation

$$(8.1) \quad L(f) = \int_0^{\infty} f(t)e^{-pt} dt = F(p).$$

This is an example of an *integral transform* (also see Fourier transforms, Chapter 7, Section 12, and Hilbert transforms, Chapter 14, page 698). If we start with a function $f(t)$, multiply by a function of t and p , and find a definite integral with respect to t , we have a function $F(p)$ which is called an integral transform of $f(t)$. There are many named integral transforms which you may discover in tables and computer. Observe the notation for Laplace transforms in (8.1): we shall consistently use a small letter for the function of t , and the corresponding capital letter for the transform which is a function of p , for example $f(t)$ and $F(p)$, or $g(t)$ and $G(p)$, etc. Also note from (8.1) that since we integrate from 0 to ∞ , $F(p)$ is the same no matter how $f(t)$ is defined for negative t . However, it is desirable to define $f(t) = 0$ for $t < 0$ (see footnote, page 447; also see Bromwich integral, page 696).

It is very convenient to have a table of corresponding $f(t)$ and $F(p)$ when we are using Laplace transforms to solve problems. Let us calculate some of the entries in the table of Laplace transforms at the end of the chapter (pages 469 to 471). Note that numbers preceded by L ($L1$, $L2$, $\dots$, $L35$) refer to entries in the Laplace transform table.

► **Example 1.** To obtain $L1$ in the table, we substitute $f(t) = 1$ into (8.1) and find

$$(8.2) \quad F(p) = \int_0^{\infty} 1 \cdot e^{-pt} dt = -\frac{1}{p} e^{-pt} \Big|_0^{\infty} = \frac{1}{p}, \quad p > 0.$$

We have assumed $p > 0$ to make e^{-pt} zero at the upper limit; if p is complex, as it may be, then the real part of p must be positive ($\operatorname{Re} p > 0$), and this is the restriction we have stated in the table for $L1$.

► **Example 2.** For $L2$, we have

$$(8.3) \quad \begin{aligned} f(t) &= e^{-at}, \\ F(p) &= \int_0^{\infty} e^{-(a+p)t} dt = \frac{1}{p+a}, \quad \operatorname{Re}(p+a) > 0. \end{aligned}$$

We could continue in this way to obtain the function $F(p)$ corresponding to each $f(t)$ by using (8.1) and evaluating the integral. However, there are some easier methods which we now illustrate. First observe that the Laplace transform of a sum of two functions is the sum of their Laplace transforms; also the transform of

$cf(t)$ is $cL(f)$ when c is a constant:

$$\begin{aligned}
 L[f(t) + g(t)] &= \int_0^\infty [f(t) + g(t)]e^{-pt} dt \\
 (8.4) \qquad &= \int_0^\infty f(t)e^{-pt} dt + \int_0^\infty g(t)e^{-pt} dt = L(f) + L(g), \\
 L[cf(t)] &= \int_0^\infty cf(t)e^{-pt} dt = c \int_0^\infty f(t)e^{-pt} dt = cL(f).
 \end{aligned}$$

In mathematical language, we say that the Laplace transform is *linear* (or is a *linear operator*—see Chapter 3, Section 7).

► **Example 3.** Now let us verify *L3*. In (8.3), replace the a by $-ia$; then we have

$$\begin{aligned}
 f(t) &= e^{iat} = \cos at + i \sin at, \\
 (8.5) \qquad F(p) &= \frac{1}{p - ia} = \frac{p + ia}{p^2 + a^2}, \quad \operatorname{Re}(p - ia) > 0.
 \end{aligned}$$

Remembering (8.4), we can write (8.5) as

$$(8.6) \qquad L(\cos at + i \sin at) = L(\cos at) + iL(\sin at) = \frac{p}{p^2 + a^2} + i \frac{a}{p^2 + a^2}.$$

Similarly, replacing a by ia in (8.3), we get

$$(8.7) \qquad L(\cos at - i \sin at) = \frac{p}{p^2 + a^2} - i \frac{a}{p^2 + a^2}, \quad \operatorname{Re}(p + ia) > 0.$$

Adding (8.6) and (8.7), we get *L4*; by subtracting, we get *L3*.

► **Example 4.** To verify *L11*, start with *L4*, namely

$$(8.8) \qquad L(\cos at) = \int_0^\infty e^{-pt} \cos at \, dt = \frac{p}{p^2 + a^2}.$$

Differentiate (8.8) with respect to the parameter a to get

$$\int_0^\infty e^{-pt} (-t \sin at) \, dt = \frac{p(-2a)}{(p^2 + a^2)^2}$$

or

$$\int_0^\infty e^{-pt} t \sin at \, dt = \frac{2pa}{(p^2 + a^2)^2}$$

which is *L11*. Ways of finding other entries in the table are outlined in the problems.

► PROBLEMS, SECTION 8

1. For integral k , verify *L5* and *L6* in the Laplace transform table. *Hint:* From *L2*, you can write: $\int_0^\infty e^{-pt} e^{-at} \, dt = 1/(p + a)$. Differentiate this equation repeatedly with respect to p . (See Chapter 4, Section 12, Example 4, page 235.) Also note *L32*. For the Γ function results in *L5* and *L6*, see Chapter 11, Problem 5.7.
2. By using *L2*, verify *L7* and *L8* in the Laplace transform table.

3. Using either $L2$, or $L3$ and $L4$, verify $L9$ and $L10$.
4. By differentiating the appropriate formula with respect to a , verify $L12$.
5. By integrating the appropriate formula with respect to a , verify $L19$.
6. By replacing a in $L2$ by $a + ib$ and then by $a - ib$, and adding and subtracting the results [as in (8.6) and (8.7)], verify $L13$ and $L14$.
7. Verify $L15$ to $L18$, by combining appropriate preceding formulas using (8.4).

Find the inverse transforms of the functions $F(p)$ in Problems 8 to 13.

8. $\frac{1+p}{(p+2)^2}$ *Hint: Use $L6$ and $L18$.*
9. $\frac{5-2p}{p^2+p-2}$ *Hint: Use $L7$ and $L8$.*
10. $\frac{2p-1}{p^2-2p+10}$ *Hint: You can use $L7$ and $L8$ with complex a and b , but $L13$ and $L14$ are more direct.*
11. $\frac{3p+2}{3p^2+5p-2}$
12. $\frac{3p+10}{p^2-25}$
13. $\frac{6-p}{p^2+4p+20}$
14. Show that a combination of entries $L3$ to $L10$, $L13$, $L14$ and $L18$ in the table, will give the inverse transform of any function of the form

$$\frac{Ap+B}{Cp^2+Ep+F}, \quad \text{where } A, B, C, E, \text{ and } F \text{ are constants.}$$

15. Prove $L32$ for $n = 1$. *Hint: Differentiate equation (8.1) with respect to p .*
16. Use $L32$ and $L3$ to obtain $L11$.
17. Use $L32$ and $L11$ to obtain $L(t^2 \sin at)$.
18. Use $L31$ to derive $L21$.

Table entries $L28$ and $L29$ are known as *translation* or *shifting* theorems. Do Problems 19 to 27 about them.

19. Prove the general formula $L29$ using (8.1).
20. Use $L29$ to verify $L6$, $L13$, $L14$, and $L18$.
21. Use $L29$ and $L11$ to obtain $L(te^{-at} \sin bt)$ which is not in the table.
22. Obtain $L(te^{-at} \cos bt)$ as in Problem 21.
23. Use the results which you have obtained in Problems 21 and 22 to find the inverse transform of $(p^2 + 2p - 1)/(p^2 + 4p + 5)^2$.
24. Sketch on the same axes graphs of $\sin t$, $\sin(t - \pi/2)$, and $\sin(t + \pi/2)$, and observe which way the graph shifts. *Hint: You can, of course, have your calculator or computer plot these for you, but it's simpler and much more useful to do it in your head. Hint: What values of t make the sines equal to zero? For an even simpler example, sketch on the same axes $y = t$, $y = t - \pi/2$, $y = t + \pi/2$.*
25. Use $L28$ to find the Laplace transform of

$$f(t) = \begin{cases} \sin(t - \pi/2), & t > \pi/2, \\ 0, & t < \pi/2. \end{cases}$$

26. Use $L28$ and $L4$ to find the inverse transform of $pe^{-p\pi}/(p^2 + 1)$.
27. Find the transform of

$$f(t) = \begin{cases} \sin(x - vt), & t > x/v, \\ 0, & t < x/v, \end{cases}$$

where x and v are constants.

► 9. SOLUTION OF DIFFERENTIAL EQUATIONS BY LAPLACE TRANSFORMS

We are going to discuss the solution of linear differential equations with constant coefficients (see Sections 5 and 6). Laplace transforms can reduce such an equation to an algebraic equation and so simplify solving it. Also, since Laplace transforms automatically use given values of initial conditions, we find immediately a desired particular solution without the extra step of determining constants to satisfy the initial conditions. Discontinuous forcing functions are messy to deal with by Section 6 methods; the Laplace transform method handles them easily.

We are going to take Laplace transforms of the terms in differential equations; to do this we need to know the transforms of derivatives $y' = dy/dt$, $y'' = d^2y/dt^2$, etc. To find $L(y')$, we use the definition (8.1) and integrate by parts, as follows

$$\begin{aligned} (9.1) \quad L(y') &= \int_0^\infty y'(t)e^{-pt} dt = e^{-pt}y(t) \Big|_0^\infty - (-p) \int_0^\infty y(t)e^{-pt} dt \\ &= -y(0) + pL(y) = pY - y_0 \end{aligned}$$

where for simplicity we have written $L(y) = Y$ and $y(0) = y_0$. To find $L(y'')$, we think of y'' as $(y')'$, and substitute y' for y in (9.1) to get

$$L(y'') = pL(y') - y'(0).$$

Using (9.1) again to eliminate $L(y')$, we finally have

$$(9.2) \quad L(y'') = p^2L(y) - py(0) - y'(0) = p^2Y - py_0 - y'_0.$$

Continuing this process, we obtain the transforms of the higher-order derivatives (Problem 1 and L35).

We are now ready to solve differential equations. We illustrate the method by some examples.

► **Example 1.** Solve $y'' + 4y' + 4y = t^2e^{-2t}$ with initial conditions $y_0 = 0$, $y'_0 = 0$.

We take the Laplace transform of each term in the equation, using L35 and L6 in the table of Laplace transforms. We get

$$p^2Y - py_0 - y'_0 + 4pY - 4y_0 + 4Y = L(t^2e^{-2t}) = \frac{2}{(p+2)^3}.$$

But the initial conditions are $y_0 = y'_0 = 0$. Thus we have

$$(p^2 + 4p + 4)Y = \frac{2}{(p+2)^3} \quad \text{or} \quad Y = \frac{2}{(p+2)^5}.$$

Now we want y , which is the inverse Laplace transform of Y . We look in the table for the inverse transform of $2/(p+2)^5$. By L6, we get

$$y = \frac{2t^4e^{-2t}}{4!} = \frac{t^4e^{-2t}}{12}.$$

This is much simpler than the general solution; we have obtained just the solution satisfying the given initial conditions.

► **Example 2.** Solve $y'' + 4y = \sin 2t$, subject to the initial conditions $y_0 = 10$, $y'_0 = 0$.

Using the table, take the Laplace transform of each term of the equation to get

$$p^2Y - py_0 - y'_0 + 4Y = L(\sin 2t) = \frac{2}{p^2 + 4}.$$

Then we substitute the initial conditions and solve for Y as follows:

$$\begin{aligned}(p^2 + 4)Y - 10p &= \frac{2}{p^2 + 4}, \\ Y &= \frac{10p}{p^2 + 4} + \frac{2}{(p^2 + 4)^2}.\end{aligned}$$

Finally, taking the inverse transform using $L4$ and $L17$, we have the desired solution:

$$y = 10 \cos 2t + \frac{1}{8}(\sin 2t - 2t \cos 2t) = 10 \cos 2t + \frac{1}{8} \sin 2t - \frac{1}{4}t \cos 2t.$$

► **Example 3.** Solve $y'' + 4y' + 13y = 20e^{-t}$, $y_0 = 1$, $y'_0 = 3$.

We take the transform of each term and solve for Y as follows:

$$\begin{aligned}p^2Y - p - 3 + 4pY - 4 + 13Y &= \frac{20}{p + 1}, \\ Y &= \frac{1}{p^2 + 4p + 13} \left(\frac{20}{p + 1} + p + 7 \right) = \frac{p^2 + 8p + 27}{(p + 1)(p^2 + 4p + 13)}.\end{aligned}$$

Since this Y is not in our table, we can either use a larger table, or use partial fractions to split Y into fractions which are in our table (which you can do by computer) or find the inverse transform by computer. We find:

$$Y = \frac{2}{p + 1} + \frac{-p + 1}{p^2 + 4p + 13} = \frac{2}{p + 1} + \frac{3}{(p + 2)^2 + 9} - \frac{p + 2}{(p + 2)^2 + 9}$$

and by $L2$, $L13$, and $L14$,

$$y = 2e^{-t} + e^{-2t} \sin 3t - e^{-2t} \cos 3t.$$

Sets of simultaneous differential equations can also be solved by using Laplace transforms. Here is an example.

► **Example 4.** Solve the set of equations

$$\begin{aligned}y' - 2y + z &= 0, \\ z' - y - 2z &= 0,\end{aligned}$$

subject to the initial conditions $y_0 = 1$, $z_0 = 0$.

We shall call $L(z) = Z$ and $L(y) = Y$ as before. We take the Laplace transform of each of the equations to get

$$\begin{aligned}pY - y_0 - 2Y + Z &= 0, \\ pZ - z_0 - Y - 2Z &= 0.\end{aligned}$$

After substituting the initial conditions and collecting terms, we have

$$\begin{aligned}(p-2)Y + Z &= 1, \\ Y - (p-2)Z &= 0.\end{aligned}$$

We solve this set of algebraic equations simultaneously for Y and Z (by any of the methods usually used for a pair of simultaneous equations—elimination, determinants, etc.). For example, we may multiply the first equation by $(p-2)$ and add the second to get

$$[(p-2)^2 + 1]Y = p-2 \quad \text{or} \quad Y = \frac{p-2}{(p-2)^2 + 1}.$$

We find y by looking up the inverse transform of Y using L14. We get

$$y = e^{2t} \cos t.$$

Similarly, solving for Z and looking up the inverse transform, we find

$$Z = \frac{1}{(p-2)^2 + 1}, \quad z = e^{2t} \sin t$$

. Alternatively, we could find z from the first differential equation by substituting the y solution:

$$z = 2y - y' = 2e^{2t} \cos t + e^{2t} \sin t - 2e^{2t} \cos t = e^{2t} \sin t.$$

Solving linear differential equations with constant coefficients is not the only use of Laplace transforms. As you will see in Chapter 13, Section 10, we may solve some kinds of partial differential equations by Laplace transforms. Also a table of Laplace transforms can be used to evaluate definite integrals of the type $\int_0^\infty e^{-pt} f(t) dt$.

► **Example 5.** By L15 with $a = 3$ and $p = 2$, we have

$$\int_0^\infty e^{-2t}(1 - \cos 3t) dt = \frac{3^2}{2(2^2 + 3^2)} = \frac{9}{26}.$$

Actually, there is more to the subject than this. Although we are discussing in this chapter the use of Laplace transforms as a tool, they also can play a more theoretical role in applied problems. It is often possible to find desired information about a problem directly from the Laplace transform of the solution without ever finding the solution. Thus the use of Laplace transforms may lead to a better understanding of a problem or a simpler method of solution. (Compare the use of matrices, for example, or the use of Fourier transforms.)

► PROBLEMS, SECTION 9

1. Continuing the method used in deriving (9.1) and (9.2), verify the Laplace transforms of higher-order derivatives of y given in the table (L35).

By using Laplace transforms, solve the following differential equations subject to the given initial conditions.

2. $y' - y = 2e^t, \quad y_0 = 3$

3. $y'' + 4y' + 4y = e^{-2t}, \quad y_0 = 0, y'_0 = 4$
4. $y'' + y = \sin t, \quad y_0 = 1, y'_0 = 0$
5. $y'' + y = \sin t, \quad y_0 = 0, y'_0 = -\frac{1}{2}$
6. $y'' - 6y' + 9y = te^{3t}, \quad y_0 = 0, y'_0 = 5$
7. $y'' - 4y' + 4y = 4, \quad y_0 = 0, y'_0 = -2$
8. $y'' + 16y = 8 \cos 4t, \quad y_0 = y'_0 = 0$
9. $y'' + 16y = 8 \cos 4t, \quad y_0 = 0, y'_0 = 8$
10. $y'' - 4y' + 4y = 6e^{2t}, \quad y_0 = y'_0 = 0$
11. $y'' - 4y = 4e^{2t}, \quad y_0 = 0, y'_0 = 1$
12. $y'' - y = e^{-t} - 2te^{-t}, \quad y_0 = 1, y'_0 = 2$
13. $y'' + y = 5 \sinh 2t, \quad y_0 = 0, y'_0 = 2$
14. $y'' - 4y' = -4te^{2t}, \quad y_0 = 0, y'_0 = 1$
15. $y'' + 9y = \cos 3t, \quad y_0 = 0, y'_0 = 6$
16. $y'' + 9y = \cos 3t, \quad y_0 = 2, y'_0 = 0$
17. $y'' + 5y' + 6y = 12, \quad y_0 = 2, y'_0 = 0$
18. $y'' - 4y = 3e^{-t}, \quad y_0 = 1, y'_0 = -3$
19. $y'' + y' - 5y = e^{2t}, \quad y_0 = 1, y'_0 = 2$
20. $y'' - 8y' + 16y = 32t, \quad y_0 = 1, y'_0 = 2$
21. $y'' + 4y' + 5y = 26e^{3t}, \quad y_0 = 1, y'_0 = 5$
22. $y'' + 2y' + 5y = 10 \cos t, \quad y_0 = 2, y'_0 = 1$
23. $y'' + 2y' + 5y = 10 \cos t, \quad y_0 = 0, y'_0 = 3$
24. $y'' - 2y' + y = 2 \cos t, \quad y_0 = 5, y'_0 = -2$
25. $y'' + 4y' + 5y = 2e^{-2t} \cos t, \quad y_0 = 0, y'_0 = 3$
26. $y'' + 2y' + 10y = -6e^{-t} \sin 3t, \quad y_0 = 0, y'_0 = 1$

Solve the following sets of equations by the Laplace transform method.

27. $y' + z' - 3z = 0 \quad y_0 = y'_0 = 0$
 $y'' + z' = 0 \quad z_0 = \frac{4}{3}$
28. $y' + z = 2 \cos t \quad y_0 = -1$
 $z' - y = 1 \quad z_0 = 1$
29. $y' + z' - 2y = 1 \quad y_0 = z_0 = 1$
 $z - y' = t$
30. $y' + 2z = 1 \quad y_0 = 0$
 $2y - z' = 2t \quad z_0 = 1$
31. $y'' + z'' - z' = 0 \quad y_0 = 0, y'_0 = 1$
 $y' + z' - 2z = 1 - e^t \quad z_0 = 1, z'_0 = 1$
32. $z' + 2y = 0 \quad y_0 = z_0 = 0$
 $y' - 2z = 2$

$$\begin{aligned} 33. \quad & y' - z' - y = \cos t & y_0 = -1 \\ & y' + y - 2z = 0 & z_0 = 0 \end{aligned}$$

Evaluate each of the following definite integrals by using the Laplace transform table.

$$34. \quad \int_0^\infty e^{-2t} \sin 3t \, dt = \frac{3}{13}. \text{ Hint: In (8.1), let } p = 2, f(t) = \sin 3t; \text{ use } L3 \text{ with } a = 3.$$

$$35. \quad \int_0^\infty t e^{-t} \sin 5t \, dt$$

$$36. \quad \int_0^\infty \frac{e^{-3t} \sin 2t}{t} \, dt$$

$$37. \quad \int_0^\infty t^5 e^{-2t} \, dt$$

$$38. \quad \int_0^\infty e^{-t} (1 - \cos 2t) \, dt$$

$$39. \quad \int_0^\infty \frac{e^{-t} - e^{-2t}}{t} \, dt$$

$$40. \quad \int_0^\infty \frac{e^{-2t} - e^{-2et}}{t} \, dt$$

$$41. \quad \int_0^\infty \frac{1}{t} e^{-2t} \sin(t\sqrt{2}) \, dt$$

$$42. \quad \int_0^\infty \frac{1}{t} e^{-t\sqrt{3}} \sin 2t \cos t \, dt$$

► 10. CONVOLUTION

In solving differential equations by Laplace transforms in Section 9, we found Y and then found the inverse transform y either in a table or by computer. We had no way of writing a formula for y . We now want to consider another way of finding inverse transforms. (Also see Bromwich integral, Chapter 14, page 696.)

Let us first see why the method we are going to discuss in this section is useful. Consider differential equations of the kind discussed in Sections 5 and 6, namely linear second-order equations with constant coefficients. Recall that such equations describe the vibrations or oscillations of either a mechanical or an electrical system. If the right-hand side of the equation is a function of t , called the *forcing function*, then the differential equation describes forced vibrations.

► **Example 1.** Let us solve the following representative equation by Laplace transforms, assuming that the system is initially at rest and that the force $f(t)$ starts being applied at $t = 0$.

$$(10.1) \quad Ay'' + By' + Cy = f(t), \quad y_0 = y'_0 = 0.$$

We take the Laplace transform of each term, substitute the initial conditions, and solve for Y as follows:

$$(10.2) \quad Ap^2Y + BpY + CY = L(f) = F(p), \quad Y = \frac{1}{Ap^2 + Bp + C} F(p).$$

Note that Y is a product of two functions of p . We know the inverse transform of $F(p)$, namely $f(t)$. The factor $T(p) = (Ap^2 + Bp + C)^{-1}$ (called the *transfer function*) can always be written as

$$T(p) = \frac{1}{A(p+a)(p+b)}$$

by factoring the quadratic expression in the denominator. Hence by $L7$ (or $L6$ if $a = b$) we can find the inverse transform of $T(p)$ for any problem. Then y

[the inverse transform of Y in (10.2)] is the inverse transform of a product of two functions whose inverse transforms we know. We are going to show how to write y as an integral (that is, we are going to verify *L34* in the table).

Let $G(p)$ and $H(p)$ be the transforms of $g(t)$ and $h(t)$. We want the inverse transform of the product $G(p)H(p)$. By the definition (8.1)

$$(10.3) \quad G(p)H(p) = \int_0^\infty e^{-pt} g(t) dt \cdot \int_0^\infty e^{-pt} h(t) dt.$$

Let us rewrite (10.3) replacing t by different dummy variables of integration so that we can write the product of the two integrals as a double integral. We then have

$$(10.4) \quad \begin{aligned} G(p)H(p) &= \int_0^\infty e^{-p\sigma} g(\sigma) d\sigma \cdot \int_0^\infty e^{-p\tau} h(\tau) d\tau \\ &= \int_0^\infty \int_0^\infty e^{-p(\sigma+\tau)} g(\sigma) h(\tau) d\sigma d\tau. \end{aligned}$$

Now we make a change of variables; in the σ integral (that is, with τ fixed), let $\sigma + \tau = t$. Then $\sigma = t - \tau$, $d\sigma = dt$, and the range of integration with respect to t is from $t = \tau$ (corresponding to $\sigma = 0$) to $t = \infty$ (corresponding to $\sigma = \infty$). Making these substitutions into (10.4), we get

$$(10.5) \quad G(p)H(p) = \int_{\tau=0}^\infty \int_{t=\tau}^\infty e^{-pt} g(t-\tau) h(\tau) dt d\tau.$$

Next we want to change the order of integration. From Figure 10.1, we see that the double integral in (10.5) is over the triangle in the first quadrant below the line $t = \tau$. The t integral ranges from the line $t = \tau$ to $t = \infty$ (indicated by a horizontal strip of width $d\tau$ from $t = \tau$ to ∞) and then the τ integral sums over the horizontal strips from $\tau = 0$ to $\tau = \infty$ covering the whole infinite triangle. Let us integrate with respect to τ first; τ then ranges from 0 to the line $\tau = t$ [indicated by a vertical strip in Figure 10.1] and then the t integral sums over the vertical strips from $t = 0$ to ∞ . Making this change in (10.5), we get

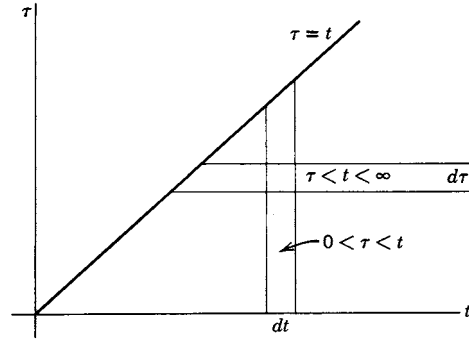


Figure 10.1

$$(10.6) \quad \begin{aligned} G(p)H(p) &= \int_{t=0}^\infty \int_{\tau=0}^t e^{-pt} g(t-\tau) h(\tau) d\tau dt \\ &= \int_0^\infty e^{-pt} \left[\int_0^t g(t-\tau) h(\tau) d\tau \right] dt \\ &= L \left[\int_0^t g(t-\tau) h(\tau) d\tau \right]. \quad (\text{See } L34.) \end{aligned}$$

The last step follows from the definition (8.1) of a Laplace transform.

Definition of Convolution The integral

$$(10.7) \quad \int_0^t g(t-\tau)h(\tau) d\tau = g * h$$

is called the *convolution* of g and h (or the *resultant* or the *Faltung*). Note the abbreviation $g * h$ for the convolution integral, and do not confuse the symbol $*$, written *on* the line, with a star used as a superscript meaning complex conjugate. It is easy to show (Problem 1) that $g * h = h * g$; this result and (10.6) and (10.7) give L34 in the table.

Now let's see how to use (10.6) or L34 to solve the kind of problem indicated in (10.1) and (10.2).

► **Example 2.** Solve $y'' + 3y' + 2y = e^{-t}$, $y_0 = y'_0 = 0$.

Taking the Laplace transform of each term, substituting the initial conditions, and solving for Y , we get

$$\begin{aligned} p^2Y + 3pY + 2Y &= L(e^{-t}), \\ Y &= \frac{1}{p^2 + 3p + 2}L(e^{-t}). \end{aligned}$$

Since we are intending to use the convolution integral, we do not bother to look up the transform of e^{-t} . We do want, however, the inverse transform of $1/(p^2 + 3p + 2)$; by L7, this is $e^{-t} - e^{-2t}$, so we have

$$Y = L(e^{-t} - e^{-2t})L(e^{-t}) = G(p)H(p),$$

with $g(t) = e^{-t} - e^{-2t}$ and $h(t) = e^{-t}$. We now use L34 to find y . Observe from L34 that we may use either $g(t-\tau)h(\tau)$ or $g(\tau)h(t-\tau)$ in the integral. It is well to choose whichever form is easier to integrate; usually it is best to put $(t-\tau)$ in the simpler function [here $h(t)$]. Then we have

$$\begin{aligned} y &= \int_0^t g(\tau)h(t-\tau) d\tau = \int_0^t (e^{-\tau} - e^{-2\tau})(e^{-(t-\tau)}) d\tau \\ &= e^{-t} \int_0^t (1 - e^{-\tau}) d\tau = e^{-t}(\tau + e^{-\tau}) \Big|_0^t \\ &= e^{-t}(t + e^{-t} - 1) = te^{-t} + e^{-2t} - e^{-t}. \end{aligned}$$

It is not always as easy to evaluate the convolution integral as it was in this example. However, let us observe that, at the very worst, we can always write the solution to a forced vibrations problem [equation (10.1)] as an integral (which can, if necessary, be evaluated numerically). This is true because, as we showed just after (10.2), we can always find the inverse transform of the transfer function $T(p)$, and so have Y as a product of two functions whose inverse transforms we know. Then y is given by the convolution (10.7) of the forcing function $f(t)$ and the inverse transform of the the transfer function. Also note (Problem 16) that a combination of L6, L7, L8 and L18 will handle any terms arising in a problem with nonzero initial conditions.

Fourier Transform of a Convolution We have shown that the Laplace transform of the convolution of two functions is the product of their Laplace transforms. There is a similar theorem for Fourier transforms; let us see what it says. Let $g_1(\alpha)$ and $g_2(\alpha)$ be the Fourier transforms of $f_1(x)$ and $f_2(x)$. By analogy with equations (10.3), (10.4), (10.5), and (10.6), we might expect the product $g_1(\alpha) \cdot g_2(\alpha)$ to be the Fourier transform of something; let's investigate this idea. Assuming that $\int_{-\infty}^{\infty} |f_1(x)f_2(x)|dx$ is finite, then by the definition of a Fourier transform [Chapter 7, equation (12.2)], we have

$$(10.8) \quad \begin{aligned} g_1(\alpha) \cdot g_2(\alpha) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} f_1(v)e^{-i\alpha v} dv \cdot \frac{1}{2\pi} \int_{-\infty}^{\infty} f_2(u)e^{-i\alpha u} du \\ &= \left(\frac{1}{2\pi}\right)^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-i\alpha(v+u)} f_1(v)f_2(u) dv du. \end{aligned}$$

[We have used different dummy integration variables as in (10.4).] Next we make the change of variables $x = v + u$, $dx = dv$, in the v integral, to get

$$(10.9) \quad \begin{aligned} g_1(\alpha)g_2(\alpha) &= \left(\frac{1}{2\pi}\right)^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-i\alpha x} f_1(x-u)f_2(u) dx du \\ &= \left(\frac{1}{2\pi}\right)^2 \int_{-\infty}^{\infty} e^{-i\alpha x} \left[\int_{-\infty}^{\infty} f_1(x-u)f_2(u) du \right] dx. \end{aligned}$$

If we define the convolution of $f_1(x)$ and $f_2(x)$ by

$$(10.10) \quad f_1 * f_2 = \int_{-\infty}^{\infty} f_1(x-u)f_2(u) du,^\dagger$$

then (10.9) becomes

$$(10.11) \quad g_1 \cdot g_2 = \frac{1}{2\pi} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} f_1 * f_2 e^{-i\alpha x} dx \right] = \frac{1}{2\pi} \cdot \text{Fourier transform of } f_1 * f_2.$$

In other words,

$$(10.12) \quad g_1 \cdot g_2 \text{ and } \frac{1}{2\pi} f_1 * f_2 \text{ are a pair of Fourier transforms.}$$

Because of the symmetry of the $f(x)$ and $g(\alpha)$ integrals, there is a similar result relating $f_1 \cdot f_2$ and the convolution of g_1 and g_2 . We find that (Problem 19)

$$(10.13) \quad g_1 * g_2 \text{ and } f_1 \cdot f_2 \text{ are a pair of Fourier transforms.}$$

As discussed in Chapter 7, after (12.2) and after (12.10), various references differ

[†]Note that (10.10) is really the same as (10.7) if we agree that, for Laplace transforms, $f(t) = 0$ when $t < 0$ (see the first paragraph of Section 8, page 437). For then in (10.7), $h(\tau) = 0$ for $\tau < 0$ and $g(t - \tau) = 0$ for $\tau > t$, so the integral would not really be different if written with infinite limits (in fact, it is sometimes written that way).

in the position of the factor $1/(2\pi)$. Some authors include factors of $1/(2\pi)$ or $1/\sqrt{2\pi}$ in the convolution definition (10.10); this definition as well as Chapter 7, equation (12.2), affects (10.12) and (10.13). Check the notation in any reference you are using.

► PROBLEMS, SECTION 10

1. Show that $g * h = h * g$ as claimed in L34. *Hint:* Let $u = t - \tau$ in (10.7).
2. Use L34 and L2 to find the inverse transform of $G(p)H(p)$ when $G(p) = 1/(p + a)$ and $H(p) = 1/(p + b)$; your result should be L7.

Use the convolution integral to find the inverse transforms of:

- | | |
|--|---|
| 3. $\frac{p}{(p^2 - 1)^2} = \frac{p}{p^2 - 1} \cdot \frac{1}{p^2 - 1}$ | 4. $\frac{1}{(p + a)(p + b)^2}$ |
| 5. $\frac{p}{(p + a)(p + b)^2}$ | 6. $\frac{1}{(p + a)(p^2 - b^2)}$ |
| 7. $\frac{p}{(p + a)(p^2 - b^2)}$ | 8. $\frac{1}{(p + a)(p + b)(p + c)}$ |
| 9. $\frac{2}{p^3(p + 2)}$ | 10. $\frac{1}{p(p^2 + a^2)^2}$ |
| 11. $\frac{p}{(p^2 + a^2)(p^2 + b^2)}$ | 12. $\frac{1}{p(p^2 + a^2)(p^2 + b^2)}$ |

Hint: In Problems 11 and 12 use $2 \sin \theta \cos \phi = \sin(\theta + \phi) + \sin(\theta - \phi)$.

13. Use the Laplace transform table to find $f(t) = \int_0^t e^{-\tau} \sin(t - \tau) d\tau$. *Hint:* In L34, let $g(t) = e^{-t}$ and $h(t) = \sin t$, and find $G(p)H(p)$ which is the Laplace transform of the integral you want. Break the result into partial fractions and look up the inverse transforms.

Use the convolution integral (see Example 2) to solve the following differential equations.

14. $y'' + 5y' + 6y = e^{-2t}$, $y_0 = y'_0 = 0$.
15. $y'' + 3y' - 4y = e^{3t}$, $y_0 = y'_0 = 0$.
16. Consider solving an equation like (10.1) but with nonzero initial conditions.
 - (a) Write the corrected form of (10.2), writing the transfer function in factored form as indicated just after (10.2). Consider the extra terms in Y which arise from the initial conditions; show that the inverse transforms of such terms can always be found from L6, L7, L8, and L18.
 - (b) Find the explicit form of the inverse transform of the transfer function for $a \neq b$ (use L7), and so write the general solution of (10.2) with nonzero initial conditions as a convolution integral plus the terms which you found in (a).
17. Solve the differential equation $y'' - a^2y = f(t)$, where

$$f(t) = \begin{cases} 0, & t < 0, \\ 1, & t > 0, \end{cases} \quad \text{and } y_0 = y'_0 = 0.$$

Hint: Use the convolution integral as in the example.

18. A mechanical or electrical system is described by the differential equation $y'' + \omega^2 y = f(t)$. Find y if

$$f(t) = \begin{cases} 1, & 0 < t < a, \\ 0, & \text{otherwise,} \end{cases} \quad \text{and} \quad y_0 = y'_0 = 0.$$

Hint: Use the convolution integral carefully. Consider $t < a$ and $t > a$ separately, remembering that $f(t) = 0$ for $t > a$. Show that

$$y = \begin{cases} \frac{1}{\omega^2}(1 - \cos \omega t), & t < a, \\ \frac{1}{\omega^2}[\cos \omega(t - a) - \cos \omega t], & t > a. \end{cases}$$

Sketch the motion if $a = \frac{1}{3}T$ where T is the period for free vibrations of the system; if $a = \frac{3}{2}T$; if $a = \frac{1}{10}T$.

19. Following the method of equations (10.8) to (10.12), show that $f_1 f_2$ and $g_1 * g_2$ are a pair of Fourier transforms.

► 11. THE DIRAC DELTA FUNCTION

In mechanics we consider the idea of an impulsive force such as a hammer blow which lasts for a very short time. We usually do not know the exact shape of the force function $f(t)$, and so we proceed as follows. Let the impulsive force $f(t)$ lasting from $t = t_0$ till $t = t_1$ be applied to a mass m ; then by Newton's second law we have

$$(11.1) \quad \int_{t_0}^{t_1} f(t) dt = \int_{t_0}^{t_1} m \frac{dv}{dt} dt = \int_{v_0}^{v_1} m dv = m(v_1 - v_0).$$

This says that the integral of $f(t)$ [called the impulse of $f(t)$] is equal to the change in the momentum of m , and we note that the result is independent of the shape of $f(t)$ but depends only on the area under the $f(t)$ curve. If this area is 1, we call the impulse a *unit impulse*. If $t_1 - t_0$ is very small, we may simply ignore the motion of m during this small time, and say only that the momentum jumped from mv_0 to mv_1 during the time $t_1 - t_0$. If $v_0 = 0$, the graph of the momentum as a function of time would be as in Figure 11.1, where we have simply omitted the (unknown) part of the graph between t_0 and t_1 . We note that if $t_1 - t_0$ is very small, the graph in Figure 11.1 is almost the unit step function (L24). Let us imagine making $t_1 - t_0$ smaller and smaller while keeping the jump in mv always 1.

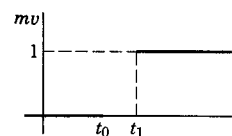


Figure 11.1

In Figures 11.2, 11.3 and 11.4 we have sketched some possible sequences of functions $f_n(t)$ which would do this. We could draw many other similar sets of graphs; the essential requirement is that $f(t)$ should become taller and narrower (that is, that the force should become more intense but act over a shorter time) in such a way that the impulse [area under the $f(t)$ curve] remains 1. We might then consider the limiting case in which Figure 11.1 has a jump of 1 at t_0 ; the force $f(t)$ required to produce this result would have to be infinite and act instantaneously. Also from equation (11.1), we see that the function $f(t)$ is the slope of the mv graph; thus we are asking for $f(t)$ to be the derivative of a step function at the jump. We see immediately that no ordinary function has these properties. However, we also note

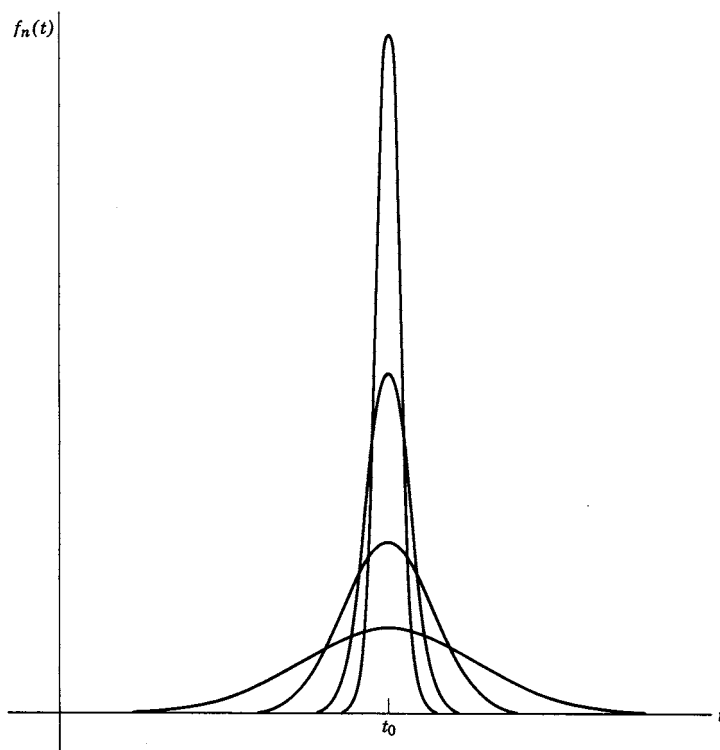


Figure 11.2

that we are not so much interested in $f(t)$ as in the results it produces. Figure 11.1 with a jump at t_0 makes perfectly good sense; for any $t > t_0$ we could choose a sufficiently tall and narrow $f_n(t)$ so that mv would already have its final value. We shall see that it is convenient to introduce a symbol $\delta(t - t_0)$ to represent the force which produces a jump of 1 in mv at t_0 ; $\delta(t - t_0)$ is called the *Dirac delta function* although it is not an ordinary function as we have seen. (It may properly be called a *generalized function* or a *distribution*, and is one of a whole class of such functions.) Introducing and using this symbol is much like introducing and using the symbol ∞ . It is convenient to write equations like $1/\infty = 0$, but we must not write $\infty/\infty = 1$; that is, such symbolic equations must be abbreviations for correct limiting processes. Let us investigate, then, how we can use the δ function correctly.

► **Example 1.** Consider the differential equation

$$(11.2) \quad y'' + \omega^2 y = f(t), \quad y_0 = y'_0 = 0.$$

This equation might describe the oscillations of a mass suspended by a spring, or a simple series electric circuit with negligible resistance. Let us assume that the system is initially at rest ($y_0 = y'_0 = 0$); then suppose that, at $t = t_0$ the mass is struck a sharp blow, or a sudden short surge of current is sent through the electric circuit. The function $f(t)$ may be one of those shown in Figures 11.2 to 11.4 or another similar function. Let us solve (11.2) with $f(t)$ equal to one of the functions in Figure 11.3, that is, $f(t) = ne^{-n(t-t_0)}, t > t_0$. Using Laplace transforms $L28$

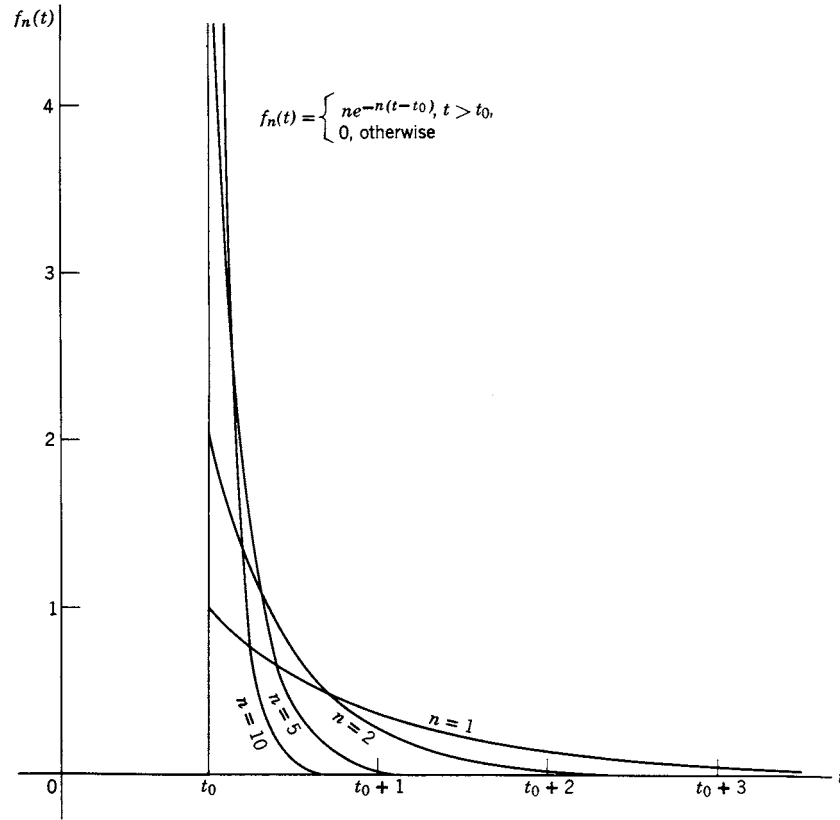


Figure 11.3

and $L2$ we find

$$(11.3) \quad \begin{aligned} (p^2 + \omega^2)Y &= L(ne^{-n(t-t_0)}) = n \cdot \frac{e^{-pt_0}}{p+n}, \\ Y &= n \cdot \frac{e^{-pt_0}}{(p+n)(p^2 + \omega^2)} = \frac{ne^{-pt_0}}{(n^2 + \omega^2)} \left[\frac{1}{p+n} + \frac{n}{p^2 + \omega^2} - \frac{p}{p^2 + \omega^2} \right]. \end{aligned}$$

(You can easily verify the partial fractions expansion in the last step.) Then by $L28$ with $a = t_0$, $L2$ with $a = n$, and $L3$, $L4$, with $a = \omega$, we find:

$$(11.4) \quad y = n \left(\frac{e^{-n(t-t_0)}}{n^2 + \omega^2} + \frac{n \sin \omega(t-t_0)}{(n^2 + \omega^2)\omega} - \frac{\cos \omega(t-t_0)}{n^2 + \omega^2} \right), \quad t > t_0.$$

(Of course, $y = 0$ for $t < t_0$.) By making $f(t)$ sufficiently narrow and peaked (that is, by making n large enough), we can make the first and third terms in y negligible, and the coefficient of $\sin \omega(t-t_0)$ approximately equal to $1/\omega$. Thus the solution is approximately

$$(11.5) \quad y = \frac{1}{\omega} \sin \omega(t-t_0), \quad t > t_0,$$

for a unit impulse of very short duration at $t = t_0$. (We have shown this only for the functions of Figure 11.3; however, the same result would be found for other sets of functions, such as those in Figure 11.4, for example—see Problem 5.)

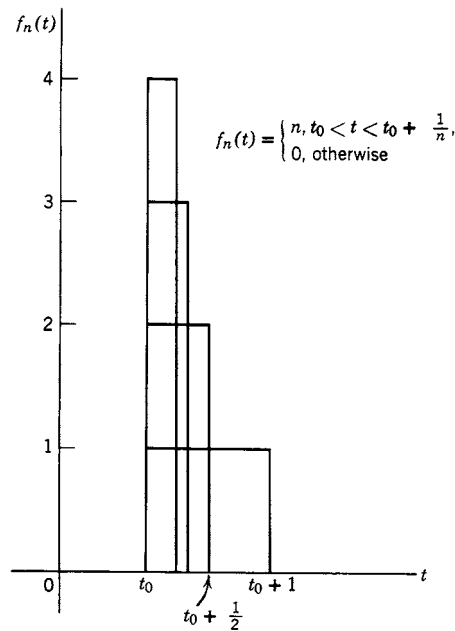


Figure 11.4

Now we would like to be able to find (11.5) without finding (11.4), in fact, without choosing a specific set of functions $f_n(t)$. Our discussion above suggests that we try using the symbol $\delta(t - t_0)$ for $f(t)$ on the right-hand side of (11.2). In solving the equation, we would then like to take the Laplace transform of $\delta(t - t_0)$.

Laplace Transform of a δ Function Let us investigate whether we can make sense out of the Laplace transform of $\delta(t - t_0)$. More generally, let us try to attach meaning to the integral $\int \phi(t)\delta(t - t_0) dt$, where $\phi(t)$ is any continuous function and $\delta(t - t_0)$ is the symbol indicating an impulse at t_0 . We consider the integrals $\int \phi(t)f_n(t - t_0) dt$, where the functions $f_n(t - t_0)$ are more and more strongly peaked at t_0 as n increases (Figure 11.5), but the area under each graph is 1. When $f_n(t - t_0)$ is so narrow that $\phi(t)$ is essentially constant [equal to $\phi(t_0)$] over the width of $f_n(t - t_0)$, the integral becomes nearly $\phi(t_0) \int f_n(t - t_0) dt = \phi(t_0) \cdot 1 = \phi(t_0)$; that is, the sequence of integrals $\int \phi(t)f_n(t - t_0) dt$ tends to $\phi(t_0)$ as n tends to infinity. It then seems reasonable to say that

$$(11.6) \quad \int_a^b \phi(t)\delta(t - t_0) dt = \begin{cases} \phi(t_0), & a < t_0 < b, \\ 0, & \text{otherwise.} \end{cases}$$

Equation (11.6) is the defining equation for the δ function; when we operate with δ functions, we use them in integrals and (11.6) tells us the value of the integral. The integral in (11.6) is not a Riemann integral; it is just a very useful symbol indicating that we have found the limit of $\int \phi(t)f_n(t - t_0) dt$ as $n \rightarrow \infty$. You may then ask how we can carry out familiar operations like integration by parts. When you treat an integral containing a δ function as an ordinary integral, you can, if you like, think

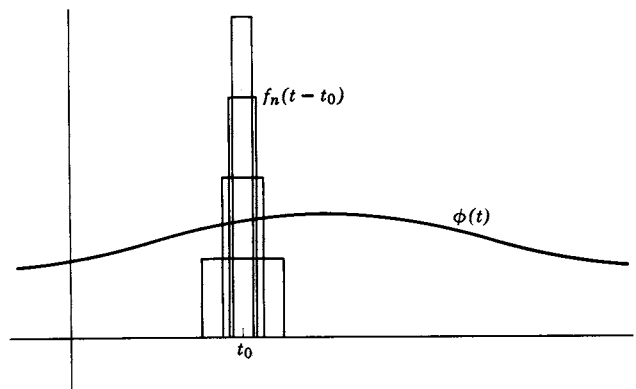


Figure 11.5

that you are really working with the functions $f_n(t - t_0)$ and then taking the limit at the end. Of course, all this needs mathematical justification which exists but is beyond our scope. (For two different mathematical developments of generalized functions, see Lighthill, and Chapter 9 of Folland.) Our purpose is just to make understandable the δ function formulas which are so useful in applications.

► **Example 2.** We can now easily find the Laplace transform of $\delta(t - t_0)$. In the notation used in L27 (which we are about to derive), we have, using (8.1),

$$(11.7) \quad L[\delta(t - a)] = \int_0^\infty \delta(t - a)e^{-pt} dt = e^{-pa}, \quad a > 0,$$

since, by 11.6, the integral of the product of $\delta(t - a)$ and a function “picks out” the value of the function at $t = a$. Now let us use our results to obtain (11.5) more easily.

► **Example 3.** Solve

$$(11.8) \quad y'' + \omega^2 y = \delta(t - t_0), \quad y_0 = y'_0 = 0.$$

Taking Laplace transforms and using (11.7), we get

$$(11.9) \quad (p^2 + \omega^2)Y = L[\delta(t - t_0)] = e^{-pt_0}.$$

Then

$$(11.10) \quad Y = \frac{e^{-pt_0}}{p^2 + \omega^2}$$

and, by L3 and L28,

$$(11.11) \quad y = \frac{1}{\omega} \sin \omega(t - t_0), \quad t > t_0,$$

as in (11.5).

Fourier Transform of a δ Function Using (11.6) and the definition of a Fourier transform [Chapter 7, equation (12.2)], we may write

$$(11.12) \quad g(\alpha) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(x-a) e^{-i\alpha x} dx = \frac{1}{2\pi} e^{-i\alpha a}.$$

Formally, then (12.2) of Chapter 7 would give for the inverse transform

$$(11.13) \quad \delta(x-a) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\alpha(x-a)} d\alpha.$$

We say “formally” because the integral in (11.13) does not converge. However, if we replace the limits $-\infty, \infty$ by $-n, n$, we obtain a set of functions (Problem 12) which, like the functions $f_n(t)$ in Figures 11.2 to 11.4, are increasingly peaked around $x = a$ as n increases, but all have area 1. In this sense, then, (11.13) is a representation of the δ function. Equations (11.12) and (11.13) are useful in quantum mechanics.

Another Physical Application of δ functions What is the density (mass per unit length) of a point mass on the x axis? Compare the concept of a point mass with our discussion of δ functions. We could think of a point mass as corresponding to the limiting case of a density function like those in Figures 11.2 to 11.4. A point mass at $x = a$ requires that the density be zero everywhere except at $x = a$ but the integral of the density function across $x = a$ should be the mass m . Thus we can write the density function for a point mass m at $x = a$ as $m\delta(x-a)$. Similarly, we can represent the charge density for point electrical charges using δ functions.

► **Example 4.** The charge density for a charge of 2 at $x = 3$, a charge of -5 at $x = 7$ and a charge of 3 at $x = -4$ would be $2\delta(x-3) - 5\delta(x-7) + 3\delta(x+4)$.

Derivatives of the δ Function To see that we can attach a meaning to the derivative of $\delta(x-a)$, we write $\int_{-\infty}^{\infty} \phi(x)\delta'(x-a) dx$ and integrate by parts to get

$$(11.14) \quad \int_{-\infty}^{\infty} \phi(x)\delta'(x-a) dx = \phi(x)\delta(x-a) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \phi'(x)\delta(x-a) dx = -\phi'(a).$$

The integrated term is zero at $\pm\infty$, and we evaluated the integral using equation (11.6). Thus, just as $\delta(x-a)$ “picks out” the value of $\phi(x)$ at $x = a$ [see equation (11.6)], so $\delta'(x-a)$ picks out the negative of $\phi'(x)$ at $x = a$. Integrating by parts twice (Problem 14), we find

$$(11.15) \quad \int_{-\infty}^{\infty} \phi(x)\delta''(x-a) dx = \phi''(a).$$

Repeated integrations by parts gives the formula for the derivative of any order of the δ function (Problem 14):

$$(11.16) \quad \int_{-\infty}^{\infty} \phi(x)\delta^{(n)}(x-a) dx = (-1)^n \phi^{(n)}(a).$$

We have written the integrals in (11.14) to (11.16) with limits $-\infty$ to ∞ , but all that is necessary is that the range of integration include $x = a$; otherwise, as for (11.6), the integrals are zero.

Some Formulas Involving δ Functions Our discussion at the beginning of this section (see Figure 11.1 and $L24$ in the Laplace transform table) implied that the derivative at $x = a$ of the unit step function ought to be $\delta(x - a)$.

$$(11.17) \quad \begin{aligned} (a) \quad u(x - a) &= \begin{cases} 1, & x > a \\ 0, & x < a \end{cases} \\ (b) \quad u'(x - a) &= \delta(x - a). \end{aligned}$$

What does the $u' = \delta$ equation mean? By definition, two generalized functions (distributions) are equal, say $G_1(x) = G_2(x)$, if $\int \phi(x)G_1 dx = \int \phi(x)G_2 dx$ for any test function $\phi(x)$. Test functions are assumed to be very well behaved functions; let's assume that they are continuous with continuous derivatives of all orders and that they are identically zero outside some finite interval so that the integrated term in an integration by parts is always zero. You can think of generalized functions as being operators; given a test function $\phi(x)$, they "operate" on it to produce a value such as $\phi(0)$. Compare the differential operators in Problem 5.31. We wrote $Dx = xD + 1$ where $D = d/dx$. This would be nonsense as an elementary calculus formula, but as an operator equation to be applied to $y(x)$, it means $D(xy) = (xD + 1)y = xy' + y$ which is correct. In a similar way, two generalized functions are equal if they give the same results when they operate on any test function. Let's try this for (11.17b). We multiply u' by $\phi(x)$, integrate by parts (noting that the integrated term is zero because we require test functions to be zero for large $|x|$), substitute the values of $u(x - a)$, and integrate again to get

$$\int_{-\infty}^{\infty} \phi(x)u'(x - a) dx = - \int_{-\infty}^{\infty} \phi'(x)u(x - a) dx = - \int_a^{\infty} \phi'(x) dx = \phi(a).$$

This is indeed the value of the integral of $\phi(x)\delta(x - a)$, so $u'(x - a) = \delta(x - a)$ is a valid operator equation (generalized function equation).

Since we think of $\delta(x)$ and its derivatives as being zero except at the origin, and x is zero at the origin, it might seem plausible that $x\delta$, $x^2\delta$, $x\delta'$, etc. would be identically zero. It turns out that some of these are zero and some are not; to find out, we multiply by an arbitrary test function $\phi(x)$ and integrate. We state a few results; also see Problems 17 and 18.

$$(11.18) \quad \begin{aligned} (a) \quad x\delta(x) &= 0 \\ (b) \quad x\delta'(x) &= -\delta(x) \\ (c) \quad x^2\delta''(x) &= 2\delta(x) \end{aligned}$$

To check (b), we multiply by $\phi(x)$ and integrate using (11.14) with $\phi(x)$ replaced by $x\phi(x)$.

$$\int_{-\infty}^{\infty} x\delta'(x)\phi(x) dx = -(x\phi)' \Big|_{x=0} = -(x\phi' + \phi) \Big|_{x=0} = -\phi(0) = - \int_{-\infty}^{\infty} \delta(x)\phi(x) dx.$$

Here is another way to produce valid generalized function identities like those in (11.18). Suppose $G_1(x) = G_2(x)$; then we can show (Problem 19a) that $\frac{d}{dx}G_1(x) = \frac{d}{dx}G_2(x)$ and $x^n G_1(x) = x^n G_2(x)$. For example, if we differentiate (11.18a) we get $x\delta'(x) + \delta(x) = 0$, or $x\delta'(x) = -\delta(x)$ which is (11.18b).

We list a few more operator equations (see Problems 20 and 21).

$$\begin{aligned}
 (a) \quad & \delta(-x) = \delta(x) \text{ and } \delta(x-a) = \delta(a-x); \\
 (b) \quad & \delta'(-x) = -\delta'(x) \text{ and } \delta'(x-a) = -\delta'(a-x); \\
 (11.19) \quad (c) \quad & \delta(ax) = \frac{1}{|a|} \delta(x), \quad a \neq 0; \\
 (d) \quad & \delta[(x-a)(x-b)] = \frac{1}{|a-b|} [\delta(x-a) + \delta(x-b)], \quad a \neq b; \\
 (e) \quad & \delta[f(x)] = \sum_i \frac{\delta(x-x_i)}{|f'(x_i)|} \quad \text{if } f(x_i) = 0 \text{ and } f'(x_i) \neq 0.
 \end{aligned}$$

We first prove (c) when a is negative, say $a = -b$, $b > 0$. Let $u = -bx$, then $du = -b dx$, and the limits $x = -\infty, \infty$, become $u = \infty, -\infty$.

$$\begin{aligned}
 \int_{-\infty}^{\infty} \phi(x) \delta(-bx) dx &= \int_{\infty}^{-\infty} \phi\left(\frac{u}{-b}\right) \delta(u) \left(\frac{du}{-b}\right) = \frac{1}{b} \int_{-\infty}^{\infty} \phi\left(\frac{u}{-b}\right) \delta(u) du \\
 &= \frac{1}{b} \phi(0) = \frac{1}{|a|} \phi(0) = \frac{1}{|a|} \int_{-\infty}^{\infty} \phi(x) \delta(x) dx.
 \end{aligned}$$

From the second integral to the third, we have reversed the order of integration (one minus sign) and also changed $\frac{du}{b}$ to $\frac{du}{-b}$ (another minus sign, which cancels the first). Now, if we repeat the calculation using $a > 0$ instead of $-b$, neither of these sign reversals occurs, and so we get the result $\frac{1}{a}\phi(0)$ instead of $\frac{1}{b}\phi(0)$. But when $a > 0$, a and $|a|$ are the same. Thus we get the result stated in (11.19c).

δ functions in 2 or 3 dimensions It is now straightforward to write the defining equations in rectangular coordinates for δ functions in 2 or 3 dimensions. We have

$$(11.20) \quad \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \phi(x, y) \delta(x - x_0) \delta(y - y_0) dx dy = \phi(x_0, y_0).$$

$$\begin{aligned}
 (11.21) \quad & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \phi(x, y, z) \delta(x - x_0) \delta(y - y_0) \delta(z - z_0) dx dy dz \\
 &= \phi(x_0, y_0, z_0).
 \end{aligned}$$

As in one dimension, the delta function “picks out” the value of the test function ϕ at the “peak” of the δ function. The integrals need not be over all space, just over a region containing the point $\mathbf{r}_0$; otherwise the integral is zero. The abbreviations $\delta(\mathbf{r})$ or $\delta^3(\mathbf{r})$ are often used for $\delta(x)\delta(y)\delta(z)$, but note carefully that they do not mean functions of the vector $\mathbf{r}$, but rather functions of the components x, y, z of $\mathbf{r}$. Similarly you may see $\delta(\mathbf{r} - \mathbf{r}_0)$ or $\delta^3(\mathbf{r} - \mathbf{r}_0)$ meaning the δ function in (11.21).

In spherical coordinates, let's use f instead of ϕ to mean a test function (since ϕ is a spherical coordinate angle). By the definition of δ functions, we want $\iiint f(r, \theta, \phi) \delta(r - r_0) \delta(\theta - \theta_0) \delta(\phi - \phi_0) dr d\theta d\phi = f(r_0, \theta_0, \phi_0)$. But since we would like to use the volume element $d\tau = r^2 \sin \theta dr d\theta d\phi = r^2 dr d\Omega$, we need to write

(also see Problem 22)

$$(11.22) \quad \delta(\mathbf{r} - \mathbf{r}_0) = \frac{\delta(r - r_0)\delta(\theta - \theta_0)\delta(\phi - \phi_0)}{r^2 \sin \theta}; \quad \text{then}$$

$$\iiint f(r, \theta, \phi) \delta(\mathbf{r} - \mathbf{r}_0) d\tau = f(r_0, \theta_0, \phi_0).$$

Similarly in cylindrical coordinates, with $d\tau = r dr d\theta dz$,

$$(11.23) \quad \delta(\mathbf{r} - \mathbf{r}_0) = \frac{\delta(r - r_0)\delta(\theta - \theta_0)\delta(z - z_0)}{r}; \quad \text{then}$$

$$\iiint f(r, \theta, z) \delta(\mathbf{r} - \mathbf{r}_0) d\tau = f(r_0, \theta_0, z_0).$$

Note that we can use these formulas to write mass density or charge density functions in the various coordinate systems.

► **Example 5.** Suppose there is a unit charge or unit mass at the point $(x, y, z) = (-1, \sqrt{3}, -2)$; then in rectangular coordinates, the density is

$$\rho = \delta(x + 1)\delta(y - \sqrt{3})\delta(z + 2).$$

In cylindrical coordinates the point is $(r, \theta, z) = (2, 2\pi/3, -2)$ so in cylindrical coordinates the density is

$$\rho = \delta(r - 2)\delta(\theta - 2\pi/3)\delta(z + 2)/r.$$

In spherical coordinates, the point is $(r, \theta, \phi) = (2\sqrt{2}, 3\pi/4, 2\pi/3)$, so in spherical coordinates the density is

$$\rho = \delta(r - 2\sqrt{2})\delta(\theta - 3\pi/4)\delta(\phi - 2\pi/3)/(r \sin \theta).$$

Finally, let's verify two useful operator equations for δ functions in 3 dimensions.

$$(11.24) \quad \nabla \cdot \frac{\mathbf{e}_r}{r^2} = 4\pi\delta(\mathbf{r});$$

$$(11.25) \quad \nabla^2 \frac{1}{r} = -4\pi\delta(\mathbf{r}).$$

You can easily show (Problem 24a) that $\nabla \cdot (\mathbf{e}_r/r^2)$ is zero for any $r \neq 0$ (and undefined for $r = 0$). Also, by the divergence theorem [Chapter 6, equation (10.17)] in spherical coordinates, we find

$$\iiint_{\text{volume } \tau} \nabla \cdot \frac{\mathbf{e}_r}{r^2} d\tau = \iint_{\substack{\text{surface} \\ \text{inclosing } \tau}} \frac{\mathbf{e}_r}{r^2} \cdot \mathbf{e}_r d\sigma = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \frac{1}{r^2} r^2 \sin \theta d\theta d\phi = 4\pi.$$

Thus $\nabla \cdot (\mathbf{e}_r/r^2)$ has the properties that it is zero for all $r > 0$ but its integral over any volume including the origin $= 4\pi$; this suggests that it is equal to $4\pi\delta(\mathbf{r})$. Let's verify that this is correct. (Compare Problem 25.) Since $\nabla \cdot (\mathbf{e}_r/r^2)$ depends only on r (Problem 24a), we use a test function $f(r)$. We want to show that

$\iiint f(r) \nabla \cdot (\mathbf{e}_r/r^2) d\tau$, over any volume containing the origin, is equal to $4\pi f(0)$. For convenience we integrate over the volume inside the sphere $r = a$. (Since the integrand is zero for $r > 0$, the answer is the same for any volume containing the origin.) By Problem 11.17(e) of Chapter 6 with $\phi = f$, $\mathbf{V} = \mathbf{e}_r/r^2$, and $\mathbf{n} = \mathbf{e}_r$, we find

$$\iiint_{\text{volume } r < a} f(r) \nabla \cdot \frac{\mathbf{e}_r}{r^2} d\tau = \iint_{\text{surface } r=a} f(r) \frac{\mathbf{e}_r}{r^2} \cdot \mathbf{e}_r d\sigma - \iiint_{\text{volume } r < a} \nabla f(r) \cdot \frac{\mathbf{e}_r}{r^2} d\tau.$$

On the surface $r = a$, the integrand of the surface integral is $f(a) \frac{1}{a^2} a^2 d\Omega$, so the surface integral is $4\pi f(a)$. In the volume integral, $\nabla f(r) \cdot \mathbf{e}_r$ is the r component of $\nabla f(r)$; in spherical coordinates this is just $\partial f/\partial r$ (Chapter 6, equation 6.8). Thus the volume integral on the right-hand side is

$$\iiint_{\text{volume } r < a} \frac{\partial f}{\partial r} \frac{1}{r^2} r^2 dr d\Omega = 4\pi f(r) \Big|_0^a = 4\pi[f(a) - f(0)]$$

and we have

$$\iiint_{\text{volume } r < a} f(r) \nabla \cdot \frac{\mathbf{e}_r}{r^2} d\tau = 4\pi f(a) - 4\pi[f(a) - f(0)] = 4\pi f(0)$$

as we expected. Thus (11.24) is a valid operator equation. You can show (Problem 24b) that $\nabla(1/r) = -\mathbf{e}_r/r^2$. Since $\nabla \cdot \nabla = \nabla^2$ (that is, $\text{div grad} = \text{Laplacian}$), we have $\nabla^2(1/r) = \nabla \cdot \nabla(1/r) = -\nabla \cdot (\mathbf{e}_r/r^2)$. Thus (11.25) is also valid.

We can write (11.24) and (11.25) with the peak of the δ function shifted from the origin to $\mathbf{r}_0$. The unit vector from $\mathbf{r}_0$ to $\mathbf{r}$ can be written as $(\mathbf{r} - \mathbf{r}_0)/|\mathbf{r} - \mathbf{r}_0|$. Then we have

$$(11.26) \quad \nabla^2 \frac{1}{|\mathbf{r} - \mathbf{r}_0|} = -\nabla \cdot \frac{\mathbf{r} - \mathbf{r}_0}{|\mathbf{r} - \mathbf{r}_0|^3} = -4\pi\delta(\mathbf{r} - \mathbf{r}_0).$$

It is now interesting to note that we have seen this before without recognizing that we were dealing with a δ function. Look back to Chapter 6, Equations (6.10.19) to (6.10.25). With $\mathbf{D}$ given by (6.10.19), we have from (8.11.24) above, $\nabla \cdot \mathbf{D} = \frac{q}{4\pi} 4\pi\delta(\mathbf{r}) = q\delta(\mathbf{r})$ which we recognize as the charge density for a charge q at the origin. Thus, although we wrote (6.10.25) with ρ as the density of a charge distribution, we could now write it with $\rho = \text{charge density of a point charge} = q\delta(\mathbf{r})$, and then (6.10.25) becomes (6.10.22).

► PROBLEMS, SECTION 11

- Find the inverse Laplace transform of e^{-2p}/p^2 in the following ways:
 - using $L5$ and $L27$ and the convolution integral of Section 10;
 - using $L28$.
- Verify $L24$ in the table by using $L1$, $L27$, and the convolution integral.
- Verify $L28$ in the table by using $L27$ and the convolution integral.
- Show that $\int_{-\infty}^{\infty} f_n(t) dt = 1$ for the functions $f_n(t)$ in Figures 11.3 and 11.4.

5. Solve the differential equation $y'' + \omega^2 y = f(t)$, $y_0 = y'_0 = 0$, with $f(t)$ given by the functions in Figure 11.4, by the following methods.
- Use the convolution integral, being careful to consider separately the three intervals 0 to t_0 , t_0 to $t_0 + 1/n$, and $t_0 + 1/n$ to ∞ .
 - Write $f_n(t)$ as a difference of unit step functions as in L25, and use L25 to find $L(f_n)$. Expand $\frac{1}{p(p^2 + \omega^2)}$ by partial fractions and use L28 to find $y_n(t)$. Your result should agree with (a).
 - Let $n \rightarrow \infty$ and show that your solution in (a) and (b) tends to the same solution (11.5) obtained using the functions of Figure 11.3; that is, either set of functions gives, in the limit, the same solution (11.11) obtained by using the δ function. Note that, when you let $n \rightarrow \infty$, you do not need to consider the interval t_0 to $t_0 + 1/n$ since, if $t > t_0$, then for sufficiently large n , $t > t_0 + 1/n$.
6. (a) Let a mechanical or electrical system be described by the differential equation $Ay'' + By' + Cy = f(t)$, $y_0 = y'_0 = 0$. As in Problem 10.16b, write the solution as a convolution (assume $a \neq b$). Let $f(t)$ be one of the functions in Figure 11.4 and Problem 5. Find y and then let $n \rightarrow \infty$.
- Solve (a) with $f(t) = \delta(t - t_0)$; your result should be the same as in (a).
 - The solution y as found in (a) and (b) is called the *response* of the system to a unit impulse. Show that the response of a system to a unit impulse at $t_0 = 0$ is the inverse Laplace transform of the transfer function.

Using the δ function method, find the response (see Problem 6c) of each of the following systems to a unit impulse.

- $y'' + 2y' + y = \delta(t - t_0)$
- $y'' + 4y' + 5y = \delta(t - t_0)$
- $y'' + 2y' + 10y = \delta(t - t_0)$
- $y'' - 9y = \delta(t - t_0)$
- $\frac{d^4 y}{dt^4} - y = \delta(t - t_0)$
- Evaluate the functions $f_n(x - a)$ defined by the integral in (11.13) with limits $-n, n$. Show that $\int_{-\infty}^{\infty} f_n(x - a) dx = 1$ for all n . Sketch or computer plot graphs of several f_n 's to show that as n increases the functions $f_n(x)$ are increasingly peaked around $x = a$, and that as $|x - a|$ increases, they oscillate with decreasing amplitude.
- Using δ functions, write the following mass or charge density functions.
 - Mass 5 at $x = 2$, and mass 3 at $x = -7$.
 - Charge 3 at $x = -5$ and charge -4 at $x = 10$.
- Integrate by parts as we did for (11.14) to obtain (11.15) and (11.16).
- Use (11.6) and (11.14) to (11.16) to evaluate the following integrals. *Warning hint:* See comments just after (11.6) and (11.16) about the range of integration.
 - $\int_0^\pi \sin x \, \delta(x - \frac{\pi}{2}) dx$
 - $\int_0^\pi \sin x \, \delta(x + \frac{\pi}{2}) dx$
 - $\int_{-1}^1 e^{3x} \delta'(x) dx$
 - $\int_0^\pi \cosh x \, \delta''(x - 1) dx$

16. Verify the operator equation $\frac{d}{dx} \operatorname{sgn} x = 2\delta(x)$ where the function signum x , meaning “sign of x ,” and abbreviated $\operatorname{sgn} x$, is defined by

$$\operatorname{sgn} x = \begin{cases} 1, & x > 0, \\ -1, & x < 0. \end{cases}$$

17. Verify (11.18a) and (11.18c) by multiplying by a test function and integrating.
 18. Use equation (11.16) to generalize the operator equations (11.18) as follows:

- (a) Show that $x^m \delta^{(n)}(x) = 0$ if $m > n$; compare equation (11.18a).
 (b) Show that $x^n \delta^{(n)}(x) = (-1)^n n! \delta(x)$; compare (11.18b) and (11.18c).
 (c) Show that $x^m \delta^{(n)}(x) = (-1)^m \frac{n!}{(n-m)!} \delta^{(n-m)}(x)$, $m \leq n$.
 (d) Use the results in (a) and (b) to show that

$$(x^2 + y^2 + z^2) \nabla^2 [\delta(x) \delta(y) \delta(z)] = 6\delta(x) \delta(y) \delta(z).$$

19. (a) Show that you can differentiate a generalized function equation or multiply it by a power of x . This means to show that if $\int \phi(x) G_1(x) dx = \int \phi(x) G_2(x) dx$ for all test functions ϕ , then

$$\begin{aligned} \int \phi(x) G_1'(x) dx &= \int \phi(x) G_2'(x) dx \quad \text{and} \\ \int \phi(x) x^n G_1(x) dx &= \int \phi(x) x^n G_2(x) dx. \end{aligned}$$

Hints: For the differentiation proof, integrate by parts. For the multiplication by x^n proof, consider whether $x^n \phi(x)$ is a test function if ϕ is. See comment just after equation (11.17).

- (b) Multiply (11.18b) by x and use (11.18a). Differentiate the result and simplify to get (11.18c).
 (c) Multiply (11.18c) by x , use (11.18a), differentiate and simplify to find $x^3 \delta'''(x)$ in terms of $\delta(x)$. Check your result by Problem (18b).
 (d) Try a few more examples as in (b) and (c) and check your results by Problem 18.
 20. Verify the operator equations in (11.19) not done in text.
Hints for (a) and (b): Follow the text method of proof of (c), making the change of variable $u = -x$ or $u = a - x$. *Hints* for (c) and (d): Split the integral into a sum of integrals each including just one x_i . In (d), what is the value of $(x - a)$ when x is in the vicinity of b ? Use part (c).

21. Make use of the operator equations (11.19) and previous equations to evaluate the following integrals.

$$\begin{array}{ll} \text{(a)} \int_0^3 (5x - 2) \delta(2 - x) dx & \text{(b)} \int_0^\infty \phi(x) \delta(x^2 - a^2) dx \\ \text{(c)} \int_{-1}^1 \cos x \delta(-2x) dx & \text{(d)} \int_{-\pi/2}^{\pi/2} \cos x \delta(\sin x) dx \end{array}$$

22. You may find the spherical coordinate δ function written as

$$\delta(\mathbf{r} - \mathbf{r}_0) = \delta(r - r_0) \delta(\cos \theta - \cos \theta_0) \delta(\phi - \phi_0) / r^2.$$

Show that this equation is equivalent to (11.22). *Hints:* You want to show that $\delta(\cos \theta - \cos \theta_0) = \delta(\theta - \theta_0) / \sin \theta$. See (11.19e). Also note that it doesn't really matter whether we write $r^2 \sin \theta$ or $r_0^2 \sin \theta_0$ in the denominator of (11.22) since the δ functions are zero unless $r = r_0$ and $\theta = \theta_0$.

23. Write a formula in rectangular coordinates, in cylindrical coordinates, and in spherical coordinates for the density of a unit point charge or mass at the point with the given rectangular coordinates:

- (a) $(-5, 5, 0)$ (b) $(0, -1, -1)$
 (c) $(-2, 0, 2\sqrt{3})$ (d) $(3, -3, -\sqrt{6})$

24. (a) Show that $\nabla \cdot (\mathbf{e}_r/r^2) = 0$ for $r > 0$. *Hint:* You can do this in rectangular coordinates, but it is easier in spherical coordinates. See Chapter 6, equation (7.9). Show that $\nabla \cdot [\mathbf{e}_r F(r)]$, for any $F(r)$, is a function of r only.

(b) Show that $\nabla(1/r) = -\mathbf{e}_r/r^2$. See Chapter 6, equation (6.8).

25. Let

$$F(x) = \begin{cases} x - 2, & x > 0, \\ 0, & x < 0. \end{cases}$$

Show that $F''(x) = 0$ for all $x \neq 0$, and $\int_{-\infty}^{\infty} F''(x) dx = 1$, which leads you to think that $F''(x)$ might $= \delta(x)$. Show in two ways, as outlined in (a) and (b), that this is not true.

- (a) Show that $\int_{-\infty}^{\infty} \phi(x) F''(x) dx = \phi(0) + 2\phi'(0)$, where ϕ is any test function. Then by (11.6) and (11.14), what is $F''(x)$?
 (b) Show that $F(x) = (x - 2)u(x)$ where $u(x)$ is the unit step function in (11.17). Differentiate this equation twice and simplify using (11.17) and (11.18). Compare your result in (a).
 (c) As in (a) and (b), find $G''(x)$ in terms of δ and δ' if

$$G(x) = \begin{cases} 3x + 1, & x > 0, \\ 2x - 4, & x < 0. \end{cases}$$

► 12. A BRIEF INTRODUCTION TO GREEN FUNCTIONS

Let's do some examples to see what a Green function is and how we can use it to solve ordinary differential equations. Also see Chapter 13, Section 8, for an application to partial differential equations. (You might find it interesting to read "The Green of Green Functions", *Physics Today*, December 2003, 41–46.)

► **Example 1.** We reconsider the differential equation (11.2), namely

$$(12.1) \quad y'' + \omega^2 y = f(t), \quad y_0 = y'_0 = 0$$

where $f(t)$ is some given forcing function. Using (11.6), we can write

$$(12.2) \quad f(t) = \int_0^\infty f(t') \delta(t' - t) dt',$$

that is, we can think of the force $f(t)$ as (a limiting case of) a whole sequence of impulses. (You might reflect that, on the molecular level, air pressure is the force per unit area due to a tremendous number of impacts of individual molecules.) Now suppose that we have solved (12.1) with $f(t)$ replaced by $\delta(t' - t)$, that is, we find the response of the system to a unit impulse at t' . Let us call this response $G(t, t')$, that is, $G(t, t')$ is the solution of

$$(12.3) \quad \frac{d^2}{dt^2} G(t, t') + \omega^2 G(t, t') = \delta(t' - t).$$

Then, given some forcing function $f(t)$, we try to find a solution of (12.1) by “adding up” the responses of many such impulses. We shall show that this solution is

$$(12.4) \quad y(t) = \int_0^\infty G(t, t') f(t') dt'.$$

Substituting (12.4) into (12.1) and using (12.3) and (12.2), we find

$$\begin{aligned} y'' + \omega^2 y &= \left(\frac{d^2}{dt^2} + \omega^2 \right) y = \left(\frac{d^2}{dt^2} + \omega^2 \right) \int_0^\infty G(t, t') f(t') dt' \\ &= \int_0^\infty \left(\frac{d^2}{dt^2} + \omega^2 \right) G(t, t') f(t') dt' = \int_0^\infty \delta(t' - t) f(t') dt' = f(t). \end{aligned}$$

Thus (12.4) is a solution of (12.1).

The function $G(t, t')$ is called a *Green function* (or Green's function). The Green function is the response of the system to a unit impulse at $t = t'$. Solving (12.3) with initial conditions $G = 0$ and $dG/dt = 0$ at $t = 0$, we find (Problem 1)

$$(12.5) \quad G(t, t') = \begin{cases} 0, & 0 < t < t', \\ \frac{1}{\omega} \sin \omega(t - t'), & 0 < t' < t. \end{cases}$$

Then (12.4) gives the solution of (12.1) with $y_0 = y'_0 = 0$, namely

$$(12.6) \quad y(t) = \int_0^t \frac{1}{\omega} \sin \omega(t - t') f(t') dt'.$$

(The upper limit is $t' = t$ since $G = 0$ for $t' > t$.) Thus, given a forcing function $f(t)$, we can find the response $y(t)$ of the system (12.1) by integrating (12.6) (see Problems 2 to 5). Similarly for other differential equations we can find the solution in terms of an appropriate Green function (see Problems 6 to 8).

► **Example 2.** As we will see later (Chapter 13, Section 8), in using Green functions in three-dimensional problems, we usually want a solution which is zero on the boundary of some region. In order to have a similar problem here, let us ask for a solution of

$$(12.7) \quad y'' + y = f(x)$$

such that $y = 0$ at $x = 0$ and at $x = \pi/2$. A physical interpretation of this problem may be useful. If a string is stretched along the x axis from $x = 0$ to $x = \pi/2$, and then caused to vibrate by a force proportional to $-f(x) \sin \omega t$, then $|y(x)|$ in (12.7) gives the amplitude of small vibrations.

We first find a solution of [compare (12.3)]

$$(12.8) \quad \frac{d^2}{dx^2} G(x, x') + G(x, x') = \delta(x' - x)$$

satisfying $G(0, x') = G(\pi/2, x') = 0$; this solution is the Green function for our problem. Then [compare (12.4)]

$$(12.9) \quad y(x) = \int_0^{\pi/2} G(x, x') f(x') dx'$$

gives a solution of (12.7) satisfying the conditions $y(0) = y(\pi/2) = 0$ (Problem 9).

To construct the desired Green function, we first note that for any $x \neq x'$, the equation (12.8) becomes

$$(12.10) \quad \frac{d^2}{dx^2} G(x, x') + G(x, x') = 0, \quad x \neq x'.$$

The solutions of (12.10) are $\sin x$ and $\cos x$; we observe that $\sin x = 0$ at $x = 0$ and $\cos x = 0$ at $x = \pi/2$. Thus we try to find a Green function of the form

$$(12.11) \quad G(x, x') = \begin{cases} A(x') \sin x, & 0 < x < x' < \pi/2, \\ B(x') \cos x, & 0 < x' < x < \pi/2. \end{cases}$$

The next step may be clarified by thinking about the string problem. If the string is oscillated by a concentrated force at x' [see (12.8)], then the amplitude of the vibration given by (12.11) is shown in Figure 12.1. At $x = x'$, $G(x, x')$ is continuous, that is, from (12.11)

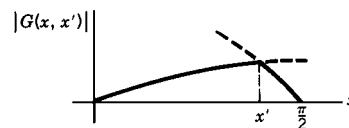


Figure 12.1

$$(12.12) \quad A(x') \sin x' = B(x') \cos x'.$$

However (see Figure 12.1), the slope changes abruptly at x' . From (12.11), we find

$$(12.13) \quad \frac{d}{dx} G(x, x') = \begin{cases} A(x') \cos x, & x < x', \\ -B(x') \sin x, & x > x'. \end{cases}$$

Change in $\frac{dG}{dx}$ at x' is $-B(x') \sin x' - A(x') \cos x'$.

We can evaluate this change in dG/dx by integrating (12.8) from $x = x' - \epsilon$ to $x = x' + \epsilon$ and letting $\epsilon \rightarrow 0$. Since $\int d^2 G/dx^2 = dG/dx$, we find

$$\frac{dG}{dx} \Big|_{x'-\epsilon}^{x'+\epsilon} + \int_{x'-\epsilon}^{x'+\epsilon} G(x, x') dx = \int_{x'-\epsilon}^{x'+\epsilon} \delta(x' - x) dx = 1,$$

or, letting $\epsilon \rightarrow 0$:

$$\left(\text{Change in slope } \frac{dG}{dx} \text{ at } x' \right) \text{ is } 1.$$

Then from (12.13)

$$(12.14) \quad -B(x') \sin x' - A(x') \cos x' = 1.$$

We solve (12.12) and (12.14) for $A(x')$ and $B(x')$ (Problem 10) and get

$$(12.15) \quad A(x') = -\cos x', \quad B(x') = -\sin x'.$$

Thus we have

$$(12.16) \quad G(x, x') = \begin{cases} -\cos x' \sin x, & 0 < x < x' < \pi/2, \\ -\sin x' \cos x, & 0 < x' < x < \pi/2. \end{cases}$$

Then from (12.9), the solution of (12.7) with $y(0) = y(\pi/2) = 0$ is

$$(12.17) \quad y(x) = -\cos x \int_0^x (\sin x') f(x') dx' - \sin x \int_x^{\pi/2} (\cos x') f(x') dx'.$$

► **Example 3.** If $f(x) = \csc x$, we find from (12.17):

$$\begin{aligned} y(x) &= -\cos x \int_0^x \sin x' \csc x' dx' - \sin x \int_x^{\pi/2} \cos x' \csc x' dx' \\ &= (-\cos x)(x) - (\sin x)(\ln \sin x') \Big|_x^{\pi/2} = -x \cos x + (\sin x)(\ln \sin x). \end{aligned}$$

It is interesting to note that we can use the Green function method to obtain a particular solution of a nonhomogeneous differential equation (nonzero right-hand side) when we know the solutions of the corresponding homogeneous equation (zero right-hand side). (See Problems 14 to 18.) In (12.17) each integral gives a function of x minus a constant (from the constant limits); these constants times $\sin x$ and $\cos x$ give a solution of the homogeneous equation. Thus the remaining terms give a particular solution of the nonhomogeneous equation. We can write this particular solution in a simple form by changing $\int_x^{\pi/2}$ to $-\int_{\pi/2}^x$, dropping the constant limits and writing indefinite integrals. Then a particular solution $y_p(x)$ of (12.7) is given by

$$(12.18) \quad y_p(x) = -\cos x \int (\sin x) f(x) dx + \sin x \int (\cos x) f(x) dx.$$

► **Example 4.** By the same methods used above, you can verify (Problem 14) that a solution of the differential equation

$$(12.19) \quad y'' + p(x)y' + q(x)y = f(x)$$

with $y(a) = y(b) = 0$ is given by

$$(12.20) \quad y(x) = y_2(x) \int_a^x \frac{y_1(x')f(x')}{W(x')} dx' + y_1(x) \int_x^b \frac{y_2(x')f(x')}{W(x')} dx',$$

where $y_1(x)$ and $y_2(x)$ are solutions of the homogeneous equation with $y_1(a) = 0$, $y_2(b) = 0$, and W is the Wronskian of $y_1(x)$ and $y_2(x)$ [See Chapter 3, equation (8.5)]. Just as in (12.18), we find that a particular solution y_p of (12.19) is

$$(12.21) \quad y_p(x) = y_2(x) \int \frac{y_1(x)f(x)}{W(x)} dx - y_1(x) \int \frac{y_2(x)f(x)}{W(x)} dx.$$

The particular solution (12.18) and (12.21) are exactly the same as those obtained by the method of *variation of parameters* (see Problem 14b) but the Green function method may seem less arbitrary.

► PROBLEMS, SECTION 12

1. Solve (12.3) if $G = 0$ and $dG/dt = 0$ at $t = 0$ to obtain (12.5). *Hint:* Use L28 and L3 to find the inverse transform.

In Problems 2 and 3, use (12.6) to solve (12.1) when $f(t)$ is as given.

2. $f(t) = \sin \omega t$
3. $f(t) = e^{-t}$

4. Use equation (12.6) to solve Problem 10.18.
5. Obtain (12.6) by using the convolution integral to solve (12.1).
6. For Problem 10.17, show (as in Problem 1) that the Green function is

$$G(t, t') = \begin{cases} 0, & 0 < t < t', \\ (1/a) \sinh a(t - t'), & 0 < t' < t. \end{cases}$$

Thus write the solution of Problem 10.17 as an integral [similar to (12.6)] and evaluate it.

7. Use the Green function of Problem 6 to solve

$$y'' - a^2 y = e^{-t}, \quad y_0 = y'_0 = 0.$$

8. Solve the differential equation $y'' + 2y' + y = f(t)$, $y_0 = y'_0 = 0$, where

$$f(t) = \begin{cases} 1, & 0 < t < a, \\ 0, & t > a. \end{cases}$$

As in Problems 6 and 7, find the Green function for the problem and use it in equation (12.4). Consider the cases $t < a$ and $t > a$ separately.

9. Following the proof of (12.4), show that (12.9) gives a solution of (12.7).
10. Solve (12.12) and (12.14) to get (12.15). *Hint:* Use Cramer's rule (Chapter 3, Section 3); note that the denominator determinant is the Wronskian [Chapter 3, equation (8.5)] of the functions $\sin x$ and $\cos x$.

In Problems 11 to 13, use (12.17) to find the solution of (12.7) with $y(0) = y(\pi/2) = 0$ when the forcing function is given $f(x)$.

11. $f(x) = \sin 2x$

12. $f(x) = \sec x$

13. $f(x) = \begin{cases} x, & 0 < x < \pi/4 \\ \pi/2 - x, & \pi/4 < x < \pi/2. \end{cases}$

Hint: Write separate formulas for $y(x)$ for $x < \pi/4$ and $x > \pi/4$.

14. (a) Given that $y_1(x)$ and $y_2(x)$ are solutions of (12.19) with $f(x) = 0$, and that $y_1(a) = 0$, $y_2(b) = 0$, find the Green function [as in (12.11) to (12.16)] and so obtain the solution (12.20). Then find the particular solution (12.21) as discussed for (12.18) and (12.21).
- (b) The method of *variation of parameters* is an elementary way of finding a particular solution of (12.19) when you know the solutions of the homogeneous equation. Show as follows that this method leads to the same result (12.21) as the Green function method. Start with the known solution of the homogeneous equation, say $y = c_1 y_1 + c_2 y_2$ and allow the "constants" to be functions of x to be determined so that y satisfies (12.19). (The c 's are the "parameters" which are to be "varied" in the expression "variation of parameters".) You want to find y' and y'' to substitute into (12.19). First find y' and set the sum of the terms involving derivatives of the c 's equal to zero. Differentiate the rest of y' again to get y'' . Now substitute y , y' and y'' into (12.19) and use the fact that y_1 and y_2 both satisfy the homogeneous equation [that is, (12.19) with $f(x) = 0$]. You should have the two equations:

$$\begin{aligned} c'_1 y_1 + c'_2 y_2 &= 0, \\ c'_1 y'_1 + c'_2 y'_2 &= f(x). \end{aligned}$$

Solve this pair of equations for c'_1 and c'_2 [say by determinants, and note that the denominator determinant is the Wronskian as in (12.20) and (12.21)]. Write the indefinite integrals for c_1 and c_2 , and write $y = c_1 y_1 + c_2 y_2$ to get (12.21).

In Problems 15 to 18, use the given solutions of the homogeneous equation to find a particular solution of the given equation. You can do this either by the Green function formulas in the text or by the method of variation of parameters in Problem 14b.

15. $y'' - y = \operatorname{sech} x$; $\sinh x, \cosh x$
16. $x^2 y'' - 2xy' + 2y = x \ln x$; x, x^2
17. $y'' - 2(\csc^2 x)y = \sin^2 x$; $\cot x, 1 - x \cot x$
18. $(x^2 + 1)y'' - 2xy' + 2y = (x^2 + 1)^2$; $x, 1 - x^2$

► 13. MISCELLANEOUS PROBLEMS

Identify each of the differential equations in Problems 1 to 24 as to type (for example, separable, linear first order, linear second order, etc.), and then solve it.

1. $x^2 y' - xy = 1/x$
2. $x(\ln y)y' - y \ln x = 0$
3. $y''' + 2y'' + 2y' = 0$
4. $\frac{d^2 r}{dt^2} - 6\frac{dr}{dt} + 9r = 0$
5. $(2x - y \sin 2x) dx = (\sin^2 x - 2y) dy$
6. $y'' + 2y' + 2y = 10e^x + 6e^{-x} \cos x$
7. $3x^3 y^2 y' - x^2 y^3 = 1$
8. $x^2 y'' - xy' + y = x$
9. $dy - (2y + y^2 e^{3x}) dx = 0$
10. $u(1 - v) dv + v^2(1 - u) du = 0$
11. $(y + 2x) dx - x dy = 0$
12. $xy'' + y' = 4x$
13. $y'' + 4y' + 5y = 26e^{3x}$
14. $y'' + 4y' + 5y = 2e^{-2x} \cos x$
15. $y'' - 4y' + 4y = 6e^{2x}$
16. $y'' - 5y' + 6y = e^{2x}$
17. $(2x + y) dy - (x - 2y) dx = 0$
18. $(x \cos y - e^{-\sin y}) dy + dx = 0$
19. $\sin^2 x dy + [\sin^2 x + (x + y) \sin 2x] dx = 0$
20. $y'' - 2y' + 5y = 5x + 4e^x(1 + \sin 2x)$
21. $y' + xy = x/y$
22. $(D - 2)^2(D^2 + 9)y = 0$
23. $\sin \theta \cos \theta dr - \sin^2 \theta d\theta = r \cos^2 \theta d\theta$
24. $x(yy'' + y'^2) = yy'$ Hint: Let $u = yy'$.

In Problems 25 to 28, find a particular solution satisfying the given conditions.

25. $3x^2 y dx + x^3 dy = 0$, $y = 2$ when $x = 1$.
26. $xy' - y = x^2$, $y = 6$ when $x = 2$
27. $y'' + y' - 6y = 6$, $y = 1$, $y' = 4$ when $x = 0$
28. $yy'' + y'^2 + 4 = 0$ $y = 3$, $y' = 0$ when $x = 1$
29. If 10 kg of rock salt is placed in water, it dissolves at a rate proportional to the amount of salt still undissolved. If 2kg dissolve during the first 10 minutes, how long will it be until only 2kg remain undissolved?

30. A mass m falls under gravity (force mg) through a liquid whose viscosity is decreasing so that the retarding force is $-2mv/(1+t)$, where v is the speed of m . If the mass starts from rest, find its speed, its acceleration, and how far it has fallen (in terms of g) when $t = 1$.
31. The acceleration of an electron in the electric field of a positively charged sphere is inversely proportional to the square of the distance between the electron and the center of the sphere. Let an electron fall from rest at infinity to the sphere. What is the electron's velocity when it reaches the surface of the sphere?
32. Suppose that the rate at which you work on a hot day is inversely proportional to the excess temperature above 75° . One day the temperature is rising steadily, and you start studying at 2 p.m. You cover 20 pages the first hour and 10 pages the second hour. At what time was the temperature 75° ?
33. Compare the temperatures of your cup of coffee at time t
- (a) if you add cream and let the mixture cool;
 - (b) if you let the coffee and cream sit on the table and mix them at time t .

Hints: Assume Newton's law of cooling (Problem 2.27) for both coffee and cream (where it is a law of heating). Combine n' units of cream initially at temperature T'_0 with n units of coffee initially at temperature T_0 , and find the temperature at time t in (a) and in (b) assuming that the air temperature remains a constant T_a , and that the proportionality constant in the law of cooling is the same for both coffee and cream.

34. A flexible chain of length l is hung over a peg with one end of the chain slightly longer than the other. Assuming that the chain slides off with no friction, write and solve the differential equation of motion to show that $y = y_0 \cosh t\sqrt{2g/l}$, $0 < y < l/2$, where $2y$ is the difference in length of the two ends, and $y = y_0$ when $t = 0$.
35. A raindrop falls through a cloud, increasing in size as it picks up moisture. Assume that its shape always remains spherical. Also assume that the rate of increase of its volume with respect to distance fallen is proportional to the cross-sectional area of the drop at any time (that is, the mass increase $dm = \rho dV$ is proportional to the volume $\pi r^2 dy$ swept out by the drop as it falls a distance dy). Show that the radius r of the drop is proportional to the distance y the drop has fallen if $r = 0$ when $y = 0$. Recall that when m is not constant, Newton's second law is properly stated as $(d/dt)(mv) = F$. Use this equation to find the distance y which the drop falls in time t under the force of gravity, if $y = \dot{y} = 0$ at $t = 0$. Show that the acceleration of the drop is $g/7$ where g is the acceleration of gravity.
36. (a) A rocket of (variable) mass m is propelled by steadily ejecting part of its mass at velocity u (constant with respect to the rocket). Neglecting gravity, the differential equation of the rocket is $m(dv/dm) = -u$ as long as $v \ll c$, $c =$ speed of light. Find v as a function of m if $m = m_0$ when $v = 0$.
- (b) In the relativistic region (v/c not negligible), the rocket equation is

$$m \frac{dv}{dm} = -u \left(1 - \frac{v^2}{c^2} \right).$$

Solve this differential equation to find v as a function of m . Show that $v/c = (1-x)/(1+x)$, where $x = (m/m_0)^{2u/c}$.

- 37.** The differential equation for the path of a planet around the sun (or any object in an inverse square force field) is, in polar coordinates,

$$\frac{1}{r^2} \frac{d}{d\theta} \left(\frac{1}{r^2} \frac{dr}{d\theta} \right) - \frac{1}{r^3} = -\frac{k}{r^2}.$$

Make the substitution $u = 1/r$ and solve the equation to show that the path is a conic section.

- 38.** Use *L15* and *L31* to find the Laplace transform of $(1 - \cos at)/t$.
39. Use *L32* and *L9* to find the Laplace transform of $t \sinh at$. Verify your result by finding its inverse transform using the convolution integral.

Use the Laplace transform table to evaluate:

$$40. \int_0^\infty t^3 e^{-4t} \sinh 2t \, dt \qquad 41. \sum_{n=0}^\infty (-1)^n \int_n^{n+1} t e^{-2t} \, dt$$

Find the inverse Laplace transform of:

$$42. \frac{p}{(p+a)^3} \qquad 43. \frac{p^2}{(p^2+a^2)^2} \qquad 44. \frac{1}{(p^2+a^2)^3}$$

- 45.** Prove the following shifting or translation theorems for Fourier transforms. If $g(\alpha)$ is the Fourier transform of $f(x)$, then

- (a) the Fourier transform of $f(x-a)$ is $e^{-i\alpha a} g(\alpha)$;
 (b) the Fourier transform of $e^{i\beta x} f(x)$ is $g(\alpha - \beta)$.

Compare Problems 8.19 to 8.27.

- 46.** Use the table of Laplace transforms to find the sine and cosine Fourier transforms of e^{-x} ; of $x e^{-x}$.

Solve Problems 47 and 48 either by Laplace transforms and the convolution integral or by Green functions.

$$47. y'' + y = \sec^2 t \qquad 48. y'' + y = t \sin t$$

Table of Laplace Transforms

	$y = f(t), t > 0$ $[y = f(t) = 0, t < 0]$	$Y = L(y) = F(p) = \int_0^\infty e^{-pt} f(t) dt$	
L1	1	$\frac{1}{p}$	$\operatorname{Re} p > 0$
L2	e^{-at}	$\frac{1}{p+a}$	$\operatorname{Re} (p+a) > 0$
L3	$\sin at$	$\frac{a}{p^2 + a^2}$	$\operatorname{Re} p > \operatorname{Im} a $
L4	$\cos at$	$\frac{p}{p^2 + a^2}$	$\operatorname{Re} p > \operatorname{Im} a $
L5	$t^k, k > -1$	$\frac{k!}{p^{k+1}}$ or $\frac{\Gamma(k+1)}{p^{k+1}}$	$\operatorname{Re} p > 0$
L6	$t^k e^{-at}, k > -1$	$\frac{k!}{(p+a)^{k+1}}$ or $\frac{\Gamma(k+1)}{(p+a)^{k+1}}$	$\operatorname{Re} (p+a) > 0$
L7	$\frac{e^{-at} - e^{-bt}}{b-a}$	$\frac{1}{(p+a)(p+b)}$	$\operatorname{Re} (p+a) > 0$ $\operatorname{Re} (p+b) > 0$
L8	$\frac{ae^{-at} - be^{-bt}}{a-b}$	$\frac{p}{(p+a)(p+b)}$	$\operatorname{Re} (p+a) > 0$ $\operatorname{Re} (p+b) > 0$
L9	$\sinh at$	$\frac{a}{p^2 - a^2}$	$\operatorname{Re} p > \operatorname{Re} a $
L10	$\cosh at$	$\frac{p}{p^2 - a^2}$	$\operatorname{Re} p > \operatorname{Re} a $
L11	$t \sin at$	$\frac{2ap}{(p^2 + a^2)^2}$	$\operatorname{Re} p > \operatorname{Im} a $
L12	$t \cos at$	$\frac{p^2 - a^2}{(p^2 + a^2)^2}$	$\operatorname{Re} p > \operatorname{Im} a $
L13	$e^{-at} \sin bt$	$\frac{b}{(p+a)^2 + b^2}$	$\operatorname{Re} (p+a) > \operatorname{Im} b $
L14	$e^{-at} \cos bt$	$\frac{p+a}{(p+a)^2 + b^2}$	$\operatorname{Re} (p+a) > \operatorname{Im} b $
L15	$1 - \cos at$	$\frac{a^2}{p(p^2 + a^2)}$	$\operatorname{Re} p > \operatorname{Im} a $
L16	$at - \sin at$	$\frac{a^3}{p^2(p^2 + a^2)}$	$\operatorname{Re} p > \operatorname{Im} a $
L17	$\sin at - at \cos at$	$\frac{2a^3}{(p^2 + a^2)^2}$	$\operatorname{Re} p > \operatorname{Im} a $

Table of Laplace Transforms (continued)

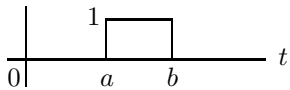
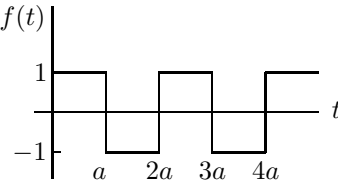
	$y = f(t), t > 0$ $[y = f(t) = 0, t < 0]$	$Y = L(y) = F(p) = \int_0^\infty e^{-pt} f(t) dt$	
L18	$e^{-at}(1 - at)$	$\frac{p}{(p+a)^2}$	$\operatorname{Re} (p+a) > 0$
L19	$\frac{\sin at}{t}$	$\arctan \frac{a}{p}$	$\operatorname{Re} p > \operatorname{Im} a $
L20	$\frac{1}{t} \sin at \cos bt,$ $a > 0, b > 0$	$\frac{1}{2} \left(\arctan \frac{a+b}{p} + \arctan \frac{a-b}{p} \right)$	$\operatorname{Re} p > 0$
L21	$\frac{e^{-at} - e^{-bt}}{t}$	$\ln \frac{p+b}{p+a}$	$\operatorname{Re} (p+a) > 0$ $\operatorname{Re} (p+b) > 0$
L22	$1 - \operatorname{erf} \left(\frac{a}{2\sqrt{t}} \right), \quad a > 0$ (See Chapter 11, Section 9)	$\frac{1}{p} e^{-a\sqrt{p}}$	$\operatorname{Re} p > 0$
L23	$J_0(at)$ (See Chapter 12, Section 12)	$(p^2 + a^2)^{-1/2}$	$\operatorname{Re} p > \operatorname{Im} a ;$ or $\operatorname{Re} p \geq 0$ for real $a \neq 0$
L24	$u(t-a) = \begin{cases} 1, & t > a > 0 \\ 0, & t < a \end{cases}$ (unit step, or Heaviside function)	$\frac{1}{p} e^{-pa}$	$\operatorname{Re} p > 0$
L25	$f(t) = u(t-a) - u(t-b)$ 	$\frac{e^{-ap} - e^{-bp}}{p}$	All p
L26	$f(t)$ 	$\frac{1}{p} \tanh \left(\frac{1}{2} ap \right)$	$\operatorname{Re} p > 0$
L27	$\delta(t-a), a \geq 0$ (See Section 11)	e^{-pa}	
L28	$f(t) = \begin{cases} g(t-a), & t > a > 0 \\ 0, & t < a \end{cases}$ $= g(t-a)u(t-a)$	$e^{-pa}G(p)$ [$G(p)$ means $L(g)$.]	
L29	$e^{-at}g(t)$	$G(p+a)$	

Table of Laplace Transforms (continued)

	$y = f(t), t > 0$ $[y = f(t) = 0, t < 0]$	$Y = L(y) = F(p) = \int_0^\infty e^{-pt} f(t) dt$
L30	$g(at), a > 0$	$\frac{1}{a} G\left(\frac{p}{a}\right)$
L31	$\frac{g(t)}{t}$ (if integrable)	$\int_p^\infty G(u) du$
L32	$t^n g(t)$	$(-1)^n \frac{d^n G(p)}{dp^n}$
L33	$\int_0^t g(\tau) d\tau$	$\frac{1}{p} G(p)$
L34	$\int_0^t g(t-\tau)h(\tau) d\tau = \int_0^t g(\tau)h(t-\tau) d\tau$ (convolution of g and h , often written as $g * h$; see Section 10)	$G(p)H(p)$
L35	Transforms of derivatives of y (see Section 9): $L(y') = pY - y_0$ $L(y'') = p^2Y - py_0 - y'_0$ $L(y''') = p^3Y - p^2y_0 - py'_0 - y''_0$, etc. $L(y^{(n)}) = p^nY - p^{n-1}y_0 - p^{n-2}y'_0 - \dots - y_0^{(n-1)}$	